

Certified finite-model gauge calculations: spectral bounds, local corrections, Wilson readouts, and controlled heat evolution

First manuscript draft for author review

22 September 2026

Research checkpoint: Round26

Abstract

We collect the derivations, computational certificates, counterexamples, and unresolved obligations of a gauge-theory research workbench. Its surviving record begins with regulated Maxwell–Dirac and gravitational consistency studies and develops into finite-graph and fixed-spacing $SU(2)$ calculations. The main results include full-representation one-square spectral certificates, shared-link Haar contractions, conditional local stability, projected-memory and boundary-response formulas, canonical slow-time Wilson readouts, and controlled physical heat evolution. The latest cycle constructs a graph-preserving Gaussian correction with an explicit full-source residual, an explicitly enumerated 293-state physical compression on an 18-vertex graph, and an all-time evaluated heat approximation with true-output-relative error below 3.7×10^{-5} on a specified preparation class. A restricted compact-time filter also gives per-source contraction with finite original support-cardinality-weighted norms. We explain the brief derivations, parameter domains, changes from earlier results, possible applications, and failure modes, and provide calculators, plots, a contribution catalog, and a skeptical audit. The record contains 121 study or loop entries across Rounds 3–26; separate Round1/2 packages are absent. The double-star marker identifies a contribution newly derived or implemented in this workbench, not established priority in the literature. The results concern explicitly different models and do not construct continuum Yang–Mills theory or establish its mass gap.

Draft status. Author names and affiliations remain to be supplied. No arXiv submission has been made. Separate model-agent review is not external peer review or formal verification.

Contents

1	Starting point, question, and evidence boundary	4
1.1	Coverage and reproducible checkpoint	4
1.2	What the two stars mean	4
1.3	Research design and attribution	5
2	Models, notation, and nontransfer rules	5
3	Relation to established results	6
4	From semiclassical backreaction to a controlled finite gauge theory	7
4.1	The original closure problem and its finite electromagnetic realization	7
4.2	What gravitational and scalar extensions actually add	10
4.3	Spectral logic before a nonabelian calculation	11
4.4	Gauge reduction, infinite representation tails, and certified dynamics	12
5	Finite Haar constraints and connected lattice integrals	16
5.1	Compact moments, response, and complete interval coverage **	16
5.2	Closed-surface gluing and the first shared face **	18
5.3	Recoupling, sufficient coordinates, and surrounding weights **	18
6	Physical spectral bounds and an explicit infinite-volume exception	19
6.1	Finite untruncated graph bounds and complete complements **	19
6.2	Dressed strips, literal boundaries, and summable perturbations **	21
6.3	Physical GNS space and a nonzero Wilson fluctuation **	22
7	Locality, profile limits, and stationary observations	22
7.1	An exact spatial budget and its endpoint **	23
7.2	Stationarity improves the time window **	24
8	Static information, clocks, and exact physical reduction	24
8.1	Conditional dynamics and finite-rate identifiability **	24
8.2	A failed section and a valid maximal-tree map **	25
8.3	Magnetic leakage, projected memory, and exact channels **	26
9	Homogeneous stability: explicit progress and retained obstructions	27
9.1	A matched theorem dictionary and a controlled rotation **	27
9.2	Generated supports, interacting inverses, and the remaining source **	28
10	Full sources, endpoint observables, and certified heat: Rounds 23–26	30
10.1	An actual homogeneous source and its regulated inverses	30
10.2	A fixed physical graph: from memory to computed relative heat	34
10.3	Canonical endpoint observables and an exact reached spectrum	42

11 What changed, and what the results can be used for	48
11.1 From closed retained dynamics to complete channel loading	48
11.2 From a useful heat bound to an executable all-time result	49
11.3 From a source correction to a nonlinear stability problem	49
11.4 From a resonant state to an informative observable	49
11.5 Evidence networks and nonphysical geometry	50
12 Verification, stress tests, and interpretation of the figures	50
12.1 What the new checks challenge	51
12.2 Arithmetic, analytic scope, and sampling	51
12.3 Known limits of this audit	51
13 Next planned goals and remaining proof obligations	57
A Contribution catalog, applications, and remaining obligations	60
B Complete historical record	82
C Reproduction and manuscript build	91

1 Starting point, question, and evidence boundary

The initial motivation was to turn an exploratory Einstein–QED workbench into a program addressing precise nonabelian gauge questions. The surviving early calculations concern a regulated homogeneous Maxwell–Dirac mean field and reduced gravitational consistency, rather than a construction of either quantum gravity or continuum QED. Subsequent work asks what a finite $SU(2)$ gauge system can establish exactly, which approximations can be certified, and what additional premises would be needed to connect such results to the four-dimensional Yang–Mills existence and mass-gap problem. The latter requires a nontrivial continuum quantum theory satisfying appropriate axioms and a positive physical mass gap; a finite lattice gap alone is insufficient [1].

This paper is a research compendium and first manuscript, not a claim that the final target has been reached. It preserves failed claims because they determine the next mathematical question. The transition from a matrix with a positive gap to an infinite representation space requires a tail bound. The transition from a finite graph to an infinite lattice requires uniform estimates. The transition from a static probability measure to a Hamiltonian requires a justified state-space and generator map. Each of these transitions appears explicitly in the history.

1.1 Coverage and reproducible checkpoint

The source checkpoint is repository commit 40960f39a3dcaa6b2735d47adaa0e3b40e6a0f02, with Git tree 169b2db4e8017f5c4e783ba5f42e4f96522ad2b1. The repository is https://github.com/occult-kranti/yang_mills_workbench. The historical ledger records 111 entries through Round25, comprising 35 named study entries and 76 explicitly numbered physics loops. Round26 adds ten loops. This gives 121 entries and 86 explicit loops across 24 recorded rounds, Rounds3–26. An entry is an audit unit, not necessarily a distinct discovery. There are no independently identifiable Round1/2 packages in this checkpoint; their contents cannot be reconstructed from a round number.

The paper groups repeated refinements into mathematical contributions while Appendix B retains every recorded entry. The exact contracts, complete longer proofs, source files, archived outputs, and historical reviews remain in the pinned repository. A concise derivation here is not a replacement for a missing historical proof. Where only a numerical study or conditional argument is available, it is described at that level.

1.2 What the two stars mean

The marker ****** accompanies a workbench contribution and its central newly derived or specialized equation. It means *newly derived, computed, repaired, or implemented in this research record*. It does not mean a new law of nature, a historically first theorem, or a completed literature-priority search. Established ingredients—Haar integration, Schur complements, spectral projection, Duhamel formulas, Rayleigh–Ritz and residual bounds, and Lieb–Robinson estimates—remain established mathematics. Their use in a new calculation does not make the underlying method an invention.

The appendix classifies contributions as exact identities, conditional bounds, certified arithmetic, numerical diagnostics, obstructions, and workflow artifacts. Proposed future goals are not starred results. Literature comparison identifies the nearest checked sources and remaining differences; scientific priority is unverified throughout this first draft.

1.3 Research design and attribution

The workflow pairs a forward calculation from frozen premises with a reverse reconstruction of the premises sufficient for a desired conclusion. An advisor chooses the next experiment from the first result and skeptical objection. Separate model agents wrote many forward, reverse, and skeptical records; these share the same project, inherited premises, and model infrastructure. This is not independent experimental replication. Round23 T2/U1/U2 and Round25 AA1/AA2 retain their same-author review attribution. Replays, software tests, admission repairs, and manuscript checks are not counted as new physics loops.

Newton’s analysis/synthesis motivates the question “which assumptions are necessary to reconstruct this observable?” Tesla’s mechanism and loading approach motivates “which omitted channel changes the proposed device or resonance?” These are modern methodological adaptations of selected historical sources; the inspected Opticks discussion and Tesla patent are historical context, not mathematical authority [2, 3]. They supply no premise in an operator proof. The double Fibonacci spiral was a way to organize investigation nodes; no tested physical significance or golden-ratio coupling has been established. No historical or esoteric analogy is inserted as a term in the gauge Hamiltonian.

2 Models, notation, and nontransfer rules

Keep a positive lattice spacing a , physical energy reference E_* , α/E_* , and $\hbar$ fixed unless a stated construction changes them. A change of notation cannot serve as physical scale matching. Dimensionless eigenvalue inequalities become physical energy inequalities only after multiplication by their own energy unit.

Table 1: Distinct model families. Symbols reused in historical reports are local to the stated model.

Family	Generator or object	Scope and scale
Early mean field	Finite-regulator Maxwell–Dirac variables and declared scalar/gravity reductions	Classical mean-field evolution; cutoff and closure assumptions remain.
Static compact integrals	Normalized Haar density with specified Euclidean action coefficients	Coupling derivatives are static responses, not physical time evolution.
Finite physical gauge graph	$H = \alpha L$, with $L = K + \lambda(20 - S)$ in the latest graph	18 vertices, 33 links, 20 faces; all vertex Gauss constraints; dimensionless heat coordinate $\sigma = \alpha t/\hbar$.
Homogeneous complete blocks	Initial interacting operator G ; $M = 7 \tau $	Energy unit $\delta = \alpha/8$; original $ \tau \leq 5/1664$; stronger AE2 result uses $ \tau \leq 1/7000$.
Canonical summable profile	Declared q -dependent full-link family	Fixed spacing; $q \uparrow 1$; physical observation time $T_q = (z/\eta)(\hbar/\alpha)(1 - q)^{-3}$.

In finite-link gauge systems the ambient Hilbert space is a product of $L^2(\text{SU}(2))$ spaces and the physical space is the invariant subspace under the independent vertex gauge actions. A finite number of links does not make this representation space finite-dimensional. A retained projector P must be built from actual physical vectors; $Q = I - P$ denotes its full complement. On the latest graph, $P_{21} = \mathbf{1}_{[0,9/2)}(K)$ is the complete strict low-electric-energy sector. Its 293-state residual enrichment reduces K but is not a complete ambient energy cutoff. The 293-coordinate representation has a nontrivial Gram matrix, so coordinate Euclidean norms cannot replace physical norms.

For positive self-adjoint L with ground energy ϵ we use centered heat $e^{-\sigma(L-\epsilon)}$. For $A = PLP$

with its own lowest eigenvalue μ , the correct retained comparison is $e^{-\sigma(A-\mu)P}$. These two centers are generally different. Neither heat semigroup is a real-time unitary $e^{-itH/\hbar}$; an estimate for one is not silently transferred to the other.

An observable is a full operator, together with its state and readout. Equality $B\Omega = C\Omega$ identifies a vacuum-created vector but does not imply $B = C$, equal operator norms, or transferable finite-parameter error constants. An exact identity, a norm estimate, and a numerical evaluation are separate results. These distinctions are repeatedly used below.

3 Relation to established results

The earlier QED models draw on established, regime-specific methods: constant-field electric-probe response with a stated finite subtraction [4], homogeneous mean-field pair production with adiabatic current renormalization [5], and locally covariant current and stress renormalization with explicit background restrictions [6]. These sources do not themselves provide the workbench's evaluated finite-mode formulas or a general strong-field Bianchi-I closure.

The lattice and Hamiltonian frameworks originate with Wilson and Kogut–Susskind [7, 8]. For the particular reductions studied here, the physical single-plaquette problem already has a Mathieu-function formulation [9, Appendix B], and the open two-plaquette electric spectrum has explicit low-level tabulations [10, Appendix C]. Maximal-tree gauge fixing also has a developed operator treatment: its electric generators require transported derivatives, beyond the kinematic identification of gauge orbits [11, Appendix C]. Thus these reductions and low levels are comparison targets, not claims of discovery. The possible increment is the explicit graph arithmetic, Gram identities, complete residual accounting, and inspectable bounds recorded below. The Gegenbauer polynomials and variational tools used in those calculations are classical [12, 13].

Product-vacuum stability already permits infinite-dimensional onsite spaces under appropriate gap and bounded-interaction hypotheses [14]. Local stability of ground-state expectations is a related statement whose local perturbation need not preserve the perturbed global gap [15, Theorem 3]. Explicit qubit thresholds [16] and finite-dimensional local block diagonalization [17] have narrower hypotheses than some models here. However, a broader construction does cover separable onsite spaces and relatively form-bounded interactions [18]. Its relevance prevents any claim that unbounded-interaction stability is absent from the literature. Applying it quantitatively still requires matching domains, interaction geometry, weighted norms, and smallness constants. This compendium does not supply an evaluated general homogeneous threshold by citing that theorem.

Locality and approximation likewise have established frameworks. Nachtergaele–Sims treat unbounded onsite terms with controlled bounded intersite interactions [19]; their operator topologies must be retained when comparing time continuity and spatial limits. Exponential-in-volume tails offer a further lead under a finite-qudit setup [20], not an admitted extension to these rotor factors. Rigorous quantum-number truncation estimates [21] require their specified initial-state and Hamiltonian assumptions. They do not automatically certify an arbitrary retained block or its numerically centered heat evolution.

Finally, strong-coupling Wilson-measure and stochastic-generator results [22] concern explicitly specified measures and clocks. They do not identify an auxiliary Markov gap with the canonical physical-time gap. Continuum reconstruction requires a full compatible Schwinger-function hierarchy with the corrected regularity and other axioms [23]; the four-dimensional existence and physical mass-gap target remains distinct [1]. This is a selected literature comparison, not an exhaustive priority search. Scientific novelty of the model-specific certificates has not been established.

4 From semiclassical backreaction to a controlled finite gauge theory

The early program asked how a quantum current, electromagnetic backreaction, and gravitational constraints could be calculated consistently. The surviving record begins with Round 3. It contains imported prescribed-background results and descriptions of earlier research fronts, but no independently identifiable Round 1 or Round 2 package. We therefore do not reconstruct their missing contracts or assign them new results. The present section accounts for all 31 named studies in Rounds 3–12, using the Round 25 history inventory and the underlying mathematical reports. These studies precede the explicit paired loop contracts introduced later. Here ^{**} marks a contribution to this workbench, including an implementation or an independently derived obstruction; it does not establish priority over the mathematical literature.

4.1 The original closure problem and its finite electromagnetic realization

Initial scope. The organizing Einstein–QED action contained Einstein gravity, Maxwell and Dirac terms, possible scalar-dependent coefficients, and curvature counterterms. After integrating out the electron, the intended causal object was a closed-time-path functional $\Gamma[g_+, A_+; g_-, A_-; \rho_0]$. Varying one branch before identifying the branches defines state-dependent current and stress. This is a specification of a problem, not its solution. A local vacuum-persistence rate is not a causal current; an Euler–Heisenberg expansion cannot be added independently to the same full electron determinant. The four contextual fronts—charged de Sitter discharge, magnetized pair creation, Cauchy-horizon dynamics, and field mixing—have different geometries and boundary data. No theorem identifies their solutions. Sources: `evidence/qeg-research/round3/chapters/01_master_action.md` and `evidence/qeg-research/round3/chapters/02_fronts.md`.

Matched finite Maxwell–Dirac closure ^{}.** Self-consistent field/mode initial-value formulations and adiabatic regularization have established precedents [5]. The executable reduction fixes a flat homogeneous geometry, a static magnetic field, finitely many Landau levels, and a finite positive momentum quadrature. This early reduction uses Heaviside–Lorentz natural units, $c = \hbar = 1$; later gauge-graph comparisons restore their explicitly stated physical scales. With $s = mt$, $a = eA_z/m$, $x = eE/m^2$, and $b = |eB|/m^2$, set $p_i = k_i - a$, $M_i^2 = 1 + 2bn_i$, $\omega_i^2 = M_i^2 + p_i^2$, and $h_i = (M_i, 0, p_i)$. The positive weights are $w_i = b(2 - \delta_{n_i 0})w_{k_i}/(4\pi^2)$. A prime in this subsection means d/ds . Bloch evolution and three finite sums are

$$\begin{aligned} a' &= -x, & r'_i &= 2h_i \times r_i, \\ S &= \sum_i w_i(r_{iz} + p_i/\omega_i), & C &= \sum_i \frac{w_i M_i^2}{4\omega_i^5}, & D &= \sum_i \frac{5w_i M_i^2 p_i}{8\omega_i^7}. \end{aligned} \quad (1)$$

The fixed magnetic matching coefficient and divided field equation are

$$\begin{aligned} \chi_b &= \frac{e^2}{12\pi^2} \left[b - \log(2b) - \psi \left(1 + \frac{1}{2b} \right) \right], \\ Z &= 1 + \chi_b - e^2 C, & Zx' &= F - e^2(S + Dx^2). \end{aligned} \quad \text{**} \quad (2)$$

The susceptibility comparison uses the constant-field, second-order electric-probe and all-orders magnetic-field setting of [4]; the evolving current is obtained from the retained modes. Here F is the prescribed source. The mode subtraction is $c_i = M_i^2 x'/(4\omega_i^5) - 5M_i^2 p_i x^2/(8\omega_i^7)$, with matching restored once. It follows by solving $2h_i \times r_i^{(2)} = (r_i^{(1)})'$, with $r_i^{(1)} = (0, -M_i x/(2\omega_i^3), 0)$, and fixing the longitudinal component by $|r_i| = 1$; the companion energy subtraction $M_i^2 x^2/(8\omega_i^5)$ differentiates to xc_i . The finite matching is the difference between the full-momentum magnetic

Landau sum and the zero-field coefficient $e^2 \log(1 + 2bN)/(12\pi^2)$, followed by $N \rightarrow \infty$. In particular $S + Dx^2$ is a numerator, whereas the full renormalized matter current in these conventions is $J = S - Cx' + Dx^2 + (\chi_b/e^2)x'$. Confusing the two changes the response.

Define $U = \sum_i w_i(h_i \cdot r_i + \omega_i)$. Direct differentiation uses $p'_i = x$, $h_i \cdot (h_i \times r_i) = 0$, and $C' = -2Dx$ to give $U' = xS$, $Z' = 2e^2Dx$, and therefore

$$W = \frac{1}{2}Zx^2 + e^2U, \quad W' = xF. \quad ** \quad (3)$$

This is an exact finite-model identity, not a continuum error estimate. For physical $|r_i| \leq 1$, $U \geq 0$. If $Z(a) \geq z_* > 0$ for all a , then $W \geq z_*x^2/2$ and, by applying the differential inequality first to $\sqrt{W + \varepsilon}$ and then letting $\varepsilon \downarrow 0$,

$$\sqrt{W(s)} \leq \sqrt{W(0)} + \frac{1}{\sqrt{2z_*}} \int_0^s |F(u)| du. \quad (4)$$

The spins remain in compact balls, x is bounded on each finite interval, and $a = a_0 - \int x$ cannot diverge there. Local Lipschitz continuity of the finite vector field proves global classical continuation for continuous drives, as assumed in the historical theorem. An initially mixed mode satisfying $|r_j| \leq 1 - \delta$ also gives $U \geq w_j\delta|k_j - a|$, hence bounds a for an integrable drive. That additional premise does not hold for every pure-vacuum production run.

The historical source-off calculation reached $x(50) \simeq -0.03757$ with 4096 momentum nodes, Landau levels 0–8, and window $K = 40$. The recorded 2048-to-4096 history difference 3.26×10^{-9} and separate Landau endpoint difference 5.06×10^{-8} are refinement diagnostics. The advisor's 10^{-18} absolute benchmark failed at approximately 1.82×10^{-18} ; it remains failed. Independent spinor/Bloch and matching controls established a further methodological point: a consistently wrong prescription may conserve its own energy. The separate massless lowest-Landau-level comparator obeys $x'' + \Omega_B^2 x = F'$, $\Omega_B^2 = e^2 b / (2\pi^2)$, in its own declared scaling; it is not obtained by dividing the massive variables by zero. Sources: `evidence/qeg-research/round3/theory.md`, `evidence/qeg-research/round3/README.md`, and `evidence/qeg-research/round5/theorem_advisor.md`.

For the weak probe about a stationary vacuum, eliminating the transverse components of the linearized Bloch equation supplies the oscillator denominator $4\omega_i^2 - (\nu + i0)^2$. The same zero-frequency subtraction gives

$$\chi(b, \nu) = \chi_b + e^2 \sum_i \frac{w_i M_i^2}{\omega_i^3} \left[\frac{1}{4\omega_i^2 - (\nu + i0)^2} - \frac{1}{4\omega_i^2} \right]. \quad ** \quad (5)$$

This tests matching independently of the nonlinear work identity. In an instantaneous eigenbasis, the mode current also contains a coherence: $r_{iz} + p_i/\omega_i = 2p_i f_i/\omega_i - M_i v_i^{\text{coh}}/\omega_i$. Retaining the occupation f_i alone cannot reproduce that current.

Causal tangent and the quantity a stability theorem controls **. Let $v = \delta a$, $u = \delta x$, $q_i = -v$, $\eta_i = \delta r_i$, and $f = \delta F$, with grid and matching held fixed. Differentiation gives

$$\begin{aligned} \delta S &= \sum_i w_i (\eta_{iz} + M_i^2 q_i / \omega_i^3), \\ \delta C &= - \sum_i \frac{5w_i M_i^2 p_i q_i}{4\omega_i^7}, & \delta D &= \sum_i \frac{5w_i M_i^2 (M_i^2 - 6p_i^2) q_i}{8\omega_i^9}, \\ v' &= -u, & u' &= \frac{f - e^2(\delta S + \delta D x^2 + 2Dxu) + e^2 \delta C x'}{Z}, \\ \eta'_i &= 2h_i \times \eta_i + 2q_i e_z \times r_i. \end{aligned} \quad ** \quad (6)$$

The fundamental-matrix integral gives causal support and finite-time Gronwall bounds. The first variation δW , satisfying $(\delta W)' = uF + xf$, is signed and cannot be used as a positive norm. At a stationary pure vacuum, restrict to variations preserving the Bloch norm, so $h_i \cdot \eta_i = 0$, and define $t_i = \partial_{p_i}(h_i/\omega_i)$ and $\xi_i = \eta_i + q_i t_i$. Then $Z_0 u' = f - e^2 \sum_i w_i \xi_{iz}$, $\xi'_i = 2h_i \times \xi_i + u t_i$, and

$$E_2 = Z_0 u^2 + e^2 \sum_i w_i \omega_i |\xi_i|^2, \quad E'_2 = 2uf. \quad ** \quad (7)$$

Indeed $h_i \cdot \xi_i = 0$ and $\omega_i \xi_i \cdot t_i = \xi_{iz}$ cancel the exchange terms. This controls u, ξ , not every raw tangent. For one mode with $M = w = e^2 = 1$, $p_0 = 0$, and $Z_0 = 3/4$, $u = 1$, $v = -s$, $\xi = (0, -1/2, 0)$, $\eta = (0, -1/2, -s)$ solve the unforced tangent while $E_2 = 1$. The secular raw tangent defeats a stronger stability claim. Rounds 4 and 6 also retained source-center and acceptance repairs rather than overwriting earlier failures. Sources: `evidence/qeg-research/round4/response_contract.md` and `evidence/qeg-research/round6/extensions.md`.

Finite-window positivity and its cofinal obstruction **. For continuous momentum in $[-K, K]$, write

$$C_{N,K}(a) = \frac{b}{16\pi^2} \sum_{n=0}^N d_n [P_{M_n}(K-a) + P_{M_n}(K+a)],$$

$$P_M(y) = \frac{u - u^3/3}{M^2}, \quad u = \frac{y}{\sqrt{M^2 + y^2}}. \quad ** \quad (8)$$

Since $P'_M = M^2/(M^2 + y^2)^{5/2}$ is even and decreases with $|y|$, C is even and strictly decreases for $a > 0$. Its maximum is at zero. This statement is about an integral: a two-node Gauss rule with nodes $\pm K/\sqrt{3}$ has, at a node, a lowest-level contribution $e^2 C_{\text{disc}} \geq \alpha_{\text{em}} b K / (4\pi)$, which can exceed the continuous bound $\alpha_{\text{em}} b / (3\pi)$. Independently, at $b = 10, K = 20$, positivity of every quadrature weight and exactness on constants, together with $\alpha_{\text{em}} = e^2 / (4\pi) < 1/137$, suffice for the rational estimate

$$e^2 C_{\text{disc}} < \frac{67743204500}{274510827927} < \frac{1}{4}, \quad Z > \frac{3}{4}, \quad ** \quad (9)$$

uniformly in every finite Landau cutoff. The estimate uses mass bounds for the first levels and the integral majorant for the decreasing tail. It verifies a premise of (4), without certifying a trajectory.

At infinite momentum window the integral is elementary:

$$C_{N,\infty} = \frac{b}{12\pi^2} \left(1 + 2 \sum_{n=1}^N \frac{1}{1+2bn} \right), \quad Z_{N,\infty} = 1 - \frac{e^2}{12\pi^2} \left[\log(2b) + \psi \left(N + 1 + \frac{1}{2b} \right) \right]. \quad ** \quad (10)$$

The latter tends to $-\infty$. Positivity makes $C_{N,K}(0)$ increase in both cutoffs: choose a finite N exceeding any prescribed threshold, then a finite K approximating that integral. Every cofinal cutoff path eventually exceeds both. Thus unchanged matching cannot retain a positive uniform coefficient margin. This obstructs this divided-coefficient proof route, not continuum QED or every renormalization prescription. A per-run reference choice $\chi_{N,K} = z_{\text{ref}} - 1 + e^2 C_{N,K}(0)$ yields $Z = z_{\text{ref}} + e^2 [C(0) - C(a)] \geq z_{\text{ref}}$ for the continuous window; it changes the matching and does not inherit that inequality for arbitrary quadrature. A scalar contribution uniformly bounded over the permitted field domain cannot cancel the cofinal divergence at fixed a , or uniformly bounded a . Sources: `evidence/qeg-research/round6/extensions.md` and `evidence/qeg-research/round7/renormalization_map.md`.

The same estimates give the coefficient-only tail $e^2 \sup_a |C_{\infty,K} - C_{N,K}| \leq \alpha_{\text{em}} K / [\pi \sqrt{1 + 2bN}]$, provided previously retained quadratures are unchanged. This does not bound omitted dynamical

modes. Whenever a finite cutoff has $Z(0) < 0$, continuity and $Z(a) \rightarrow 1 + \chi_b > 0$ produce singular coefficient surfaces at opposite values of a ; no argument here proves that a physical trajectory reaches one.

4.2 What gravitational and scalar extensions actually add

An anisotropic current–stress interface **. For $ds^2 = -dt^2 + a_\perp^2(dx^2 + dy^2) + a_\parallel^2 dz^2$, define $h = \dot{a}_\perp/a_\perp$, $k = \dot{a}_\parallel/a_\parallel$, and $\theta = 2h + k$. Maxwell evolution gives $\dot{E} + 2hE = -j$, $\dot{B} + 2hB = 0$, with field stresses $(\rho, p_\perp, p_\parallel)_{\text{EM}} = (u, u, -u)$, $u = (E^2 + B^2)/2$. The spatial Einstein equations $\dot{h} = (\Lambda - \kappa p_\parallel - 3h^2)/2$ and $\dot{k} = \Lambda - \kappa p_\perp - h^2 - k^2 - hk - \dot{h}$ imply, by differentiation,

$$\begin{aligned} C_g &= h^2 + 2hk - \kappa\rho - \Lambda, & Q &= \dot{\rho} + 2h(\rho + p_\perp) + k(\rho + p_\parallel), \\ \dot{C}_g &= -\theta C_g - \kappa Q, & C_g(t) &= \frac{V(0)C_g(0) - \kappa \int_0^t V(s)Q(s) ds}{V(t)}, & V &= a_\perp^2 a_\parallel. \end{aligned} \quad ** \quad (11)$$

The coefficient is $-\theta$ for these off-constraint equations, not the -2θ initially suggested in review. Running κ, Λ adds $-\dot{\kappa}\rho - \dot{\Lambda}$. Moreover $(k - h) + \theta(k - h) = \kappa(p_\parallel - p_\perp)$. The replacement $(\delta p_\perp, \delta p_\parallel) = (kf, -2hf)$ leaves the energy equation unchanged but changes the shear source by $-\kappa\theta f$. Exact mean-energy conservation therefore cannot identify both pressures.

At a fixed comoving regulator the rescaled Dirac modes have $H_n = m\beta + P\alpha_z + c_n\alpha_x$, $P = (k_z - qA_z)/a_\parallel$, $c_n = \sqrt{2n|qB_0|}/a_\perp$. Writing $\mathcal{D} = |qB_0|/(4\pi^2 V)$ and $\mathcal{S}[X]$ for the occupied-mode trace and sum, variations of this same Hamiltonian give

$$(\rho_m, j, p_\parallel, p_\perp) = \mathcal{D}(\mathcal{S}[H], q\mathcal{S}[\alpha_z], \mathcal{S}[P\alpha_z], \tfrac{1}{2}\mathcal{S}[c_n\alpha_x]). \quad ** \quad (12)$$

Using $\dot{\mathcal{D}} = -\theta\mathcal{D}$, $\dot{P} = qE - kP$, and $\dot{c}_n = -hc_n$ proves $\dot{\rho}_m + 2h(\rho_m + p_\perp) + k(\rho_m + p_\parallel) = Ej$. The two occupied columns at $n \geq 1$ already include the spin multiplicity. A rotation of $P\sigma_3 + m\sigma_1 - sc_n\sigma_2$ to real transverse mass adds the connection $s(m\dot{c}_n - c_n\dot{m})/[2(m^2 + c_n^2)]$ to P ; omitting it explains why the naive effective-mass replacement fails in the imported expanding background comparator. These are bare regulated quantities, not renormalized Einstein sources.

A finite counterterm $-c \int \sqrt{-g}F^2/4$ changes the current by $c(\dot{E} + 2hE)$ and the stress by $c(u, u, -u)$; its Ward exchange cancels only when both are changed. The classical Ohmic and later scalar-gravity fixtures test this interface with their own matter. They do not compute a common covariant quantum current and directional stress. The archived physical electron scale estimate $\kappa\rho_{\text{EM}}/m_e^2 \simeq 2.40 \times 10^{-39}$ at $E/E_c = 1, B/B_c = 100$ also rules out interpreting the fixture's visibly large gravitational response as an electron-coupling prediction. Sources: `evidence/qeg-research/round3/gravity.md`, `evidence/qeg-research/round3/evidence/prior_curved_sector_results.json`, and `evidence/qeg-research/round5/research_bridge.md`. The locally covariant Dirac analysis [6] provides a renormalization framework with explicit background qualifications; its direct free stress-conservation discussion does not supply the strong-field Bianchi-I mode sums required here.

Two retained logical controls distinguish conservation from validity. A random energy equal to $\bar{E}/p$ with probability p , and zero otherwise, has the same mean as the eigenvalue $\bar{E}$ but variance $\bar{E}^2(p^{-1} - 1)$. Mean stress alone therefore does not determine stress noise. Also exact local evolution $e^{-i\eta\sigma_x} \otimes e^{-i\eta\sigma_x}|00\rangle$ stays separable, whereas normalizing its first-order truncation gives

$$\text{Var}(E) = \bar{E}^2(p^{-1} - 1), \quad \mathcal{C}_{\text{truncated}} = \frac{2\eta^2}{1 + 2\eta^2}, \quad \mathcal{C}_{\text{exact}} = 0. \quad ** \quad (13)$$

The latter apparent witness is at the discarded order. These project counterexamples ** do not diagnose gravitational breakdown or a physical entanglement experiment. For the inherited

Cauchy-horizon profile $\phi \sim |V|^\beta$, integrating $|\partial_V \phi|^2 \sim |V|^{2\beta-2}$ gives the familiar threshold $\beta > 1/2$, with logarithmic divergence at equality; it is an energy regularity test, not a solved backreacted horizon.

Action-derived scalar completion **. A dynamical coefficient requires its own equation and energy. Round 7 chose $f(\phi) = 1 + g^2\phi^2$, scalar kinetic energy $y^2/2$, and $V(\phi) = \nu^2\phi^2/2$, with $y = \phi'$. Variation of $-f(\phi)F^2/4$, with the finite-mode current unchanged, gives

$$\begin{aligned} Zx' &= F - e^2(S + Dx^2) - f_\phi yx, & y' &= -\nu^2\phi + \frac{1}{2}f_\phi(x^2 - b^2), \\ \widetilde{W} &= \frac{1}{2}Zx^2 + e^2U + \frac{1}{2}y^2 + \frac{1}{2}(\nu^2 + g^2b^2)\phi^2, & \widetilde{W}' &= xF. \end{aligned} \quad ** \quad (14)$$

The scalar and field exchange terms cancel directly. Since $f \geq 1$, the finite $Z > 3/4$ certificate is retained, and the same continuation argument applies; the uncoupled zero-frequency scalar is handled as an affine solution. By contrast, merely prescribing $\chi(s)$ in the old equations produces the unaccounted power $\chi'x^2/2$. The separate classical Bianchi-I model with $f = e^{2c\varphi}$ closes its own scalar/Maxwell exchanges. It supplies no identification with quantum stress. Renormalization-scale running, cutoff changes, and a dynamical scalar are different operations. Sources: `evidence/qeg-research/round7/research_report.md` and `evidence/qeg-research/round7/renormalization_map.md`.

4.3 Spectral logic before a nonabelian calculation

False plateaux and conditional spectral bridges **. The move to pure Yang–Mills changes the theory and its target [1]. Its useful inheritance is explicit control of assumptions, not an abelian mass-gap result. For a nonzero finite positive spectral measure supported on $[0, \infty)$, $C(t) = \int e^{-Et} d\mu(E)$, let $E_* = \inf \text{supp } \mu$. Then $C(t+h) \leq e^{-hE_*}C(t)$; logarithmic convexity follows from Hölder. Consequently the effective mass $m_h(t) = h^{-1} \log[C(t)/C(t+h)]$ decreases to E_* from above. For $t > 0$, differentiating gives

$$-\partial_t \log C = \langle E \rangle_t, \quad \partial_t \langle E \rangle_t = -\text{Var}_t(E). \quad (15)$$

Positive measure in every $[E_*, E_* + \varepsilon]$ fixes the asymptotic exponential rate; convexity then fixes the secant limit. Thus an early plateau near 1 in the executed $e^{-t} + 10^{-12}e^{-0.1t}$ example is not a lower bound on the threshold 0.1. A lighter orthogonal channel may be invisible even at infinite time. The periodic massless free-field diagnostic explicitly excluded its zero mode; its lowest nonzero momentum is not the full gap.

Two standard spectral-theorem arguments were made explicit. First, if $H_n \rightarrow H$ strongly in resolvent on a common Hilbert space and all $\text{spec}(H_n) \cap (0, m) = \emptyset$ for the same $m > 0$, then compactly supported continuous f in that interval satisfies $f(H) = \lim f(H_n) = 0$. Positive but nonuniform gaps are insufficient: $H_n e_k = \max(1/n, 1/k)e_k$, $k \geq 1$, and $H_n e_0 = 0$, converge in norm to a gapless diagonal operator. Second, if a reconstructed positive Hamiltonian already exists and centered states ψ_O span densely in $\Omega^\perp$, a common bound $\langle \psi_O, e^{-tH} \psi_O \rangle \leq A_O e^{-mt}$ at all sufficiently large times implies

$$e^{-(m-\varepsilon)t} \|P_H([0, m-\varepsilon])\psi_O\|^2 \leq A_O e^{-mt} \implies H|_{\Omega^\perp} \geq m. \quad (16)$$

Density removes every lower channel. These known implications do not construct the continuum Hilbert space or supply the common physical rate. The Round 5 and Round 8 planners replay finite reviewed implication libraries; they are not proof-assistant kernels. The Round 8 delayed-source refinement met its declared adjacent-grid working scale, while 21 symbolic checks were blocked by an unavailable dependency and remain unexecuted in that audit. Sources: `research/round8/spectral-proof.md`, `research/round8/advisor-dossier.md`, and `research/round8/README.md`.

Finite Haar hierarchy, and what a single identity cannot certify **. For the finite Wilson measure $d\mu_W = Z^{-1}e^{-S_W}dH$ with $S_W = \beta_W \sum_p (1 - \text{Tr } U_p/2)$, left Haar differentiation yields $\langle L_a f - f L_a S_W \rangle = 0$. Conversely, on a finite product of connected compact groups, if every smooth insertion satisfies every such identity under a probability measure ν , substitute $f = e^S \varphi$. Then $\int L_a \varphi e^S d\nu = 0$. Integrating the generated flows shows $e^S \nu$ is translation invariant; uniqueness of Haar measure gives $\nu = Z^{-1}e^{-S}H$, without assuming absolute continuity in advance.

For an independent link and an unnormalized staple A , write $UA = a_0 I + i\mathbf{a} \cdot \boldsymbol{\sigma}$. The generator $\exp(it\sigma_a/2)$ gives $L_a S_W = \beta_W a_a/2$ and $L_a^2 S_W = \beta_W a_0/4$, hence

$$\langle \beta_W^2 |\mathbf{a}|^2 - 3\beta_W a_0 \rangle = 0. \quad (17)$$

The staple is not a unit quaternion. The identity assumes the varied link occurs once in each incident plaquette. On one $SU(2)$ variable the wrong law $(\delta_I + \delta_{-I})/2$ passes this residual exactly but gives $\langle L_a(u_0 u_a) - u_0 u_a L_a S \rangle = 1/2$. The complete hierarchy characterizes the law; one measured residual does not. The six finite four-dimensional lattice chains at $\beta_W = 0, 0.5, 2.2$ tested the declared nonabelian sampler and local identities. The $\beta_W = 2.2$ autocorrelation evidence was insufficient. No chain supplied a continuum gap. Sources: `research/round9/advisor/advisor.md` and `research/round9/lattice/RESULTS.md`.

An exact auxiliary transfer operator and the missing norm **. For central convolution by $e^{\beta \cos \theta}/Z_\beta$ on full $L^2(SU(2))$, the class Haar measure is $(2/\pi) \sin^2 \theta d\theta$. With $n = 2j + 1$, character integration gives $c_j = I_{n-1} - I_{n+1} = 2nI_n/\beta$; Schur orthogonality divides by n , yielding the complete spectrum

$$r_j = \frac{I_{2j+1}(\beta)}{I_1(\beta)}, \quad \text{mult}(r_j) = (2j+1)^2, \quad \Delta = -a_t^{-1} \log \frac{I_2(\beta)}{I_1(\beta)}. \quad (18)$$

For $\beta > 0$, the positive integral representation identifies I_{n+1}/I_n with a mean of $t \in (-1, 1)$ tilted by $e^{\beta t}$; pairing $t, -t$ proves this mean is strictly between zero and one. The constant is the unique stationary state. At fixed spin, $-\log r_j = 2j(j+1)/\beta + O(\beta^{-2})$. If $\beta a_t \rightarrow c > 0$, dominated convergence of square-summable Peter–Weyl coefficients gives $T_\beta^{\lfloor t/a_t \rfloor} \rightarrow e^{-2tc/c}$ strongly, with rotor gap $3/(2c)$. Other choices of the clock produce zero or infinite limiting gaps. Independent endpoint gauge invariance of an open link forces constants, so this rotor is not the interacting gauge transfer operator. Negative β supplies a pointwise positive kernel with a negative fundamental eigenvalue, refuting positivity by inspection alone.

An additional exact counterexample isolates an operator-norm error. On N uniformly weighted atoms let B average the first two coordinates and fix the rest, Π average all coordinates, and $v = (1, -1, 0, \dots)/\sqrt{2}$. Set $P_0 = (1-c)B + c\Pi$, $P_1 = P_0 + \delta vv^T$, with $0 < \delta < 1-c < 1$. Both are symmetric positive Markov matrices with strictly positive entries, yet for their density kernels $K_i = NP_i$,

$$\|P_1 - P_0\|_{2 \rightarrow 2} = \delta, \quad \|K_1 - K_0\|_{L^1(H \otimes H)} = \frac{2\delta}{N}. \quad ** \quad (19)$$

Thus product-space L^1 control cannot replace uniform row and column control in Schur’s test. For actual positive contractions sharing a vacuum, $\|T|_{\Omega^\perp}\| \leq e^{-ma}$ and $\|S - T\| \leq \varepsilon_a$ imply the conditional gap $-a^{-1} \log(e^{-ma} + \varepsilon_a)$, when its argument is below one. Telescoping gives $\|S^n - T^n\| \leq n\varepsilon_a$; fixed-time convergence needs $\varepsilon_a = o(a)$ by this route. Sources: `research/round9/transfer/PROOF.md` and `research/round9/advisor/advisor.md`.

4.4 Gauge reduction, infinite representation tails, and certified dynamics

One-square certification **. For an open four-link square, all vertex Gauss laws reduce the physical space to class functions on $SU(2)$. Assume $\alpha > 0$, $\lambda \geq 0$. The known character and

radial reductions give

$$H_{nm} = [\alpha n(n+2) + \lambda] \delta_{nm} - \frac{\lambda}{2} (\delta_{n,m+1} + \delta_{n,m-1}), \quad n, m \geq 0, \quad (20)$$

The four link Casimirs contribute $4j(j+1) = n(n+2)$, while $2x\chi_{n/2} = \chi_{(n+1)/2} + \chi_{(n-1)/2}$, with the negative-index character absent, fixes the off-diagonal factor. Equivalently, the Hamiltonian is $-\alpha\partial_\theta^2 - \alpha + \lambda(1 - \cos\theta)$ on Dirichlet functions $u = \sqrt{2/\pi} \sin\theta f$ on $(0, \pi)$. The Mathieu representation is established prior work [9]; the workbench's contribution is the explicit certified error contract. Compact resolvent and uniqueness for the second-order Dirichlet equation imply a simple ground state and a positive gap for fixed $\alpha > 0$. Retain $n < N$, let $A = P_N H P_N$, $\tau = \alpha N(N+2)$, and choose a certified $\mu_1(A) < R < \tau$. The sole cross-boundary entry is $-\lambda/2$. Young's inequality and tail positivity give

$$B = A - \frac{(\lambda/2)^2}{\tau - R} |N-1\rangle\langle N-1|, \quad \mu_k(B) \leq E_k(H) \leq \mu_k(A), \quad k = 0, 1. \quad ** \quad (21)$$

Exact rational Sturm inertia, including zero and endpoint policies, encloses these eigenvalues. After dropping the scalar λI , bounded multiplication proves $|\delta(\kappa) - \delta(\kappa')| \leq 2|\kappa - \kappa'|$ for $\delta = \Delta/\alpha$, $\kappa = \lambda/\alpha$. A certified interval cover then gave $\Delta \geq 0.999999\alpha$ on $0 \leq \kappa \leq 10$. Point samples alone would not prove this. A two-dimensional trial space localized in $0 < \theta < w$ gives $\Delta \leq 4\pi^2\alpha/w^2 + \lambda w^2/2 - \alpha$, hence $\Delta \leq 2\sqrt{2}\pi\sqrt{\alpha\lambda} - \alpha \rightarrow 0$ at fixed $\lambda > 0$ as $\alpha \downarrow 0$, with the chosen $w = (8\pi^2\alpha/\lambda)^{1/4} \leq \pi$ once $\alpha/\lambda \leq \pi^2/8$. This path is not a matched continuum trajectory. Source: `research/round10/advisor/advisor.md`.

The interacting two-square quotient and its complete electric spectrum **. The canonical Hamiltonian framework is established [8]. The seven-link, six-vertex open graph has physical space $L^2(SU(2)^2)^{\text{Ad,diag}}$. Tree gauge leaves loops U, V based at the shared vertex. This planar spatial patch with continuous time is a finite $2 + 1$ -dimensional regulator; increasing its polynomial cutoff neither adds spatial cells nor produces the four-dimensional target. The complete orbit coordinates are $x = \text{Tr } U/2$, $y = \text{Tr } V/2$, $z = \text{Tr}(UV)/2$, with

$$\Omega = \{|x|, |y| \leq 1 : 1 - x^2 - y^2 - z^2 + 2xyz \geq 0\}, \quad d\mu = \frac{2}{\pi^2} dx dy dz. \quad ** \quad (22)$$

To derive it, write the $SU(2)$ quaternions as $(x, \mathbf{a})$ and $(y, \mathbf{b})$. The three traces fix their Gram matrix, and two vector pairs with that Gram matrix differ by a proper rotation. The scalar marginals are semicircular and the relative angle has uniform cosine; the Jacobian to $z = xy - \mathbf{a} \cdot \mathbf{b}$ cancels both square roots. This proves the measure as well as orbit classification, including central pairs. The pair (x, y) alone loses the relative orientation.

The shared derivative acts as $L_U - R_V$, so the kinetic operator is $K = 3(C_U + C_V) - \sum_a (L_U^a - R_V^a)^2$, rather than two independent rotors. In trace coordinates,

$$K = -\text{div}(A\nabla), \quad A = \begin{pmatrix} 1 - x^2 & (z - xy)/4 & 3(y - xz)/4 \\ (z - xy)/4 & 1 - y^2 & 3(x - yz)/4 \\ 3(y - xz)/4 & 3(x - yz)/4 & 3(1 - z^2)/2 \end{pmatrix}. \quad ** \quad (23)$$

Its domain is inherited from invariant $H^2(SU(2)^2)$, equivalently the closure of the polynomial core, not Dirichlet data on the quotient boundary. The checks $Kz = 9z/2$ and $K(xy) = 13xy/2 - z/2$ detect the shared link. For a monomial $x^a y^b z^c$, its degree-preserving coefficient is

$$\epsilon_{abc} = a^2 + b^2 + \frac{3}{2}c^2 + \frac{1}{2}ab + \frac{3}{2}c(a+b) + 2a + 2b + 3c, \quad m_d = \frac{5}{8}d^2 + 2d + \frac{3}{8}(d \bmod 2). \quad ** \quad (24)$$

All other terms lower degree. Symmetry makes each orthogonal polynomial shell reduce K ; the diagonal quotient coefficients are its eigenvalues. Density of polynomials proves completeness.

Writing $j = (a + c)/2, k = (b + c)/2, \ell = (a + b)/2$ gives $\epsilon = 3j(j + 1) + 3k(k + 1) + \ell(\ell + 1)$, independently identifying the spin-network spectrum. Its low electric levels agree with the independent two-plaquette construction in [10]; this is a comparison of those levels, not an identification of all intermediate gauge-fixed formulas. At fixed $d = a + b + c$, the minimum has $c = 0$ and a, b as equal as possible, giving m_d . Thus a degree cutoff D has dimension $\binom{D+3}{3}$ and exact omitted threshold m_{D+1} . Sources: `research/round10/skeptic/coordinate-note.md` and `research/round11/advisor/advisor.md`.

Interacting gap enclosures and the cost of a global bound **. For $H = \alpha K + \lambda_1(1 - x) + \lambda_2(1 - y)$, with $\alpha > 0, \lambda_1, \lambda_2 \geq 0$, bounded perturbation and positivity improvement on the connected compact group give compact resolvent and a unique positive physical ground state. With $P = P_D, M = -\lambda_1 x - \lambda_2 y$, and $\tau = \alpha m_{D+1}$, the exact boundary form is finite:

$$CC^* = PM(P_{D+1} - P_D)MP, \quad B = PHP - \frac{CC^*}{\tau - R}, \quad (25)$$

$$\mu_k(B) \leq E_k(H) \leq \mu_k(PHP), \quad k = 0, 1. \quad **$$

Here $\mu_1(PHP) < R < \tau$. Multiplication raises degree by at most one; Young's inequality then proves the same form comparison as (21). In raw monomials, inertia must be computed for $H_{\text{form}} - rG$, using the Haar Gram matrix G . Its entries are exact: if $\mu_{2r} = \text{Catalan}(r)/4^r$, $\mu_{2r+1} = 0$, and $J(n, h) = \sum_{q=0}^h (-1)^q \binom{h}{q} \mu_{n+2q}$, then

$$\langle x^a y^b z^c \rangle = \sum_{h=0}^{\lfloor c/2 \rfloor} \frac{\binom{c}{2h}}{2h+1} J(a+c-2h, h) J(b+c-2h, h). \quad (26)$$

This follows from the same relative-angle integration. The gap is 2-Lipschitz in the ℓ^1 distance of $(\lambda_1/\alpha, \lambda_2/\alpha)$. Exact cover certificates gave a floor exceeding 0.96α on $[0, 2]^2$, with recorded lower endpoint 0.963936567779819α . At $(\alpha, \lambda_1, \lambda_2, \rho) = (1, 2, 3, 1)$, the recorded full-space gap interval was $[3.362386061099721, 3.362408390681595]$.

An analytic estimate exposed the volume issue before larger computation. The $\text{SU}(2)$ heat kernel at Casimir time 4 lies between $3/4$ and $5/4$: its nonconstant character sum is bounded by $4e^{-3} + \sum_{n \geq 2} (2/25)^n < 1/4$. For the two-square kinetic operator, time $4/3$ therefore gives kernel bounds $9/16, 25/16$. Feynman–Kac bounds a positive ground state's ratio by $(25/9)e^{8(\lambda_1+\lambda_2)/(3\alpha)}$. The ground-state transform and free physical Poincaré constant 3 give

$$\Delta \geq \frac{243}{625} \alpha e^{-16(\lambda_1+\lambda_2)/(3\alpha)}. \quad ** \quad (27)$$

On any finite connected open hypercubic graph containing a square, the free physical gap is exactly 3α : a nonzero spin support has no degree-one vertex, hence contains at least four edges, each costing at least $3/4$; a fundamental square attains the bound. Repeating the global heat comparison for E links gives $\Delta_\Gamma \geq 3\alpha(3/5)^{2E} e^{-16 \sum_p \lambda_p/\alpha}$. Its deterioration is not a proof of gap closure. For q disconnected copies, tensor eigenvectors give the exact same first gap as one copy while the global density-ratio bound decays exponentially in q .

There is also a direct measure obstruction. For the one-square Hamiltonian,

$$\frac{[4C + \kappa(1 - x)]e^{\theta x}}{e^{\theta x}} = \kappa - \theta^2 + (3\theta - \kappa)x + \theta^2 x^2. \quad ** \quad (28)$$

It is constant only for $\theta = \kappa = 0$. The square root of a nontrivial Wilson Gibbs density is thus not this Hamiltonian's ground state. A Langevin Poincaré theorem for the former cannot silently bound the latter; the fixed-spacing strong-coupling results of [22] retain their own measure,

normalization, and generator hypotheses. For a declared Hamiltonian convention, $\alpha = g_H^2/(2a)$, $\lambda = 2/(g_H^2 a)$, so $\lambda/\alpha = 4/g_H^4$; neither a fixed graph nor a compact coupling square follows a growing-volume weak-coupling continuum limit. Sources: `research/round11/advisor/advisor.md` and `research/round12/volume/volume-bridge.md`.

A possible replacement must control the actual ground state. The conditional Bakry–Émery bound $\text{Ric} - 2\text{Hess} \log \phi_\Gamma \geq c_* g$, uniformly in volume, would give $\Delta_\Gamma \geq \alpha c_*$ through the ground-state generator $\Delta + 2\nabla \log \phi_\Gamma \cdot \nabla$. No such bound was established. Similarly, transferring $A_j \geq m_*(I - P_j)$ to a continuum operator needs a common realization, strong convergence of rank-one vacuum projections, and a limiting semigroup strongly continuous at zero. The example $A_j = j(I - P)$ has $e^{-tA_j} \rightarrow P$ for each $t > 0$, but no strongly continuous limit on the full space. Lower-gap control alone can leave only a vacuum; nontriviality is a separate obligation.

A full-Hilbert-space dynamical error bound **. The earlier driven one- and two-square histories were finite-Galerkin studies. Round 12 added the omitted representation error. Let $H(t) = \alpha(t)K_\rho + \lambda_1(t)(1 - X) + \lambda_2(t)(1 - Y)$, $K_\rho = 3(C_U + C_V) - \rho \sum (L_U - R_V)^2$, with fixed $\rho > 0$. Remove the common scalar phase. The electric propagator preserves every degree shell and the remaining bounded interaction has norm at most $w = |\lambda_1| + |\lambda_2|$, with shell bandwidth one. Real locally integrable coefficients suffice for mild unitary evolution. For normalized initial support in P_{d_0} , let ψ_D be exact Galerkin evolution and $A(t) = \int_0^t w(s) ds$. Duhamel in the finite tail block gives inductively $\|(P_D - P_{n-1})\psi_D\| \leq A^{n-d_0}/(n - d_0)!$. The only full-equation defect is $r_D = \Pi_{D+1} W \Pi_D \psi_D$. One more integral yields

$$\|\psi(t) - \psi_D(t)\| \leq \min \left\{ 2, \frac{A(t)^{D-d_0+1}}{(D - d_0 + 1)!} \right\}. \quad ** \quad (29)$$

No exponential factor is needed because both block propagators are unitary. Equal action bounds do not identify different protocols. For a computed path, add its preparation error and the integral of $\|i\tilde{\psi}_D - H_D \tilde{\psi}_D\|$. Norm conservation alone does not bound that residual. The boundary norm has the exact Gram expression $\|r_D\|^2 = c^*[N - MG^{-1}M]c$, where M, N are the form matrices of $P_D W P_D, P_D W^2 P_D$.

For a finite step of length h , midpoint $B = H_D(m)$, $C = H'_D(m)$, and $M_2 = \sup \|H''_D\|$, a Taylor comparison followed by two unitary Duhamel integrals gives

$$\|U_D - e^{-ihB}\| \leq h^3 M_2/24 + h^3 \|[B, C]\|/12 + h^4 \|C\|^2/32. \quad ** \quad (30)$$

The first term integrates the quadratic Taylor remainder. The centered linear integral cancels its constant interaction-picture part, leaving the commutator term; the second Duhamel remainder is half the squared integrated linear norm. Compressed X, Y need not commute. Evaluating the exponential is a separate error: for a finite selfadjoint B , its order- q Taylor remainder is at most $(h \|B\|)^{q+1}/(q+1)!$. The archived exact rational two-step state used degree 3, Taylor order 100, $(\lambda_1, \lambda_2) = (1/20, 1/10)$ then $(1/10, 1/20)$, each for duration 1, and $\alpha = \rho = 1$. With $e = (114/5)^{101}/101!$, its total bound was $27/80000 + 2e + e^2$, including the product error and the unnormalized computed state. This certifies that declared protocol, not all historical drives. Bounded-observable errors are at most $2\|O\|$ times a normalized state error; energy needs its own work identity and cannot use a fictitious global norm of H . Sources: `research/round12/advisor/advisor.md` and `research/round12/solver/output/exact_step_certificate.json`.

Closure tests that retain the missing fluctuation **. For the one-square Euclidean measure proportional to $e^{\kappa x} \sqrt{1 - x^2} dx$, integration of $\partial_x[(1 - x^2)^{3/2} e^{\kappa x} f]$ gives, with $m_n = \langle x^n \rangle$,

$$nm_{n-1} - (n+3)m_{n+1} + \kappa(m_n - m_{n+2}) = 0, \quad 3m_1 = \kappa(1 - m_1^2 - v), \quad v \geq \frac{1}{4}e^{-2|\kappa|} > 0. \quad ** \quad (31)$$

The last inequality follows from domination of Haar density by $e^{-2|\kappa|}$ and $\text{Var}(x) = \inf_c \langle (x - c)^2 \rangle$. Discarding v while keeping the true mean leaves residual κv . At $\kappa = 0$, the next equation, not the first, enforces $m_2 = 1/4$. Adding a fluctuation symbol does not compute its next moment. The same audit checked specific displayed formulas of [arXiv:2608.05415v1](#) [24]: directed versus unordered plaquette counting, a derivative prefactor, a missing source phase, and the rank impossibility $\eta^T g \eta = I_{N_c^2 - 1}$ when $N_c^2 - 1 > D$. These are local, version-specific tests and do not refute all closures. Source: `research/round12/closure/closure-audit.md`.

Change in understanding and the next obligations. By Round 12 the reliable object had narrowed from an intended coupled Einstein–QED system to explicit finite models with named observables, domains, and error norms. The abelian branch still required a common renormalized current and anisotropic stress for a specified state, followed by fluctuation and regulator control. The nonabelian branch had genuine infinite-representation results on finite graphs, including interacting stationary and driven bounds. Its remaining tasks were connected-volume estimates for the actual physical Hamiltonian, scale matching, nontrivial continuum observables, reconstruction, and a spectral exclusion covering the whole physical sector. Neither the formal proof planner nor the number of passed checks supplied those premises. The machine-readable early ledger preserves every historical entry, retained failure, source path, and contribution mapping; the present manuscript synthesis does not claim to rerun every historical calculation.

5 Finite Haar constraints and connected lattice integrals

Rounds 13–22 developed several distinct mathematical models. Throughout this section, $x_p = \text{Tr}(U_p)/2$ denotes a normalized fundamental Wilson trace, $\chi_p = 2x_p$ its ordinary character, and C_e the positive link Casimir with eigenvalues $j(j+1)$. A finite physical Hamiltonian has the convention

$$H = \alpha \sum_e C_e - \sum_p \lambda_p x_p, \quad \alpha > 0.$$

Here α, λ_p have energy units. Euclidean coefficients κ_p , introduced in densities proportional to $\exp(\sum_p \kappa_p x_p)$, are dimensionless and do not define a physical clock. Every spectral statement below retains a fixed positive energy reference $E_\star$. The symbol ****** identifies a result derived or computed in the workbench; it does not assert priority over the literature. The audit ledger accounts for all 62 historical entries in this range, including four named Round13 studies and 58 numbered loops.

5.1 Compact moments, response, and complete interval coverage ******

The one-holonomy model uses normalized Haar pushforward

$$d\mu_\kappa(x) = Z_\kappa^{-1} e^{\kappa x} \frac{2}{\pi} \sqrt{1 - x^2} dx, \quad -1 \leq x \leq 1.$$

Writing $m_n = \int x^n d\mu_\kappa$, integration of the derivative of $x^n e^{\kappa x} (1 - x^2)^{3/2}$ gives the familiar identity

$$nm_{n-1} - (n+3)m_{n+1} + \kappa(m_n - m_{n+2}) = 0. \quad (32)$$

The first term is absent at $n = 0$. For nonzero κ , normalization and $m_1 = u$ express every moment affinely in u . At zero coupling one instead uses $m_{2j} = C_j/4^j$, $m_{2j+1} = 0$, where C_j is a Catalan number. Positivity supplies

$$(m_{i+j})_{i,j=0}^r \succeq 0, \quad (m_{i+j} - m_{i+j+2})_{i,j=0}^{r-1} \succeq 0.$$

The workbench implementation retains rational vectors whose quadratic forms are affine inequalities in u , so rejected means have exact polynomial witnesses. Its 28 finite certificates include a level-six interval at $\kappa = 1$ of width approximately $1.40631 \cdot 10^{-13}$; the analogous width at $\kappa = 5$ is approximately $6.03663 \cdot 10^{-8}$. These are enclosure widths, not sampling errors.

Completeness is a separate argument. All moment and support inequalities produce a probability measure on $[-1, 1]$, using multiplication by x as a self-adjoint contraction on the polynomial completion. All recurrence identities extend to C^1 tests and imply

$$D[(1 - x^2)\mu] = [\kappa(1 - x^2) - 3x]\mu.$$

The interior density is proportional to $e^{\kappa x} \sqrt{1 - x^2}$; the equation excludes endpoint atoms. Consequently the full hierarchy identifies the desired measure. Nested finite feasible intervals shrink to a point, without a proved rate. Successful finite computations additionally require vanishing optimization slack; the delivered level-six cap does not execute this limit.

Differentiation with respect to the coupling gives $u' = \text{Var}(x) > 0$. Combining this with (32) yields

$$\kappa u' + 3u = \kappa(1 - u^2), \quad u(0) = 0, \quad u'(0) = \frac{1}{4}.$$

This is a static response equation. The regular solution is $u = I_2(\kappa)/I_1(\kappa)$; singular modified-Bessel branches violate boundedness at the origin. Replacing $3u$ by $2u$ changes the prior to uniform measure on $[-1, 1]$, rather than improving the Haar approximation. The project also derived a useful residual certificate: fix $0 < \kappa_0 \leq \kappa$ and a differentiable approximation $p \geq 0$ throughout $[\kappa_0, \kappa]$, with $r = p' - (1 - p^2 - 3p/\kappa)$,

$$|p(\kappa) - u(\kappa)| \leq \left(\frac{\kappa_0}{\kappa}\right)^3 |p(\kappa_0) - u(\kappa_0)| + \kappa^{-3} \int_{\kappa_0}^{\kappa} s^3 |r(s)| ds. \quad ** \quad (33)$$

It follows by an integrating factor, since the error damping contains $3/\kappa + u + p \geq 3/\kappa$. A certified launch error therefore does not bound a later floating trajectory unless the entire residual is enclosed. The point-mass proposal $m_n = (1/3)^n$ at $\kappa = 9/8$ passes positivity and the first identity but has next-identity residual $8/9$. This supplies an exact counterexample to a zero-variance closure.

Rounds14–15 moved to the genuine two-holonomy density $Z^{-1} e^{x+y+\eta z} dU dV$, with $z = \text{Tr}(UV)/2$. For $C(\eta) = \text{Cov}_\eta(x, y)$, differentiating the normalized quotient gives

$$C'(\eta) = \mathbb{E}_\eta[(x - \mathbb{E}_\eta x)(y - \mathbb{E}_\eta y)(z - \mathbb{E}_\eta z)], \quad |C'| \leq 2.$$

The last inequality uses $|z - \mathbb{E}z| \leq 2$, Cauchy–Schwarz, and $\text{Var}(x), \text{Var}(y) \leq 1$. The first loop’s small- η estimate did not resolve the chosen point $(1, 1, 1/4)$. Exact total-degree Taylor integration did. For any bounded insertion and action $|S| \leq M < N + 2$, the standard tail

$$R_N(M) = \frac{M^{N+1}}{(N+1)!} \frac{1}{1 - M/(N+2)}$$

bounds numerator and partition separately. Jensen gives $Z \geq 1$ when the unweighted action mean vanishes; signed interval division retains all endpoint combinations. Point intervals $[L, U]$ extend across a cell of radius ρ as $[L - 2\rho, U + 2\rho]$. All eight initial midpoint intervals were positive, while all eight cell bounds failed. Refining only the failing cells produced 8, 16, then 21 cells, with a final minimum lower bound approximately 0.0003346525 over the whole closed interval $\eta \in [1/8, 1/4]$. Increasing already adequate point precision would not have removed the geometric transport loss.

These results support certified static response calculations and provide countermodels to incomplete closures. Their next application requires shared-link identities for larger graphs, not an identification of η with time or energy. Sources: `research/round13/moments/README.md`, `research/round13/response/response.md`, `research/round14/forward/loop2.md`, `research/round15/forward/A2/result.md`.

5.2 Closed-surface gluing and the first shared face **

On the closed cube there are eight vertices, twelve links and six faces. For χ_n , the spin- $n/2$ character of dimension $d_n = n + 1$, write $f_p = \sum_n a_{p,n} \chi_n$. Schur orthogonality forces equal labels on adjacent faces. Each edge contributes d_n^{-1} , each surviving vertex index d_n , giving the graph-specific application

$$\int \prod_{p=1}^6 f_p(U_p) dU = \sum_{n \geq 0} d_n^{-4} \prod_{p=1}^6 a_{p,n}, \quad \mathbb{E}_0 \prod_{p=1}^6 x_p = 2^{-10}. \quad ** \quad (34)$$

The identity is immediate for finite character polynomials; analytic weights require convergence or the retained Taylor remainder. If any one face weight equals one, only label zero survives and the other five class observables factorize. Thus five-face factorization coexists with a nonzero six-face moment. It would be incorrect to use this as evidence for independent face holonomies.

For $e^{\kappa x}$,

$$a_n(\kappa) = \sum_{j \geq 0} \frac{(n+1)\kappa^{n+2j}}{2^{n+2j} j! (n+1+j)!}.$$

An insertion of one x_p differentiates that face's coefficient before setting all couplings equal. It does not differentiate the common-coupling partition six times. At $\kappa = 1/4$, total degree18 first met the specified expectation precision; degree24 gave $\mathbb{E} \prod_p x_p \simeq 0.001023116359923$ and a partition excess over the independent-face product exceeding $2.4598180 \cdot 10^{-7}$. The exact fractions, including the failed coarse degrees, remain in the source certificate.

Two adjacent cubes have twelve vertices, twenty links and eleven faces. Conditioning on their shared holonomy H , each five-face disk contributes $\sum_n d_n^{-4} (\prod_{\text{disk}} a_{p,n}) \chi_n(H)$. The actual integral is therefore

$$\sum_{n,m,k} \frac{\prod_{p \in L} a_{p,n}}{d_n^4} \frac{\prod_{p \in R} a_{p,m}}{d_m^4} a_{s,k} N_{nm}^k, \quad N_{nm}^k = \mathbf{1}_{\{|n-m| \leq k \leq n+m, \ n+m+k \text{ even}\}}. \quad ** \quad (35)$$

The shared-edge incidence is three, so the old one-surface single-label rule is insufficient. Center parity makes the all-eleven distinct-trace Haar moment zero. With the shared trace squared, both fusion channels $k = 0, 2$ contribute and $\mathbb{E}_0[(\prod_{\text{outer } 10} x_p) x_s^2] = 2^{-19}$. Removing $k = 2$ halves the answer.

Keeping the same all-eleven-trace observable, Round16 compared outer couplings $1/8$ and shared coupling $1/8$ against shared coupling zero. The degree24 difference is enclosed near $2.4303274916 \cdot 10^{-7}$, with width about $4.173 \cdot 10^{-22}$. The zero shared-action baseline is nonzero: its leading numerator at common outer coupling t is $41t^5/(81 \cdot 2^{18})$. Deleting a coefficient is not deleting an insertion. These finite integrals expose graph dependence and supply later trial matrix elements; they provide no transfer Hamiltonian by themselves. Sources: `research/round15/forward/B1/derivation.md`, `research/round15/forward/B2/derivation.md`, `research/round16/forward/loop1/derivation.md`, `research/round16/forward/loop2/derivation.md`.

5.3 Recoupling, sufficient coordinates, and surrounding weights **

The four-cube $2 \times 2 \times 1$ complex has 18 vertices, 33 links and 20 faces. Integrating its central link in the adjoint representation requires the complete invariant projector on $(\mathbb{R}^3)^{\otimes 4}$. The three invariant tensors $B_1 = \delta_{ab} \delta_{cd}$, $B_2 = \delta_{ac} \delta_{bd}$, $B_3 = \delta_{ad} \delta_{bc}$ have Gram matrix with diagonal nine and off-diagonal three. The Clebsch–Gordan decomposition $1 \otimes 1 = 0 \oplus 1 \oplus 2$ proves completeness, and Haar averaging gives $P = B(B^T B)^{-1} B^T$. The inverse has diagonal $2/15$ and off-diagonal $-1/30$. For an explicitly realized tetrahedral boundary, the product of four normalized adjoint

traces has expectation $-1/405$; diagonal-only and one-channel substitutes instead give $2/405$ and $1/729$. The negative entries cannot be dropped.

Writing the central quaternion as $q \in S^3$, face traces as $q \cdot a_i$, and the action as $q \cdot b$, with $b = \sum_i \kappa_i a_i$, the partition depends only on $\|b\|$. The joint observable also depends on the directions a_i . Tetrahedral and commuting boundaries can have the same b but opposite-sign joint expectations. The complete Gram matrix of $(b, a_1, \dots, a_4)$ is sufficient: equality of Grams defines an isometry between the spans, extends to an element of $O(4)$, and preserves Haar sphere measure. This proof includes singular Grams and $b = 0$, without inverting a Gram matrix. Admissibility requires positive semidefiniteness, rank at most four, unit a_i , and both action relations

$$G_{0i} = \sum_j \kappa_j G_{ji}, \quad G_{00} = \sum_{i,j} \kappa_i \kappa_j G_{ij}.$$

An exact computational realization uses

$$M(k) = \frac{\sum_j (k_j - \delta_{ij}) G_{ij} M(k - e_i - e_j)}{|k| + 2}, \quad |k| > 0 \text{ even, } k_i > 0, \quad ** \quad (36)$$

with $M(0) = 1$ and odd moments zero. A term of zero multiplicity is omitted before its reduced multi-index is formed, so no negative exponent is evaluated. Gaussian integration by parts followed by radial separation derives this recurrence; its project contribution is the complete action/observable implementation and audit, not a new sphere-moment principle.

Integrating one actual surrounding link changes the problem again. For $x = \text{Tr } U/2$, $y = \text{Tr } V/2$, and $w = \text{Tr}(UV^\dagger)/2$, character convolution gives

$$\mathbb{E}[x^a y^b w^c] = 2^{-a-b-c} \sum_n \frac{m(a, n) m(b, n) m(c, n)}{n+1}, \quad m(a, n) = \binom{a}{(a-n)/2} - \binom{a}{(a-n)/2 - 1}.$$

In particular $\mathbb{E}[xyw] = 1/16$ and $\mathbb{E}[x^2 y^2 w^2] = 1/48$. The actual six affected faces give $S_2 = 3x + 2y + w$, with $O = (4x^2 - 1)^3(4w^2 - 1)/81$. Its normalized expectation at $\kappa = 1/64$ is enclosed near $1.95552781 \cdot 10^{-6}$. The surrounding marginal is the semicircle measure multiplied by $e^{2\kappa y} Z_U(y)$; averaging normalized inner expectations against bare Haar discards both weights.

Unfreezing a third actual link gives $S_3 = 3x + y + z + w + t$, where $t = \text{Tr}(VW^\dagger)/2$. The same V occurs in both branches: $\mathbb{E}[xzw] = 1/64$, whereas resampling it independently gives zero. Round19 integrated all required monomials through degree16, retained the full Taylor tails, and obtained a continuous result for $F(\kappa) = \mathbb{E}[Oe^{\kappa S_3}]/\mathbb{E}[e^{\kappa S_3}]$:

$$F(0) = 0, \quad F(\kappa) \geq \kappa^2/2048 \quad (|\kappa| \leq 1/8), \quad F(\kappa) > F(-\kappa) \quad (0 < \kappa \leq 1/64). \quad ** \quad (37)$$

Here $N_2 = 1/324$, $N_3 = 13/1296$ for the numerator. Subtracting absolute higher-order coefficients and its complete tail bounds N/κ^2 ; an upper partition bound gives the displayed direction of the quotient. The sign comparison uses $N(\kappa)Z(-\kappa) - N(-\kappa)Z(\kappa)$, whose leading term is $13\kappa^3/648$. This conditional three-link result still fixes thirty links. Full surrounding integration and physical dynamical matching remain separate tasks. Sources: `research/round17/forward/c1/report.md`, `research/round18/forward/c1/report.md`, `research/round18/forward/c2/report.md`, `research/round19/forward/c1/report.md`, `research/round19/forward/c2/report.md`.

6 Physical spectral bounds and an explicit infinite-volume expectation

6.1 Finite untruncated graph bounds and complete complements **

On a connected finite simple graph with Gauss law at every vertex and no external charges, a nonconstant spin network has no degree-one active vertex. Its support therefore contains a cycle.

Every occupied link costs at least $3\alpha/4$; a fundamental character on a shortest cycle attains the resulting energy. For graph girth g , $\delta_G = 3\alpha g/4$ is the exact free physical gap. A forest has only the vacuum physical sector, not a numerical first excitation.

Let $L = \sum_p |\lambda_p|$. Bounded multiplication gives $E_1(H) \geq \delta_G - L$, while the vacuum trial gives $E_0(H) \leq 0$. For square graphs, the orthonormal trial vectors $1, \chi_p$ have electric energies $0, 3\alpha$ and magnetic vacuum-to-face entries $-\lambda_p/2$. The relevant third Haar moments vanish. Diagonalizing only this arrowhead yields a ground-energy *upper* bound. Combining it with the independent full-space excited-energy bound gives

$$\Delta(H) \geq \frac{\delta + \sqrt{\delta^2 + \sum_p \lambda_p^2}}{2} - L, \quad \delta = 3\alpha. \quad ** \quad (38)$$

This is not a Ritz-gap lower bound. On the six-face cube, it expands the common-magnitude sufficient range from $r = |\lambda|/\alpha < 1/2$ to $r < 12/23$, and gives $(\sqrt{21/2} - 3)\alpha/2 > 0$ at $r = 1/2$. On the eleven-face two-cube graph it gives $r < 12/43$. At that latter boundary an actual shared-face adjoint state $\chi_2(U_s) = 4x_s^2 - 1$, of electric energy 8α , repairs the zero margin. With $v_0 = 1 + \sum_p \chi_p/22$, $v_\eta = v_0 + \eta\chi_2(U_s)$,

$$\frac{\Delta}{\alpha} \geq \frac{(6/473)\eta - (347/43)\eta^2}{45/44 + \eta^2} > 0 \quad (0 < \eta < 6/3817). \quad **$$

At $\eta = 3/3817$ the exact bound is $1388/284767457$. This amplitude maximizes the quadratic numerator, not the quotient.

The next improvement required a complete complement, rather than a larger trial space alone. Round18 classified every physical state below $9\alpha/2$: only the vacuum and eleven fundamental faces occur. For their projector P , $QH_0Q \geq (9\alpha/2)Q$, attained by a six-link rectangle. The full cross Gram is

$$\begin{aligned} (QVP)^*(QVP) &= PV^2P - (PVP)^2, & C_{00} &= C_{0p} = 0, \\ C_{pp} &= \frac{1}{4} \sum_q \lambda_q^2, & C_{pq} &= \lambda_p \lambda_q / 4 \quad (p \neq q). \end{aligned}$$

Both terms in the subtraction matter: $V\Omega$ already lies in P . For $|\lambda_p| \leq r\alpha$, $\|QVP\|^2 \leq 21\alpha^2 r^2/4$. On the codimension-one space $\Omega^\perp$, comparison with

$$\alpha \begin{pmatrix} 3 & -\sqrt{21}r/2 \\ -\sqrt{21}r/2 & 9/2 - 11r \end{pmatrix}$$

gives the full E_1 lower bound. At $r = 3/8$, it proves $\Delta/\alpha \geq (27 - 3\sqrt{70})/16 > 0$ for the entire signed box.

Round19 raised the strict electric projector to $E_{el} < 6\alpha$. Its complete dimension is 48: vacuum, eleven four-cycles and thirty-six six-cycles. At the excluded threshold 6α , 99 support/label assignments represent 107 physical channels, because eight assignments have intertwiner multiplicity two. Counting supports alone would miss this distinction. The full 48-channel cross Gram gives $\|QVR\|^2 \leq 19\alpha^2 r^2/4$, with $R = P - P_\Omega$, while $RHR \geq \alpha(3 - 2r)$ and $QH_0Q \geq \alpha(6 - 11r)$. Hence

$$\frac{\Delta(H)}{\alpha} \geq R(r) = \frac{9 - 13r - \sqrt{100r^2 - 54r + 9}}{2} > 0, \quad 0 \leq r < \frac{30 - 2\sqrt{87}}{23}. \quad ** \quad (39)$$

In particular $R(3/8) = (33 - 6\sqrt{5})/16$ and $R(7/16) = 19/32$. The failed $r = 1/2$ estimate is a failure of this row-envelope certificate, not a proved gap closure.

These calculations yield full untruncated finite-graph theorems because the omitted physical spectrum is bounded analytically. They do not give a dense volume-uniform threshold. Under the usual lattice normalization $\alpha = g_H^2/(2a)$, $\lambda = 2/(g_H^2 a)$, one has $r = 4/g_H^4$, so fixed small- r regimes do not follow a weak-bare-coupling path $g_H \rightarrow 0$ [9]. Sources: `research/round15/forward/C2/derivation.md`, `research/round17/forward/b2/report.md`, `research/round18/forward/b1/report.md`, `research/round19/forward/b1/report.md`, `research/round19/forward/b2/report.md`.

6.2 Dressed strips, literal boundaries, and summable perturbations **

A different route obtains volume-uniform results by an explicit restriction of the interaction family. Pairwise link-disjoint nonzero plaquettes factorize the *full* link Hilbert space into four-link blocks and free links. The four-link free gap before Gauss restriction is $3\alpha/4$, not the physical loop energy 3α . With $\max |\lambda_p|/\alpha \leq \rho < 3/4$, min-max and zero vacuum mean give $\Delta \geq \alpha(3/4 - \rho)$. Each unique block ground is gauge invariant: its gauge action is a continuous one-dimensional representation of a finite product of $SU(2)$, hence trivial. The physical restriction contains the same ground and inherits the lower bound without factorizing the physical Hilbert space.

Three consecutive square faces form a ten-link strip. Dress its two link-disjoint ends first and add the middle bridge with coefficient μ . The bridge contains a free Haar link in this reference, so its mean is zero and its squared norm on the reference ground is $1/4$. Thus

$$\Delta_{\text{strip}} \geq \alpha \left(\frac{3}{4} - \max(|\lambda_L|, |\lambda_R|)/\alpha \right) - |\mu| \geq \alpha/8 \quad \begin{cases} |\lambda_L|, |\lambda_R| \leq \alpha/2, \\ |\mu| \leq \alpha/8. \end{cases} \quad **$$

The zero-mean premise saves one perturbation norm; a generic bounded perturbation would lose two.

Complete strips anchored at $(4i, 2j, z)$ are link-disjoint. Round18's finite-box rule initially classified incomplete boundary strips as remainder faces; a fixed face could consequently change coefficient when the box grew. Round19 replaced that rule by literal restrictions of a single infinite assignment. All possible clipped components L, M, R, LM, MR, LMR retain the same bound $\alpha/8$, using the bare vacuum for proper clips and dressed ends for the complete strip.

In the incomplete tensor product of complete-strip grounds and free Haar links, define

$$H_{\text{ref}} = \sum_C (H_C - E_C) + \sum_{e \text{ free}} \alpha C_e, \quad H_{\text{ref}} \geq \delta(1 - P_\Omega), \quad \delta = \alpha/8.$$

This is the closed nonnegative form sum on the finite-excitation core. Give every omitted face of anchor (x, y, z) the coefficient $\nu_f = \alpha\tau 2^{-x-y-z}/24$. The exact geometric sums are

$$\sum_{\text{all}} w_f = 1, \quad \sum_{\text{selected}} w_f = 28/135, \quad \sum_{\text{omitted}} w_f = 107/135, \quad \|V\| \leq \beta = \alpha|\tau| 107/135. \quad ** \quad (40)$$

Every omitted face has a free Haar link, so $\langle \Omega, V\Omega \rangle = 0$. On $\Omega^\perp$, $H_{\text{ref}} + V \geq \delta - \beta$, whereas the vacuum trial energy is zero. If $\beta < \delta$, a spectral subspace below $\delta - \beta$ is nonempty and at most one-dimensional. This proves an isolated simple ground without invoking infinite-volume compact resolvent. At $|\tau| = 1/64$,

$$\Delta \geq 973\alpha/8640. \quad ** \quad (41)$$

All actual face coefficients can be nonzero, but the omitted ones decay with position. This is a fully supported inhomogeneous exception.

Round20 completed each finite retained perturbation to whole reference factors. If $t_L \downarrow 0$ is its exact omitted dyadic tail and $\epsilon_L = \alpha|\tau|t_L$, the correct lift is $H_L = K_L \otimes I + I \otimes H_{\text{ref,out}}$. Standard bounded-perturbation identities [13] give

$$\left\| (H - z)^{-1} - (H_L - z)^{-1} \right\| \leq \epsilon_L / |\text{Im } z|^2, \quad \|e - e_L\| \leq \epsilon_L, \quad \|P - P_L\| \leq \min(1, \epsilon_L/g_L),$$

where $g_L = \delta - \|V_L\|_{\text{budget}} > 0$. The projector bound follows by applying the complementary inverse of $H_L - e$ to its residual on the true ground. A zero exterior generator would leave escaping free excitations at zero and fails norm-resolvent convergence even at zero perturbation.

Aligned literal vertex boxes $N = 4m + 3$ coincide with full factors after subtracting the explicit reference ground energy $c_N = \sum_{C \subset \Lambda_N} E_C$. Other upper boundary phases admit local-state convergence: complete the support of both a fixed observable and an interior perturbation, compare their identical central ground states, then bound each omitted tail. This gives

$$|\omega_{\text{box}}(A) - \omega_{\infty}(A)| \leq 4 \|A\| \min(1, \epsilon_M/g_M)$$

once the box contains that common completion. It does not establish arbitrary-phase global vector or norm-resolvent convergence. Sources: `research/round18/forward/a1/report.md`, `research/round19/forward/a1/report.md`, `research/round19/forward/a2/report.md`, `research/round20/forward/d1/report.md`, `research/round20/forward/e2/report.md`.

6.3 Physical GNS space and a nonzero Wilson fluctuation **

For the dyadic model the full bounded local-factor algebra is irreducible in the chosen incomplete product. Finite-factor vacuum projections converge strongly to P_{Ω} ; an operator in the commutant is scalar on Ω , then on its dense local rank-one orbit. Every nonzero ground vector Ψ is consequently cyclic for that full algebra. The GNS null quotient maps isometrically by $[A] \mapsto A\Psi$. Its energy generator is $H - e$, since this subtraction fixes the cyclic vacuum, even though scalar shifts cancel in observable conjugation.

The physical algebra requires an additional argument. Let $\mathcal{H}_{\text{inv}}$ be the vectors invariant under every finite-support vertex gauge transformation, and $\mathcal{A}_{\text{phys}}$ the norm closure of bounded invariant local operators. For any invariant ξ , approximate it by $A_n\Psi$ using full-algebra cyclicity. Haar-average A_n over its finitely many endpoint groups. The weak operator average preserves local support and, on Ψ , is a strong vector integral satisfying $\|\mathcal{E}(A_n)\Psi - \xi\| \leq \|A_n\Psi - \xi\|$. Therefore

$$\overline{\mathcal{A}_{\text{phys}}}\Psi = \mathcal{H}_{\text{inv}}, \quad K_{\text{phys}} = (H - e)|_{\mathcal{H}_{\text{inv}}}, \quad \text{spec } K_{\text{phys}} \subset \{0\} \cup [973\alpha/8640, \infty). \quad ** \quad (42)$$

The restriction is self-adjoint on $D(H) \cap \mathcal{H}_{\text{inv}}$; gauge-commuting spectral cutoffs prove density of that domain.

This sector contains a nonvacuum vector. For the actual omitted origin xz Wilson multiplier W , the reference mean is zero and variance $1/4$. Two free Haar links per omitted face make distinct-face covariances vanish. The resulting residual and pure-state trace distance give, throughout $|\tau| \leq 1/64$,

$$\begin{aligned} d^2 &= \|P_{\Psi} - P_{\Omega}\|^2 \leq 2817/64377572 < 1/22500, & ** \\ \text{Var}_{\Psi} W &\geq \frac{1}{4} - 2d - 4d^2 \geq 5321/22500 > \frac{1}{5}. \end{aligned}$$

For $\xi = (W - \omega(W))\Psi$, spectral calculus now gives the nonzero imaginary-time correlation

$$0 < \langle \xi, e^{-tK_{\text{phys}}/\hbar} \xi \rangle \leq \text{Var}_{\Psi}(W) e^{-973\alpha t/(8640\hbar)} \quad (t \geq 0).$$

A lower spectral bound supplies this upper decay bound, not a lower exponential estimate. It also supplies no real-time magnitude decay. The result concerns the specified summable representation; equality to a homogeneous or continuum representation has not been proved. Sources: `research/round20/forward/g2/report.md`, `research/round21/forward/j1/report.md`, `research/round21/forward/j2/report.md`.

7 Locality, profile limits, and stationary observations

The Round13 locality study applied the Lieb–Robinson interaction-picture method to full $SU(2)$ links with unbounded onsite Casimirs. Common onsite unitaries preserve support; only bounded plaquette interactions enter the chain expansion. If $q_{\text{inc}} = 2(d_s - 1)$, $J = \hbar^{-1} \int j(t) dt$ is the

dimensionless integrated common magnetic envelope, and R is the shortest overlapping-plaquette chain from the observable support X to a removed boundary plaquette, then

$$\|\tau_{\Lambda'}(A) - \tau_{\Lambda}(A)\| \leq \|A\| \min \left\{ 2, \frac{|X|}{4} \sum_{k \geq R} \frac{(8q_{\text{inc}}J)^k}{k!} \right\}. \quad ** \quad (43)$$

There are at most $|X|q_{\text{inc}}(4q_{\text{inc}})^{k-1}$ chains; each commutator costs two and ordered times give $J^k/k!$. Unitary resummation is essential: unrestricted growing-support word counting would not give this chain bound. The factorial tail tends to zero at fixed J as the boundary recedes. This proves a fixed-spacing volume comparison, with no ground-state comparison or continuum velocity. The underlying strong-operator framework permits infinite-dimensional sites [19].

The summable model admits a separate direct Duhamel construction. Completing the local observable and interior perturbation to common factors gives all-phase finite-box dynamics convergence at fixed time, with error $4\|A\| |t|\epsilon_M/\hbar$. Strongly continuous implementing unitaries do not imply point-norm time continuity for every bounded local operator. On a free link the character shift $\chi_n \mapsto \chi_{n+1}$ acquires phases with arbitrarily large frequency; at times tending to zero its norm difference is exactly two. This limitation was retained when identifying the GNS generator.

7.1 An exact spatial budget and its endpoint **

Replace only the omitted weights by $w_f(q) = q^{x+y+z}/24$, $0 < q < 1$. Summing all orientations and subtracting selected strips gives

$$\mathcal{B}(q) = \frac{2 + 5q + 5q^2 + 6q^3 + 3q^4}{24(1-q)^3(1+q)^2(1+q^2)}, \quad \mathcal{B}(1/2) = 107/135, \quad (1-q)^3\mathcal{B}(q) \rightarrow 7/64. \quad ** \quad (44)$$

The same bounded perturbation proof gives $\Delta \geq \alpha[1/8 - |\tau|\mathcal{B}(q)]$. At $\tau = 1/64$ its sufficient boundary lies between 0.764003 and 0.764004. This is a boundary of the certificate, not a measured critical parameter.

Fix $0 < \eta < 1$ and follow the canonical constant-budget ray $\tau_q = \eta/[8\mathcal{B}(q)]$. Each omitted face has two free reference links, and distinct squares share at most one link. Conditional Haar integration therefore gives $\langle x_f x_g \rangle_{\Omega} = \delta_{fg}/4$, despite possible entanglement within strip factors. Hence

$$\sigma_q^2 := \|V_q \Omega\|^2 = \frac{\alpha^2 \tau_q^2}{96} \mathcal{B}(q^2), \quad \|V_q\| = \alpha\eta/8, \quad \frac{\sigma_q^2}{\alpha^2(1-q)^3} \rightarrow \frac{\eta^2}{5376}. \quad ** \quad (45)$$

The norm equality follows by localizing finitely many links near identity and then controlling the remaining absolute tail; it does not use a delta function on an infinite identity configuration. With $\bar{g} = \alpha(1-\eta)/8$, the excited threshold gives

$$\|P_q - P_{\Omega}\| \leq \sigma_q/\bar{g}, \quad -\sigma_q^2/\bar{g} \leq e_q \leq 0.$$

Thus the global ground projection returns to the selected-strip reference although the perturbation norm stays nonzero. Every fixed omitted coefficient tends to zero. This route cannot retain a nonzero homogeneous omitted coupling at fixed $\alpha/E_{\star}$.

For a bounded operator A on finitely many complete factors F , only finitely many omitted faces meet all their links $E(F)$. Writing their budget as $D_F(q) \leq |E(F)|/6$,

$$\|V_q A \Omega\| \leq \|A\| [\sigma_q + 2\alpha\tau_q D_F(q)] \rightarrow 0.$$

The local rank-one orbit is dense and $\|V_q\|$ is uniformly bounded, so $V_q \rightarrow 0$ strongly. The resolvent identity and strong Duhamel formula give strong-resolvent convergence and strong unitary convergence uniformly on compact time intervals. Neither norm-resolvent convergence nor its failure follows merely from the nonvanishing perturbation norm.

7.2 Stationarity improves the time window **

Round21 first compared connected correlations using a vector residual, obtaining sufficient windows $T_q = (\hbar/\alpha)C(1-q)^{-\gamma}$, $\gamma < 3/2$. Round22 exploited stationarity before estimating. For bounded complex local observables,

$$C_q^{A,B}(t) = \omega_q(A^* \beta_q(t)(B)) - \omega_q(A^*) \omega_q(B), \quad \beta_q(t)(B) = e^{-itH_q/\hbar} B e^{itH_q/\hbar}.$$

Both scalar energy phases cancel in this Heisenberg expression. The bounded cocycle proves the strong integral identity

$$\beta_q(t)(B) - \beta_0(t)(B) = -\frac{i}{\hbar} \int_0^t \beta_q(t-s)([V_q, \beta_0(s)(B)]) ds.$$

No differentiation of an arbitrary bounded B on an unbounded domain is needed. Reference evolution stays on the same complete factors, so $\|[V_q, \beta_0(s)(B)]\| \leq 2\alpha\tau_q \|B\| D_{FB}(q)$ for every s . Changing the state costs $2d_q$ in the first expectation and $4d_q$ in the two mean factors. Consequently

$$|C_q^{A,B}(t) - C_0^{A,B}(t)| \leq \|A\| \|B\| \left[6d_q + \frac{2\alpha\tau_q|t|}{\hbar} D_{FB}(q) \right]. \quad ** \quad (46)$$

Since $d_q = O((1-q)^{3/2})$ and $\tau_q = O((1-q)^3)$, this includes every $\gamma < 3$, with sufficient rate $O((1-q)^{\min(3/2, 3-\gamma)})$. The old $\gamma = 3/2$ endpoint is covered. Failure of the new upper certificate at $\gamma = 3$ does not establish actual nonconvergence.

For moving origin-anchored tail-block cubes of side L , all outgoing links are retained: there are $24L^3$ links and $21L^3$ incident omitted faces. Downward closure in the positive orthant makes incidence equivalent to having the face anchor inside that tail rectangle. Thus

$$D_L(q) = \mathcal{B}(q)(1-q^{4L})(1-q^{2L})(1-q^L). \quad ** \quad (47)$$

For $L = \max(1, \lfloor \ell(1-q)^{-\beta} \rfloor)$, $\ell > 0$, $\beta \geq 0$, and time windows $|t| \leq (\hbar/\alpha)C(1-q)^{-\gamma}$ with $C > 0$, $\gamma \geq 0$, the nonnegative certificate in (46) vanishes exactly when $\beta < 1$ and $\gamma < 3(1-\beta)$. This necessity concerns that certificate only. At $\beta = 0$, its constant uses $L_0 = \max(1, \lfloor \ell \rfloor)$, not ℓ . Logarithmically shortened fixed-support $\gamma = 3$ windows also pass. Translated rectangles need new incoming-face counts. Actual long-time endpoint behavior remains open. Sources: [research/round13/locality/locality.md](#), [research/round20/forward/h2/report.md](#), [research/round21/forward/m1/report.md](#), [research/round21/forward/m2/report.md](#), [research/round22/reverse/n1/report.md](#), [research/round22/reverse/n2/report.md](#).

8 Static information, clocks, and exact physical reduction

8.1 Conditional dynamics and finite-rate identifiability **

On the conditional three-link space with $\rho_\kappa \propto e^{\kappa S_3}$, a separately specified reversible generator is

$$A_{\kappa,m} = -\rho_\kappa^{-1} \operatorname{div}(\rho_\kappa m \nabla), \quad H = c A_{\kappa,m}, \quad c > 0.$$

For smooth uniformly positive mobility m , its closed form has domain H^1 , operator domain H^2 , and constant ground. This standard divergence-form construction preserves the same static law for every c and m , demonstrating why static moments cannot fix a clock. The actual shared-middle gradient produces

$$\Gamma(S_3) = \frac{1}{4} \{15 + 8y + 2x + 2z + 2r - (3x+w)^2 - (y+w+t)^2 - (z+t)^2\}, \quad r = \operatorname{Tr}(UW^\dagger)/2.$$

The mixed term involving $r - wt$ disappears under an incorrect resampling of V . Independent optimization of U, W at fixed y proves $\min S_3 = -5$, $\max S_3 = 7$. Bounded-density comparison with the Haar Casimir gap then gives $\text{gap}(cA_{\kappa,1}) \geq (3c/4)e^{-12|\kappa|} \geq c/6$ for $|\kappa| \leq 1/8$. This is a theorem for the declared conditional generator, not a derived lattice Hamiltonian.

For $m = 1 + \zeta x$, $|\zeta| < 1$, the measurable initial rate would be $r_f = -\hbar C'_f(0+) = c \int m \Gamma(f) \rho_\kappa$. At known $\kappa = 0$, the pair $x, x + x^2$ gives

$$r_x = 3c/16, \quad r_{x+x^2} = c(5/16 + \zeta/8), \quad c = 16r_x/3, \quad \zeta = 8r_{x+x^2}/c - 5/2. \quad **$$

The pair x, x^2 is parity-blind to ζ . For known $|\kappa| \leq 1/8$, choosing $g = x + x^2/4$ proves a uniform determinant lower bound $D \geq 1/7168$. Indeed, if $G(x) = (1 - x^2)/4$ and X, X' are independent copies of the actual one-coordinate marginal, symmetrization gives, for $g = x + ax^2$,

$$D = 2a \mathbb{E}\{G(X)G(X')(X - X')^2[1 + a(X + X')]\} > 0 \quad (0 < a < 1/2).$$

Comparison to Haar yields $D \geq 3a(1 - 2a)e^{-24|\kappa|}/128$; $a = 1/4$ and $e^3 < 21$ give the stated rational floor.

These inverses identify only their declared finite family. At Haar, $p_3 = x^3 - 3x/8 = C_3^{(2)}(x)/32$ is invisible to both rates. Adding $h = x^2 + x^3$ identifies $m = 1 + \zeta x + \xi p_3$ through

$$\begin{pmatrix} r_x \\ r_{x+x^2/4} \\ r_{x^2+x^3} \end{pmatrix} = \begin{pmatrix} 3/16 & 0 & 0 \\ 25/128 & 1/32 & 0 \\ 59/256 & 9/64 & 9/512 \end{pmatrix} \begin{pmatrix} c \\ c\zeta \\ c\xi \end{pmatrix}, \quad \det M = 27/262144. \quad ** \quad (48)$$

The box $|\zeta|, |\xi| \leq 1/4$ guarantees $m \geq 13/32$, hence gap at least $13c/192$ on the stated κ range. The Gegenbauer polynomial itself is classical [12]; its blind and detecting roles here are the project calculation.

More generally N initial rates impose N linear conditions on a polynomial with $N + 1$ coefficients. A nonzero null polynomial, normalized by its coefficient absolute sum, has sup norm at most one. It gives a whole positive interval of distinct mobilities with identical rates at fixed density and c . The explicit $p_5 = x^5 - 5x^3/6 + x/8 = C_5^{(2)}(x)/192$ preserves all three rates above, but changes the x -correlation curvature to

$$C''_x(0+) = \frac{c^2}{\hbar^2} \left(\frac{9}{64} + \frac{\epsilon^2}{1024} \right) \quad \text{for } m = 1 + \epsilon p_5. \quad **$$

Thus the obstruction concerns finitely many initial slopes, not complete time curves. No physical rates were measured; the inversions were tested on synthetic exact inputs. Sources: `research/round20/forward/f2/report.md`, `research/round21/forward/k2/report.md`, `research/round21/forward/l1/report.md`, `research/round21/forward/l2/report.md`.

8.2 A failed section and a valid maximal-tree map **

Round22 tested physical matching on the actual 18-vertex, 33-link graph at zero magnetic coupling. Fixing thirty links to identity leaves the old U, V, W coordinates, but fixes thirteen genuine cycles in addition to tree gauge variables. It is not a Hilbert-space reduction. If Q_0 is a face fixed to identity, then $f_n = \chi_n(Q_0)/(n + 1)$ satisfies $\|f_n\|_2 = 1/(n + 1) \rightarrow 0$, whereas its section equals one. The restriction is nonclosable. Three edge-disjoint such faces give an analogous sequence converging even in the electric graph norm. Moreover $H_E x_0 = 3\alpha x_0$ before restriction, but the conditional generator annihilates its constant image. A scalar clock cannot repair this core defect.

A designated slope fit for a physical square that restricts to x gives $c = 4\alpha$. A held-out square restricting to $w = \text{Tr}(UV^\dagger)/2$ has physical correlation $e^{-3\alpha t/\hbar}/4$, while the fitted conditional prediction is $e^{-6\alpha t/\hbar}/4$. The disagreement is strict at every finite positive time.

The repair uses a genuine maximal tree $T = \{0, \dots, 15, 26\}$, root $(0, 2, 0)$, and all sixteen chord holonomies $L_e = h_{s(e)}g_e h_{t(e)}^{-1}$. At fixed tree variables, left/right Haar invariance gives product Haar on the chords. Gauge-invariant functions become simultaneous-conjugation invariants on these sixteen variables, yielding a unitary J_{16} . The three-chord isometry J_3 embeds functions of L_{28}, L_{27}, L_{24} ; its adjoint integrates the other thirteen chords, rather than evaluating them at identity. These are the standard maximal-tree principles [11], here realized on the complete signed graph.

Every tree derivative must also be transported. If cutting tree edge q leaves subtree S_q , the induced vector field is

$$D_{q,a} = \epsilon_q \left(\sum_{s(e) \in S_q} \ell_{e,a} - \sum_{t(e) \in S_q} \rho_{e,a} \right).$$

Both endpoint occurrences remain when a chord lies inside S_q . The full generator is $-\alpha \sum_a (\sum_e \ell_{e,a}^2 + \sum_q D_{q,a}^2)$. Its quadratic form is a sum of squared gradients, bounded above and below by positive multiples of the product Casimir form. On finite Peter–Weyl blocks these comparisons give equivalence of H^1 forms and H^2 operator domains and identify the closed operators. On the selected three chords the exact reduction is

$$H_{\text{eff}}/\alpha = 3C_U + 3C_V + C_W + C_{VW}^\ell + C_{VW}^\rho + C_{UVW}^\ell + 2C_{UVW}^\rho. \quad ** \quad (49)$$

Here $C_A^\ell = -\sum_a (\sum_{j \in A} \ell_{j,a})^2$, and similarly on the right. The image reduces H_E because its orthogonal projector is Haar integration over unused original chord links and commutes with each link heat semigroup. This proves exact electric semigroup intertwining, not merely a formal compression.

The new x, y, z observables complete to physical loops of lengths six, eight, six, respectively, so their electric energies are $(9/2)\alpha, 6\alpha, (9/2)\alpha$. They are different completions from the old section’s training square. The prescribed x -slope now gives $c = 6\alpha$; its reserved curvature forces $\zeta = 0$. The held-out y -curve still differs: the physical exponent is 6α , the conditional one $9\alpha/2$. Thus exact Haar reduction succeeds while matching to the scalar-clock affine-mobility family fails. Sources: `research/round22/reverse/p1/report.md`, `research/round22/reverse/p2/report.md`.

8.3 Magnetic leakage, projected memory, and exact channels ^{**}

Keep this tree and add all twenty magnetic faces: $H_\lambda = H_E + \alpha\lambda V$, $V = \sum_{p=0}^{19} (1 - x_p)$, $\lambda \geq 0$. This λ is dimensionless, unlike the physical λ_p used earlier. With $P = J_3 J_3^*$, only one face survives conditional averaging, so $J_3^* V J_3 = 20 - \text{Tr}(VW^\dagger)/2$. Three other faces share the same omitted chord. Their nonzero conditional cross moments give

$$\begin{aligned} B_\lambda^* B_\lambda &= (\alpha\lambda)^2 M_d, & B_\lambda &= Q H_\lambda J_3, \\ d &= \frac{19}{4} + \frac{1}{2}(r + x + z) = 4 + \frac{1}{4} \|u + w + e_0\|^2 \in [4, 25/4]. \end{aligned} \quad ** \quad (50)$$

Here $r = \text{Tr}(UW^\dagger)/2$, and u, w are unit quaternions. Hence every nonzero selected vector leaks at nonzero λ ; $\|B_\lambda\| = 5\alpha\lambda/2$. Replacing d by its mean $19/4$ loses the correlated face terms. Also $H_\lambda 1$ is nonconstant, with centered squared residual $5(\alpha\lambda)^2$; the Haar reference is not an interacting ground.

Set $A_\lambda = J_3^* H_\lambda J_3$ and $T_\lambda(t) = J_3^* e^{-tH_\lambda/\hbar} J_3$. Two bounded off-diagonal Duhamel insertions prove

$$\|T_\lambda(t) - e^{-tA_\lambda/\hbar}\| \leq \min\{1, \frac{25}{8}(\alpha\lambda t/\hbar)^2\}. \quad **$$

The scaled difference tends strongly to $M_d/2$ as $t \downarrow 0$; this is not an operator-norm Taylor expansion. It already disproves autonomy of the selected compression.

The complement has the exact electric lower bound 3α , by the physical girth argument and an omitted square attaining it. Thus $C_\lambda = QH_\lambda Q \geq 3\alpha$. Eliminating the complementary component gives the exact memory and resolvent relations

$$\hbar u' = -A_\lambda u + \hbar^{-1} \int_0^t K_\lambda(t-s)u(s) ds, \quad K_\lambda(s) = B_\lambda^* e^{-sC_\lambda/\hbar} B_\lambda,$$

$$J_3^*(H_\lambda + z)^{-1} J_3 = [A_\lambda + z - \Sigma_\lambda(z)]^{-1}, \quad \Sigma_\lambda(z) = B_\lambda^*(C_\lambda + z)^{-1} B_\lambda, \quad z > 0.$$

The full return $u(s)$ remains inside the memory integral. These block identities are established methods; the actual channel content and quantified errors are the project contribution.

For the two orthonormal channels $1, 2x$, the leading kernel $(\alpha\lambda)^2 B_1^* e^{-sC_0/\hbar} B_1$ has seven exact electric energies. The 11 weights at energies $3, 9/2, 6, 15/2, 8, 17/2, 9$, in units of α , are

$$(1/8, 1/8, 3/8, 9/4, 9/8, 3/8, 3/8). \quad **$$

The 00 weight is $19/4$ at energy three, and the off-diagonal weight is $1/4$ at that same energy. To derive the table, intersect each face with the actual six-link x loop. A shared consecutive path of length $m > 0$ has spin-zero and spin-one components of squared norms $1/16, 3/16$, with energies $3(10 - 2m)/4$ and that number plus $2m$. Nine faces have disjoint edge support; the other counts for $m = 1, 2, 3$ are six, two, two. This is an exact finite spectral orbit in the full Hilbert space, not a representation cutoff.

Since $\|V\| \leq 40$, bounded Duhamel and the resolvent identity give full selected-space errors

$$\begin{aligned} \|K_\lambda(s) - K_\lambda^{(2)}(s)\| &\leq 250\alpha^3 \lambda^3 (s/\hbar) e^{-3\alpha s/\hbar}, \\ \|\Sigma_\lambda(z) - \Sigma_\lambda^{(2)}(z)\| &\leq \frac{250\alpha^3 \lambda^3}{(3\alpha + z)^2}. \end{aligned} \quad ** \quad (51)$$

These are uniform absolute kernel bounds in delay and uniform complementary self-energy bounds down to $z = 0$. They do not give uniform relative late-time accuracy or a bounded full resolvent at zero. Sources: `research/round22/reverse/q1/report.md`, `research/round22/reverse/q2/report.md`.

9 Homogeneous stability: explicit progress and retained obstructions

9.1 A matched theorem dictionary and a controlled rotation **

The qualitative homogeneous stability route is independent of the summable construction. Yarotsky's product-vacuum theorem permits unbounded onsite operators and infinite-dimensional site spaces, but its range-dependent constants remain unevaluated [14]. The Round13 outgoing-link blocking gives normalized perturbation $\epsilon \leq (4/3) \binom{d}{2} \lambda_{\max}/\alpha$. Boundary padding handles all-contained plaquettes; it does not silently identify every thermodynamic boundary limit. A sufficiently small, existential magnetic/electric ratio gives a fixed-spacing gap at least $3\alpha/8$. A qubit, two-site algorithmic theorem cannot supply its missing constants [16].

The dressed-strip reference admits a sharper dictionary. Each coarse tail cell contains 24 links: ten strip links and fourteen free links. There are 24 anchored faces, of which three are selected and 21 omitted. Their full grouped support is $G = \{0, e_x, e_y, e_z\}$. In energy unit $\delta = \alpha/8$,

$$h_b \geq 1 - P_b, \quad \phi_b = -\frac{\tau}{3} \sum_{f \in O_b} x_f, \quad \|\phi_b\| = 7|\tau|. \quad **$$

All 21 omitted faces cross a coarse block boundary. Ignoring crossing faces would erase the entire homogeneous perturbation. The matched source theorem gives an existential interval

$$|\tau| < \frac{1}{7} \min\{c_1(G), [2c_2(G)]^{-1}\}, \quad \Delta > \alpha/16,$$

including literal finite rectangles through clipped reference sites and padding. No numerical value of τ , including $1/64$, is thereby certified.

The full magnetic interaction is not purely form-bounded relative to the undressed vacuum energy. On an actual square, $\psi_t = (1 + t\chi_p)/\sqrt{1 + t^2}$ has electric energy $3\alpha t^2/(1 + t^2)$ and magnetic expectation $-\lambda t/(1 + t^2)$; their absolute ratio diverges as $t \rightarrow 0$. The diagonal compression alone does obey the incidence bound $|\langle W_{\text{diag}} \rangle| \leq (16\lambda_{\max}/3\alpha)\langle H_0 \rangle$. The missing off-diagonal term cannot be discarded.

On one four-site star, conditional Haar orthogonality gives $\|\phi\Omega\|^2 = 7\tau^2/12$. Let $v = \phi\Omega$, $u = (H_0|_Q)^{-1}v$, and $S = |u\rangle\langle\Omega| - |\Omega\rangle\langle u|$. Then $[S, H_0] = -(|v\rangle\langle\Omega| + \text{adjoint})$. The finite-rank range lies in $D(H_0)$; the exponential preserves that domain. Integrating the bounded commutator gives the exact identity

$$e^S(H_0 + \phi)e^{-S} = H_0 + Q\phi Q + R, \quad \|R\| \leq (707/60)\tau^2. \quad **$$

This instantiates a standard unitary dressing method with the actual local variance. The remainder is part of the Hamiltonian.

For simultaneous finite-volume rotations, every nonzero ordered commutator word adds a star meeting the previous union. $|G - G| = 13$, so there are at most $13^n n!$ relative words of order n , each supported on at most $4 + 3n$ sites. Rooting at a fixed site adds a factor at most $4 + 3n$. In the interaction norm $\|\Phi\|_w = \sup_x \sum_{Y \ni x} 2^{|Y|} \|\Phi_Y\|$, this yields a convergent all-order remainder majorant. The admitted common certificate is

$$\|R\|_w \leq \frac{25460736}{25}\tau^2, \quad |\tau| \leq 5/1664, \quad |\langle D \rangle| \leq 28|\tau|\langle H_0 \rangle. \quad ** \quad (52)$$

The reverse derivation separately obtained the smaller valid constant $1970176/5$; it is not substituted into the subsequently frozen common recurrence. Neither constant is a homogeneous gap threshold.

9.2 Generated supports, interacting inverses, and the remaining source **

For a residual term on an arbitrary generated support Y , split

$$R_Y = c_Y I + A_Y + Z_Y, \quad A_Y = |Q_Y R_Y \Omega_Y\rangle\langle\Omega_Y| + \text{adjoint}, \quad Z_Y = Q_Y(R_Y - c_Y I)Q_Y.$$

The subtraction inside Z_Y is necessary when the scalar is $c_Y I$. The bare homological inverse has norm at most one, and the resulting diagonal update is bounded by $2\|R_Y\|$. For exponential cardinality weight μ , an explicit root/intersection sum gives

$$\|[\Phi, \Psi]\|_{\mu-\ell} \leq \frac{4}{e\ell} \|\Phi\|_\mu \|\Psi\|_\mu, \quad \ell > 0.$$

Spending ℓ/n per nested commutator and $n! \geq (n/e)^n$ bounds the factorial-divided order by $(4\|S\|_\mu/\ell)^n \|K\|_\mu$. If $r > 0$ bounds the residual, $b \geq 0$ bounds the retained diagonal interaction, and $\ell > 4r$, the reverse majorant is

$$r' \leq \mathcal{F}(r, b; \ell) := \frac{4r}{\ell - 4r}(b + 3r/2), \quad b' \leq b + 2r, \quad ** \quad (53)$$

$$\mathcal{F}(r, b; \ell) < r \iff \ell > 4b + 10r.$$

The equivalence refers to the displayed scalar majorant, not the actual residual; $r = 0$ is the separate exact zero-residual case. Any finite total loss has $\ell_n \rightarrow 0$, while this majorant retains $b_n \geq 448|\tau| > 0$. Its positive scalar trajectory eventually leaves its convergence radius. This proves failure of that certificate, not of the true rotation algorithm or the physical gap.

An equal-weight commutator counterexample also needs its exact support convention. In the reverse structured example, regrouping to the actual support gives the special homological value $1/64$; retaining indexed unions instead gives $(9N + 16)/(64(3N + 1))$. The post-freeze gate explicitly records this distinction. It is not legitimate to improve an interaction norm by silently changing its decomposition.

Round22 then replaced the bare inverse by the actual *initial* diagonal operator. For finite Y , let $G_Y = H_Y + \sum_{b+G \subset Y} Q_b \phi_b Q_b$. Four-star incidence gives, with $a = 28|\tau| < 1$,

$$(1 - a)H_Y \leq G_Y \leq (1 + a)H_Y, \quad \left\| (G_Y|_{Q_Y})^{-1} \right\| \leq (1 - a)^{-1}.$$

The inverse maps into $D(H_Y)$, so its rank-two skew generator has a valid domain-preserving exponential. Interior cancellation is exact, but on the full finite volume

$$[S_Y, H_0 + D] + A_Y = \sum_{\substack{b+G \cap Y \neq \emptyset \\ b+G \not\subset Y}} [S_Y, D_b].$$

For the admitted actual $SU(2)$ diagnostic source consisting of a free link character at the origin, the squared boundary residual on the product vacuum is exactly

$$\|([S_Y, H_0 + D] + A_Y)\Omega\|^2 = 7\tau^2/432. \quad ** \quad (54)$$

All 21 face contributions and their vanishing distinct-face cross moments are retained. This source is admissible but is not asserted to be an actually generated term of the earlier rotation.

Finally, the infinite initial diagonal is constructed as a relative form sum on the same incomplete product. Let $K = H_0^{-1/2} D H_0^{-1/2}$ on Q_Ω ; $\|K\| \leq a < 1$. For a vacuum-exterior source $v \perp \Omega$,

$$u = H_0^{-1/2} (I + K)^{-1} H_0^{-1/2} v = \sum_{n \geq 0} H_0^{-1/2} (-K)^n H_0^{-1/2} v. \quad ** \quad (55)$$

The form equation proves $Gu = v$. Each coefficient lies in the n -collar $Y^{(n)}$ with norm at most $\|v\| a^n$; direct finite-support action also gives an H_0 -graph bound $\|H_0 u_n\| \leq \|v\| |Y| (2n - 1)^3 a^n$ for $n \geq 1$. Polynomially weighted geometric tails prove Hilbert, half-energy, H_0 -graph and G -graph convergence, including explicit finite-volume collar errors. This response regularity does not prove global equality $D(G) = D(H_0)$.

The exact global vacuum rank-two source is $A_Y \otimes P_{\text{ext}}$. Against the desired local source $A_Y \otimes I$, the remaining operator is $A_Y \otimes Q_{\text{ext}}$, of norm one on the specified exterior character test. Moreover, the prescribed collar interaction certificate $\sum_n e^{\mu(n+1)^3} a^n$ diverges for every fixed $\mu > 0$, $0 < a < 1$, although the unweighted response converges. Thus full local-source cancellation and a useful weighted decomposition remain missing. These are the precise next requirements for an all-stage homogeneous stability proof; the later diagonal updates are not covered by the initial inverse.

The domain and inverse comparisons use established form and Lie–Schwinger methods [13, 18]. Their complete source inductions and numerical stability thresholds were not imported. Sources: `research/round21/forward/i1/report.md`, `research/round21/forward/i2/report.md`, `research/round22/advisor/o1-gate.json`, `research/round22/advisor/o2-gate.json`, `research/round22/reverse/r1/report.md`, `research/round22/reverse/r2/report.md`.

10 Full sources, endpoint observables, and certified heat: Rounds 23–26

The twenty-eight investigations S1–AF2 develop three separate models. The homogeneous initial-diagonal problem uses dimensionless generators with physical energy unit $\alpha/8$; the finite-graph problem uses $L = H/\alpha$ and heat time $\sigma = \alpha t/\hbar$; the canonical summable family uses its original real-time generator and a q -dependent interaction profile. Throughout, lattice spacing a , $E_\star > 0$, $\alpha/E_\star > 0$, and $\hbar > 0$ are fixed. A source filter, collar depth, Galerkin projection, joining time, or input-overlap restriction is a proof or approximation parameter, not a new term in an action.

The marker ****** means a contribution newly derived or computed in this workbench; scientific priority is unverified. S1/S2/T1, all ten Round 24 loops, and all ten Round 26 loops have separately authored model-agent derivations and skeptical reviews. T2/U1/U2 and AA1/AA2 instead have same-author derivation, reverse reconstruction, and critical self-review. These categories describe provenance, not external mathematical peer review or experimental independence. Historical limited verdicts remain limited even where later work answers a previously missing premise. The detailed source-to-equation mapping is recorded in `audit/recent-ledger.json` accompanying this manuscript.

10.1 An actual homogeneous source and its regulated inverses

10.1.1 ****** S1: a generated cubic source and the complete crossing defect

Use the complete 24-link onsite factors and four-site stars $Y = \{0, e_1, e_2, e_3\}$ of the homogeneous construction. On one star let $H = H_{0,Y}$, $P_0 = |\Omega_Y\rangle\langle\Omega_Y|$, $Q_0 = I - P_0$, and $H \geq Q_0$. With the actual 21-face potential ϕ_0 , set

$$v = \phi_0 \Omega_Y, \quad u = H_{Q_0}^{-1} v, \quad a_0 = \langle u, v \rangle, \quad b_0 = \|u\|^2, \quad s_0^2 = \|v\|^2 = \frac{7}{12} \tau^2. \quad (56)$$

The Haar identity follows because each face has a free Haar link and distinct faces have an unmatched free link: the 21 diagonal half-trace moments are $1/4$ and every cross term vanishes. Define $S = |u\rangle\langle\Omega_Y| - |\Omega_Y\rangle\langle u|$ and $A_0 = |v\rangle\langle\Omega_Y| + \text{adjoint}$. The specified repeated-anchor term in the previous remainder is

$$C = \frac{1}{2}[S, [S, \phi_0]] - \frac{1}{6}[S, [S, A_0]], \quad w = Q_0 C \Omega_Y = -a_0 u - \frac{b_0}{3} v. \quad \mathbf{**} \quad (57)$$

The two double commutators have the same Q_0 -column $-3a_0 u - b_0 v$; their coefficients give the displayed result. In particular, $\langle u, w \rangle = -4a_0 b_0/3 < 0$ for $\tau \neq 0$, so this is an actual nonzero source, with $r := \|w\| \leq 4s_0^3/3$. It is one indexed cubic term; other terms in a regrouped cubic coefficient could cancel it. Its exact local split is $C = c_0 I + A + Q_0(C - c_0 I)Q_0$, where $c_0 = \langle \Omega_Y, C \Omega_Y \rangle$ and $A = |w\rangle\langle\Omega_Y| + \text{adjoint}$. All relative ratios divided by r below require $\tau \neq 0$, hence $r > 0$. At $\tau = 0$ the generated source vanishes; the undivided bounds still apply.

Write $M = 7|\tau| \leq 35/1664$, $D_c = Q_c \phi_c Q_c$, and $G_\Lambda = H_{0,\Lambda} + \sum_c D_c$, retaining only complete stars in a containing finite cuboid. The one-star operator $G_Y = H + D_0$ satisfies $(1 - M)H \leq G_Y \leq (1 + M)H$ as forms and $D(G_Y) = D(H)$ by bounded perturbation. Therefore

$$z = (G_Y|_{Q_0})^{-1} w, \quad K_Y = |z\rangle\langle\Omega_Y| - \text{adjoint}, \quad [K_Y, G_Y] = -A, \quad \|K_Y\| \leq \frac{r}{1 - M}. \quad (58)$$

The inverse places z in the operator domain. Identity extension preserves $D(H_{0,\Lambda})$ because the local commutator with H is bounded and exterior spectral projections commute with K_Y . The

graph-bounded exponential and its inverse preserve that domain as well. The complete extended equation is

$$[K_Y \otimes I, G_\Lambda] = -A \otimes I + F_Y, \quad F_Y = \sum_{c \text{ crossing } Y} [K_Y, D_c], \quad \|F_Y\| \leq \frac{6Mr}{1-M}. \quad ** \quad (59)$$

At the positive-octant origin there are three possible crossings, with anchors e_1, e_2, e_3 , each on a seven-site union. For the single-source decomposition, $\sum_c e^{7\mu} \| [K_Y, D_c] \| \leq 6e^{7\mu} Mr/(1-M)$. A translated interior star has twelve crossings. The reverse-origin sharper rooted count is $12 \cdot 7 = 84$, giving a family bound $168e^{7\mu} Mr_*/(1-M)$; the forward count 96 gives the valid weaker constant 192. These are one-step interaction bounds, not an all-order inverse.

An exterior mean-zero unit Haar character ξ distinguishes the actual source from its vacuum-projected replacement: $(A \otimes I)(\Omega_Y \otimes \xi) = w \otimes \xi$ while $(A \otimes P_{\text{ext}})(\Omega_Y \otimes \xi) = 0$. Also a bounded $D(G)$ -preserving exact inverse must satisfy $P_E A P_E = 0$ at every energy; a positive ground gap alone says nothing about all excited Bohr differences. This explains S1's limited gate and motivates a regulated full-source equation. Sources: `research/round23/forward/s1/report.md`, `research/round23/reverse/s1/report.md`, and `research/round23/skeptic/s1.md`.

10.1.2 **** S2 and W1–W2: retain the residual and name its topology**

For $\alpha_s^G(A) = e^{isG} A e^{-isG}$, S2 fixes $0 < \Theta \leq 1/16$ and defines

$$\begin{aligned} \mathcal{L}_\Theta(A) &= -\frac{i}{2} \int_{-\Theta}^{\Theta} \text{sgn}(s) (1 - |s|/\Theta) \alpha_s^G(A) ds, \\ \mathcal{R}_\Theta(A) &= \frac{1}{2\Theta} \int_{-\Theta}^{\Theta} \alpha_s^G(A) ds, \quad [\mathcal{L}_\Theta(A), G] = -A + \mathcal{R}_\Theta(A). \end{aligned} \quad ** \quad (60)$$

These are vectorwise strong integrals. In weak pairings on $D(G)$, integration by parts uses the kernel derivative $2\delta_0 - \Theta^{-1} \mathbf{1}_{[-\Theta, \Theta]}$. The bounded weak commutator then proves $\mathcal{L}_\Theta(A) D(G) \subset D(G)$ by self-adjointness; domain preservation of an arbitrary initial A is not assumed. Its norm and graph bounds are $\Theta r/2$ and $\Theta r/2 + 2r$. At Bohr frequency ω ,

$$\ell_\Theta(\omega) = \frac{1 - \text{sinc}(\Theta\omega)}{\omega}, \quad r_\Theta(\omega) = \text{sinc}(\Theta\omega), \quad \ell_\Theta(0) = 0, r_\Theta(0) = 1. \quad (61)$$

Thus every equal-energy block survives exactly; shortening Θ leaves more of the source.

For the rooted norm $\|\Phi\|_2 = \sup_x \sum_{X \ni x} 2^{|X|} \|\Phi_X\|$, a surviving ordered commutator word starts on one star and adds at most three sites at each step. The forward rooted count, before simplex integration, is

$$N_0 \leq 64r_*, \quad N_{n+1} \leq 64M(8+3n)N_n, \quad N_n \leq 64r_*(192M)^n(8/3)_n. \quad ** \quad (62)$$

Here the insertion cost includes $2M$ from a commutator and 2^3 from the support weight; incoming anchors and repetitions remain. The simplex contributes $|s|^n/n!$, and the inverse and residual filter moments are respectively $\Theta^{n+1}/(n+2)!$ and $\Theta^n/(n+1)!$. Consequently

$$\|\mathcal{L}_\Theta(\{A_b\})\|_2 \leq 32\Theta r_*(1 - 192M\Theta)^{-8/3}, \quad \|\mathcal{R}_\Theta(\{A_b\})\|_2 \leq 64r_*(1 - 192M\Theta)^{-8/3}, \quad (63)$$

with $192M\Theta \leq 105/416 < 1$. Independently, reverse counts at most $13^n n!$ relative words and $4 + 3n$ rooted translates. With $u = 208M\Theta$, its orbital sum is $16r_*(4-u)/(1-u)^2$; the inverse and residual bounds are $8\Theta r_*(4-u)/(1-u)^2$ and $16r_*(4-u)/(1-u)^2$. At $u = 35/128$ the rational sum is 6784/961. The full tail after depth N is

$$\sum_{n>N} (4+3n)u^n = \frac{u^{N+1}[(3N+7) - (3N+4)u]}{(1-u)^2}. \quad (64)$$

These alternative bounds concern indexed interaction decompositions, not a bounded extensive infinite sum.

The forward-origin infinite-representation identification adds a separate form argument. In the admitted product-reference representation, write $G \leftrightarrow h_0 + d$, $\kappa = 4M < 1$, and $\sum_b \|Q_b u\|^2 \leq 4h_0[u]$. Subtracting resolvent form equations for $u = (G + 1)^{-1}f$ and $u_\Lambda = (G_\Lambda + 1)^{-1}f$ gives

$$\|u_\Lambda - u\|_{\mathcal{F}} \leq \frac{\sqrt{\kappa}}{1 - \kappa} \left(M \sum_{b \text{ omitted}} \|Q_b u\|^2 \right)^{1/2} \longrightarrow 0. \quad (65)$$

Strong-resolvent convergence identifies the strong unitary limit. Stabilization of every finite connected word plus the uniform summable tail improves the single-source short-time limit to operator norm. It does not prove $D(G) = D(H_0)$ globally or a norm limit of global propagators. Sources: `research/round23/forward/s2/report.md`, `research/round23/reverse/s2/report.md`, `research/round23/skeptic/s2.md`.

W1 then evaluates the actual source's first form moment, a reverse-origin refinement separately reviewed. The ground-transform identity and the half-trace gradient identity give

$$h[F\Omega_Y] = 8 \sum_e \int |\nabla_e F|^2 |\Omega_Y|^2, \quad \sum_a |X_e^a W_f|^2 = \frac{1 - W_f^2}{4}, \quad h[v] = 14\tau^2 = 24s_0^2. \quad ** \quad (66)$$

Free-link parity also cancels cross gradients. In the spectral measure of v , $w = -f(H)v$ with decreasing positive $f(\lambda) = a_0/\lambda + b_0/3$. The covariance of λ with $f(\lambda)^2$ is nonpositive, so $h[w]/r^2 \leq 24$; Jensen's inequalities for the two inverse moments yield $r \geq s_0^3/432$. Because $G\Omega = 0$, $\mathcal{R}_\Theta(A)\Omega = \text{sinc}(\Theta G)w$. Applying $\text{sinc } x \geq 1 - x/2$ to this positive spectral measure and $G \leq (1 + 4M)H_0$ gives the actual lower bound

$$\|\mathcal{R}_\Theta(A)\| \geq [1 - 12\Theta(1 + 4M)]r \geq \frac{311}{1664}r > 0. \quad (67)$$

This prevents deleting the short-filter cubic residual. It does not establish an actual zero-frequency obstruction to every inverse. The complementary forward result is the source-mass certificate $\|\mathcal{R}_\Theta(A)\| \geq (5/6)r\sqrt{\nu([0, E])}$ when $\Theta E \leq 1$, and $\|\mathcal{R}_\Theta(A)\|/r \rightarrow 1$ as $\Theta \downarrow 0$; neither is a full-operator upper estimate. If a bounded exact inverse K were already known, integration would instead give $\|\mathcal{R}_\Theta(A)\| \leq \|K\|/\Theta$, a conditional criterion only. Sources: `research/round24/reverse/w1/report.md`, `research/round24/forward/w1/report.md`, `research/round24/skeptic/w1-review.md`.

W2 holds G fixed and sets $A_n = \mathcal{R}_\Theta^n(A)$, $K_n = \sum_{j < n} \mathcal{L}_\Theta(A_j)$. Telescoping preserves

$$[K_n, G] = -A + A_n, \quad \|K_n\| \leq n\Theta r/2, \quad (A_n)_\omega = \text{sinc}(\Theta\omega)^n A_\omega. \quad (68)$$

On the actual vacuum column, $g = 1 - 4M \geq 381/416$ supplies a gap. With the reverse bound $\rho = 1 - (\Theta g)^2/8 < 1$, a piecewise Taylor and $|\text{sinc } x| \leq 1/|x|$ argument yields

$$\|A_n \Omega\| \leq \rho^n r, \quad \|K_n \Omega - G_Q^{-1} w\| \leq \rho^n r/g, \quad \|G(K_n \Omega - G_Q^{-1} w)\| \leq \rho^n r. \quad ** \quad (69)$$

Forward independently obtains denominator 10 in $1 - (\Theta g)^2/10$. These are graph-norm convergence of a column, uniform in the containing volumes and valid in the admitted infinite representation. In each fixed finite volume, compact resolvent and dominated convergence in an energy eigenbasis additionally prove

$$\mathcal{R}_\Theta^n(A) \xrightarrow{\text{strong}} \mathcal{D}_G(A) := \sum_E P_E A P_E. \quad (70)$$

The value is not known to vanish. Compact-resolvent doublets with splittings $1/(k+1)$ and unit swap sources show why strong convergence can coexist with residual norm one and an unbounded formal inverse. In the actual model the old weighted majorants cover only $n \leq 3$ at the cap; failure at $n = 4$ is failure of that estimate, not divergence of the dynamics. W2 therefore does not close a changed-diagonal nonlinear iteration. Sources: `research/round24/forward/w2/report.md`, `research/round24/reverse/w2/report.md`, `research/round24/skeptic/w2-review.md`.

10.1.3 ** AB1–AB2: the surviving block is controlled by its boundary

Pinching the complete identity (59) gives

$$\mathcal{D}_G(A) = \mathcal{D}_G(F_Y), \quad \|\mathcal{D}_G(A)\| \leq \frac{6Mr}{1-M}, \quad \frac{6M}{1-M} \leq \frac{70}{543}. \quad ** \quad (71)$$

The diagonal map is contractive because its input and output blocks are orthogonal. As $r = O(|\tau|^3)$, this is an actual uniform finite-volume quartic upper bound, not an evaluation of a nonzero resonant block. On a spectral window $P = \mathbf{1}_{[E-\varepsilon, E+\varepsilon]}(G)$, reverse additionally obtains

$$\|PAP\| \leq \frac{(2\varepsilon + 6M)r}{1-M}, \quad (72)$$

because $\|(G - E)P\| \leq \varepsilon$. If no crossing star is retained, $F_Y = 0$ and the interior inverse is exact; otherwise zero versus nonzero crossed blocks remains unresolved. Pinching need not preserve locality.

For the Gaussian density $p_s(t) = e^{-t^2/(2s^2)}/(\sqrt{2\pi}s)$ and odd tail $h_s(t) = \text{sgn}(t) \int_{|t|}^{\infty} p_s(u) du$, define $\mathcal{R}_s(A) = \int p_s(t) \alpha_t(A) dt$ and $\mathcal{J}_s(A) = -i \int h_s(t) \alpha_t(A) dt$. Since $h'_s = \delta_0 - p_s$, $\|h_s\|_1 = s\sqrt{2/\pi}$, and $\|p'_s\|_1 = \sqrt{2/\pi}/s$, weak integration by parts proves

$$\begin{aligned} [\mathcal{J}_s(A), G] &= -A + \mathcal{R}_s(A), \quad \mathcal{J}_s(A)D(G) \subset D(G), \\ \mathcal{R}_s(A) &= \mathcal{R}_s(F_Y) + i \int p'_s(t) \alpha_t(K_Y) dt, \\ \frac{\|\mathcal{R}_s(A)\|}{r} &\leq \min \left\{ 1, \frac{6M + \sqrt{2/\pi}/s}{1-M} \right\}. \quad ** \end{aligned} \quad (73)$$

The residual and inverse spectral multipliers are $e^{-s^2\omega^2/2}$ and $(1 - e^{-s^2\omega^2/2})/\omega$, retaining the zero-frequency block. At $s = 4$ the bound is $626/1629 < 0.385$; reverse's $s = 2$ gives $1042/1629 < 2/3$. This advances W2 from a column contraction to a full-operator contraction on the fixed source and initial G . The asymptotic boundary budget is an upper bound, not a positive residual lower bound. Gaussian time support is unbounded, so the old finite-radius connected series cannot establish its locality by direct integration. Sources: `research/round26/forward/ab1/report.md`, `research/round26/reverse/ab1/report.md`, `research/round26/forward/ab2/report.md`, `research/round26/skeptic/ab1.md`, `research/round26/skeptic/ab2.md`.

10.1.4 ** AE1–AE2: locality, graph error, and the original weighted norm

AE1 uses an all-time interaction-picture commutator recurrence. Each site lies in at most four stars and a four-site star meets at most sixteen possible stars. Removing internal unitaries in the commutator recurrence leaves chains, rather than the growing-union count in (62). For the forward collar $C_R = \{x \in \mathbb{N}_0^3 : |x|_1 \leq 2R + 1\}$, every chain of at most R steps is retained. Duhamel and the chain exponential then give

$$\begin{aligned} \|\alpha_t^G(A) - \alpha_t^{G_R}(A)\| &\leq r \sum_{m \geq R+1} \frac{(32M|t|)^m}{m!} \leq r 2^{-R-1} e^{64M|t|}, \\ \|\mathcal{R}_s^G(A) - \mathcal{R}_s^{G_R}(A)\| &\leq r 2^{-R} e^{(64Ms)^2/2}. \quad ** \end{aligned} \quad (74)$$

The last step integrates the entire Gaussian using $\mathbb{E}e^{v|T|} \leq 2e^{v^2s^2/2}$. At the old cap and $s = 4$, $R = 64$ gives relative operator error below 2^{-42} but requires 374,660 complete factors. No local finite Hilbert dimension is assumed.

The independent reverse construction uses $C_R = \{0, \dots, R+1\}^3$ and $b_R = 3R^2 + 9R + 7$ crossing stars. With $v = 64M$, it bounds the residual and inverse errors by

$$\begin{aligned}\varepsilon_R &= 4rMb_Rs(1+vs)e^{v^2s^2/2}2^{-R}, \\ \kappa_R &= 4rMb_Rs^2e^{v^2s^2/2}2^{-R}, \quad \|\mathcal{J}_s - \mathcal{J}_{s,R}, G\| \leq 2\varepsilon_R.\end{aligned}\tag{75}$$

The graph conclusion is an additional calculation of every crossing commutator; small operator-norm error by itself would not prove it. At $s = 2, R = 40$, the bounds are $\varepsilon_R/r < 2 \cdot 10^{-7}$ and $\kappa_R/r < 10^{-7}$ on 74,088 factors. Telescoping collars produces exponential diameter decay, with polynomial translation multiplicities harmless. It does not establish the old support-cardinality norm: a factor $e^{\mu O(R^3)}$ overwhelms this 2^{-R} upper majorant. That failed majorant is not a divergence theorem. Sources: `research/round26/forward/ae1/report.md`, `research/round26/reverse/ae1/report.md`, `research/round26/skeptic/ae1.md`.

AE2 obtains a simultaneous statement in the original rooted weight by explicitly restricting $M \leq 10^{-3}$, equivalently $|\tau| \leq 1/7000$. At the fixed proof duration $T = 3$, use

$$p_T(t) = \frac{(1 - |t|/T)_+}{T}, \quad h_T(t) = \frac{\text{sgn}(t)}{2}(1 - |t|/T)_+^2, \quad \widehat{p}_T(\omega) = \text{sinc}^2(T\omega/2).\tag{76}$$

Its derivative and moments are $h'_T = \delta_0 - p_T$, $\|p'_T\|_1 = 2/T$, $\|h_T\|_1 = T/3$. For a translated interior source, all twelve crossings cost $24Mr_b/(1-M)$, so the boundary identity proves

$$\|\mathcal{R}_T(A_b)\| \leq \frac{24M + 2/T}{1-M}r_b \leq \frac{2072}{2997}r_b < 0.7r_b. \quad **\tag{77}$$

Origin-only three-crossing constants cannot be used for this family. Integrating the same ordered words as S2 yields, with $c = 192M$, $p = 8/3$,

$$\begin{aligned}\|\mathcal{R}_T(\{A_b\})\|_2 &\leq 128r_\star \sum_{n \geq 0} \frac{(p)_n(cT)^n}{(n+2)!}, \\ \|\mathcal{J}_T(\{A_b\})\|_2 &\leq 128Tr_\star \sum_{n \geq 0} \frac{(p)_n(cT)^n}{(n+3)!}.\end{aligned}\tag{78}$$

Here $cT \leq 72/125 < 1$, giving the common bound $64(125/53)^3r_\star$. Reverse's independent $208M$ count gives the smaller displayed sufficient budget $844000r_\star/2209 < 383r_\star$ at $208MT = 78/125$. These prove finiteness, not weighted contraction: their positive residual series already begins at the initial $64r_\star$ budget.

The same certificates require simultaneously $T > 2/(1 - 25M)$ and $T < 1/(192M)$, possible only if $409M < 1$; the reverse radius requires $441M < 1$. At the old cap $M = 35/1664$, the first lower bound is $3328/789$, while the upper bounds are $26/105$ and $8/35$. This is a precise failure of those joint certificates at the old cap. At their radius endpoint the inverse majorant is integrable but the residual majorant is not. The next missing premise is weighted contraction for generated sources with all scalar, diagonal, and domain updates, not another application of the original-source factor 0.7. Sources: `research/round26/forward/ae2/report.md`, `research/round26/reverse/ae2/report.md`, `research/round26/skeptic/ae2.md`.

10.2 A fixed physical graph: from memory to computed relative heat

The graph throughout this subsection has vertices $\{0, 1, 2\} \times \{0, 1, 2\} \times \{0, 1\}$, 33 links, 20 square faces, and all 18 vertex Gauss constraints. Its full physical Hilbert space is $\mathcal{H}_{\text{phys}} =$

$L^2(\mathrm{SU}(2)^{33}, d\mathrm{Haar})^{\mathrm{SU}(2)^{18}}$. Define

$$\begin{aligned} K &= \sum_e C_e, & L &= K + \lambda(20 - S), & S &= \sum_{p=1}^{20} W_p, \\ W_p &= \frac{1}{2} \mathrm{Tr} U_{\partial p}, & 0 \leq \lambda &\leq 10^{-2}, & \sigma &= \alpha t_{\mathrm{phys}}/\hbar. \end{aligned} \quad (79)$$

In heat estimates below, integration variables t, s and joining times are dimensionless coordinates in the same units as σ ; physical duration is $t_{\mathrm{phys}} = (\hbar/\alpha)\sigma$. The multiplication $20 - S$ lies between zero and 40. Gauge averaging commutes with the electric spectral projections; bounded perturbation preserves the self-adjoint electric domain and its smooth invariant core. The electric resolvent is compact. Its vacuum is unique and the physical electric gap is 3: a nonempty invariant active-edge support has no degree-one vertex, hence contains a cycle, whose at least four spin-half edges cost $4(3/4)$. A fundamental square character attains 3. These are full-space facts, not a finite-spin assumption.

10.2.1 ** T1–T2: a full-space memory model and separate ground centering

For the inherited infinite-rank isometry J from the selected physical space, write the full Hamiltonian in retained/complementary blocks

$$H_\lambda = \begin{pmatrix} A & B^* \\ B & C \end{pmatrix}, \quad \tilde{H}_\lambda = \begin{pmatrix} A & B^* \\ B & C_0 \end{pmatrix}, \quad C - C_0 = \alpha\lambda D, \quad 0 \leq D \leq 40Q, \quad \|B\| = \frac{5}{2}\alpha\lambda. \quad (80)$$

The diagonals have their actual inherited electric domains, $A \geq 19\alpha\lambda$ and $C_0 \geq 3\alpha$. The scalar Schur/Young inequality gives $\tilde{H}_\lambda \geq 18\alpha\lambda I$ throughout the stated interval; $H_\lambda \geq \tilde{H}_\lambda$ as forms gives the same separate spectral floor for H_λ . No order between their heat operators is inferred. Bounded perturbation establishes both on the full electric domain.

Eliminating the complementary component shows that $\tilde{T}(t) = J^* e^{-t\tilde{H}_\lambda/\hbar} J$ satisfies

$$\begin{aligned} \tilde{T}(t) &= e^{-tA/\hbar} + \hbar^{-2} \int_{0 \leq u \leq s \leq t} e^{-(t-s)A/\hbar} K_2(s-u) \tilde{T}(u) du ds, \\ K_2(v) &= B^* e^{-vC_0/\hbar} B. \end{aligned} \quad (81)$$

The full returned $\tilde{T}(u)$ retains repeated excursions. Iterating a difference of two solutions bounds it by $M_T(\|B\| t/\hbar)^{2n}/(2n)!$, proving uniqueness at each finite window. The corresponding Schur complement has minus sign $A + z - B^*(C_0 + z)^{-1}B$. A one-return substitution loses a fourth-order $(B^*B)^2$ term. The two earlier diagnostic channels cannot be made autonomous: their full second-derivative defect dominates $\alpha^2\lambda^2 \begin{pmatrix} 19/4 & 1/4 \\ 1/4 & 19/4 \end{pmatrix} > 0$. This last actual-model clarification originated in the independent T1 review.

Let $d = 3 - 18\lambda$. Duhamel for the purely complementary perturbation has one leakage factor on each side, each bounded by $(5\lambda/2)e^{-18\lambda\sigma}(1 - e^{-d\sigma})/d$. Thus

$$\begin{aligned} \|T(t) - \tilde{T}(t)\| &\leq 250\lambda^3 e^{-18\lambda\sigma} F_d(\sigma), \\ F_d(\sigma) &= \frac{\sigma(1 + e^{-d\sigma}) - 2(1 - e^{-d\sigma})/d}{d^2} \leq \min\{\sigma^3/6, \sigma/d^2\}, \\ \sup_{t \geq 0} \|T(t) - \tilde{T}(t)\| &\leq \frac{125\lambda^2}{9e(3 - 18\lambda)^2} \leq \frac{312500}{477144} \lambda^2. \end{aligned} \quad (82)$$

This is cubic in coupling at fixed σ , but its uniform quadratic bound uses decay in the unshifted energy convention.

T2 answers the centering objection rather than multiplying this estimate by uncontrolled ground exponentials. Along $L_s = \tilde{H}_\lambda/\alpha + s\lambda D$, min-max gives a simple ground $\epsilon_s \leq 40\lambda$ and excitation gap $g = 3 - 85\lambda/2 \geq 103/40$. The complementary ground inverse has denominator $d_0 = 3 - 40\lambda \geq 13/5$, so its leakage is $\ell = (5\lambda/2)/d_0$. Differentiation gives $\epsilon'_s = \lambda\langle\psi_s, D\psi_s\rangle$ and $\|G'_s\| \leq 40\lambda\ell/g$. In centered Duhamel the ground-ground term cancels exactly because $G_s(\lambda D - \epsilon'_s I)G_s = 0$. Two mixed, magnetic excited, and scalar excited terms cost respectively $80\lambda\ell/g$, $80\lambda\ell/g$, and $40\lambda\ell^2/g$. Integration in s proves

$$\sup_{t \geq 0} \left\| J^* e^{-t(H_\lambda - \epsilon_\lambda)/\hbar} J - J^* e^{-t(\tilde{H}_\lambda - \tilde{\epsilon}_\lambda)/\hbar} J \right\| \leq 60\lambda^2, \quad 0 \leq e_\lambda - \tilde{e}_\lambda \leq \frac{6250}{169}\alpha\lambda^3, \quad \|G_\lambda - \tilde{G}_\lambda\| \leq \frac{20000}{1339}\lambda^2. \quad ** \quad (83)$$

The exact cap coefficient before rounding is $1042500/17407 < 60$. The excited remainder has initial complementary component $-QG_s J$, which must remain in the bound $\|Q(e^{-u(L_s - \epsilon_s)} - G_s)J\| \leq \min\{2\ell, e^{-gu}\}$. These results concern infinite-rank selected heat, not a finite algorithm or a relative-output theorem. T2 is same-author self-review. Sources: `research/round23/forward/t1/report.md`, `research/round23/reverse/t1/report.md`, `research/round23/skeptic/t1.md`, `research/round23/forward/t2/report.md`, `research/round23/skeptic/t2.md`.

10.2.2 ** X1: two truncation theorems and an actually assembled sector

A zero-extended finite-rank heat operator has full-space error exactly one at $\sigma = 0$. X1 therefore separates two meaningful questions. For the kinetic cutoff $P_R = \mathbf{1}_{[0,R)}(K)$, $R > 3$, let $A_R = P_R L P_R$, $B_R = Q_R L P_R$, $b = 20\lambda$, and $c = 20\lambda$. Then $\|B_R\| \leq b$ and $Q_R L Q_R \geq R$. Complementary variation of constants and its return give

$$\|e^{-\sigma L} - e^{-\sigma A_R} P_R\| \leq e^{-R\sigma} + \frac{2b}{R} + \frac{b^2\sigma}{R}. \quad (84)$$

For actual ground energies ϵ, μ_R , Schur elimination and the full excited threshold 3 imply

$$0 \leq \mu_R - \epsilon \leq \frac{b^2}{R - c}, \quad \|G - G_R\| \leq D_R := \frac{b}{R - c} \left(1 + \frac{b}{3 - c}\right). \quad (85)$$

For any $0 < \sigma_0 \leq T$, separate ground centering consequently gives

$$\sup_{\sigma \geq \sigma_0} \|e^{-\sigma(L - \epsilon)} - e^{-\sigma(A_R - \mu_R)} P_R\| \leq \min\{2, \max(F_R, D_R + 2e^{-(3-c)T})\}, \quad F_R = e^{cT} \left[e^{-R\sigma_0} + 2b/R + b^2 T/R + T b^2/(R - c) \right]. \quad ** \quad (86)$$

Choosing T first and then R proves uniform operator-norm convergence on every positive delayed half-line. The certified example $\lambda \leq 0.01$, $\sigma_0 = 1$, $T = 4$, $R = 1024$ has error below 0.002; that large matrix was not assembled. A Hermitian finite coefficient perturbation of norm ζ adds at most $4\zeta/(g - 2\zeta)$ uniformly when both finite matrices use their own grounds and $2\zeta < g$. Independent rounding of a ground exponent instead produces $e^{\sigma\delta}$ growth or loss of the limiting ground.

The independent reverse projection retains all invariant Peter–Weyl blocks with degree $d = \sum_e 2j_e \leq N$. A face multiplier raises degree by at most four. For initial degree $d_0 \leq N$, set $r = \lfloor (N - d_0)/4 \rfloor + 1$ and $z = 20\lambda\sigma$. Dyson words of order below r agree exactly, including all intermediate states; therefore their uncentered difference is at most $2e^{-z} \sum_{n \geq r} z^n/n! \leq 2z^r/r!$. The true ground Neumann series has ratio $a = 20\lambda/3$ and raises degree by at most four, giving omitted ground norm $t_N = a^{\lfloor N/4 \rfloor + 1}/(1 - a)$ and energy error $\delta_N \leq 20\lambda t_N/(1 - t_N)$. The centered finite-window bound is

$$\|(e^{-\sigma(L - \epsilon)} - e^{-\sigma(A_N - \mu_N)} P_N) P_{d_0}\| \leq \min\{2, 2e^z z^r/r! + \sigma\delta_N\}. \quad ** \quad (87)$$

For $d_0 = 0, N = 20, 0 \leq \sigma \leq 1$, it is below 10^{-6} at the cap, but that larger matrix also remained unassembled. The cutoff, input, and time sets in (86) and (87) are different.

Both constructions identify the same actual rank-21 sector at $R = 9/2$ or $N = 4$: $\Omega = 1$ and $\phi_p = 2W_p$. Gauge support completeness below this energy follows from the four-cycle argument; five-edge supports cannot occur on the bipartite graph. Haar normalization and the absence of a zero parity mask among triples of faces give the complete matrix

$$(A_{21})_{00} = c, \quad (A_{21})_{0p} = -\lambda/2, \quad (A_{21})_{pq} = (3+c)\delta_{pq}, \quad \mu = c - \frac{\sqrt{9+20\lambda^2}-3}{2}. \quad ** \quad (88)$$

There are nineteen dark face eigenvectors at $3+c$ and one bright excited root $c+(3+\sqrt{9+20\lambda^2})/2$. The spin-one character $\chi_p = 4W_p^2 - 1$ is omitted, normalized, and has energy 8, with $\langle \chi_p, L\phi_p \rangle = -\lambda/2$. Thus the small matrix is a compression with genuine physical leakage. X1's parent practical-truncation gate remains limited. Sources: `research/round24/forward/x1/report.md`, `research/round24/reverse/x1/report.md`, `research/round24/skeptic/x1-review.md`.

10.2.3 ** X2: complete residual, spectral data, and evaluated late heat

Write $D = \sqrt{9+20\lambda^2}$, $w = (D-3)/2$, $h = \lambda/[2(3+w)]$, $Z = 1+20h^2$, and $f = (\Omega + h \sum_p \phi_p)/\sqrt{Z}$. The full residual is

$$F = S^2 - 5, \quad (L - \mu)f = -\frac{2\lambda h}{\sqrt{Z}}F, \quad \rho^2 = \frac{195\lambda^4}{4(3+w)D}. \quad ** \quad (89)$$

To obtain its complete norm, use $\mathbb{E}W_p^4 = 1/8$ and $\mathbb{E}W_p^2W_q^2 = 1/16$ for distinct faces, even when they share an edge: integrating an exclusive link first makes each squared factor constant. No four distinct faces have zero combined center parity. Hence

$$\mathbb{E}S^2 = 5, \quad \mathbb{E}S^4 = 295/4, \quad F = \frac{1}{4} \sum_p \chi_p + 2 \sum_{p < q} W_p W_q, \quad \|F\|^2 = 195/4. \quad (90)$$

This accounts for 20 spin-one and 190 orthogonal pair channels, not the whole omitted Hilbert space. Keeping only spin-one channels loses a factor 39 in the squared residual. The forward-origin electric refinement resolves the 62 shared-edge products into singlet/triplet norms $1/64, 3/64$, whereas 128 edge-disjoint products have energy 6. Its complete weight table is

Electric energy	9/2	6	13/2	8
Squared weight in F	31/8	32	93/8	5/4

The weights sum to $195/4$ and their first moment is 295. The independent form identity $\langle F, KF \rangle = 4\mathbb{E}S^4 = 295$ follows by integration by parts from $KS = 3S$. A link-center transformation sends every face to its negative, so the remaining odd potential moment vanishes. Thus for $\xi = -F/\sqrt{195/4}$, $\langle \xi, L\xi \rangle = \nu = 236/39 + c$.

The true ground ϵ is simple and every other eigenvalue is at least 3, by min-max. Taking the spectral expectation of $(L - \epsilon)(L - 3) \geq 0$ in f , and using the actual variational plane $\text{span}\{f, \xi\}$, gives

$$\delta_- := \frac{2\rho^2}{\nu - \mu + \sqrt{(\nu - \mu)^2 + 4\rho^2}} \leq \mu - \epsilon \leq \delta_+ := \frac{\rho^2}{3 - \mu},$$

$$\|G - |f\rangle\langle f|\| \leq p := \frac{\rho}{3 - \mu}. \quad ** \quad (91)$$

Reverse obtains the valid weaker lower displacement $\rho^2/(8 + 40\lambda - \mu + \rho)$ from complete kinetic support at energies at most 8. At $\lambda = 0.01$ the forward enclosure is $0.19983332325 \leq \epsilon \leq 0.19983333365$ and $p < 8.32 \cdot 10^{-5}$. Neither trial plane is asserted invariant.

Decomposing each heat operator into its own ground projection and excited remainder yields

$$\left\| e^{-\sigma(L-\epsilon)} - e^{-\sigma(A_{21}-\mu)} P_{21} \right\| \leq p + e^{-(3-\mu)\sigma} + e^{-(3+D)\sigma/2}. \quad (92)$$

A continuous-coupling envelope and certified scalar evaluator make this less than 0.000085 for every $\lambda \in [0, 0.01]$, $\sigma \geq 5$. Forward's distinct cap-only example begins at $\sigma = 4$ with bound 0.000103. The actual finite representation is $G_R + e^{-D\sigma}(U - G_R) + e^{-(3+D)\sigma/2}(P_{21} - U)$, where U projects onto the vacuum/symmetric-face plane. The computed rational $\hat{G}_R$ is exactly idempotent and the ground coefficient stays exactly one. Positive gap rounding and scalar exponential errors are budgeted separately. The reverse scalar algorithm uses the degree-160 positive Taylor enclosure for e^x on $[0, 96]$, reciprocates, and returns zero beyond 96. Its uniform e^{-x} error is below $3 \cdot 10^{-30}$, since the reciprocal width is bounded by $96(80!)^2/[161!(1 - 96/162)]$ and the omitted tail by $80!/96^{80}$. The combined representation allowance is below $3 \cdot 10^{-25}$. Sources: `research/round24/forward/x2/report.md`, `research/round24/reverse/x2/report.md`, `research/round24/skeptic/x2-review.md`.

10.2.4 ** Z1–Z2: a relative obstruction and a preparation-based repair

Let $E = e^{-\sigma(L-\epsilon)}$ and $E_R = e^{-\sigma(A_{21}-\mu)} P_{21}$. The relative quantity is $\|(E - E_R)x\| / \|Ex\|$ for the same nonzero retained input. Heat injectivity makes the denominator positive at every finite time, but not uniformly in time. The exact vacuum row $L\Omega = c\Omega - \lambda S$ lies in P_{21} , and (91) proves $\delta = \mu - \epsilon > 0$ at positive coupling. For $u = (L - \epsilon)\Omega$ one has $Gu = 0$ but $\langle f, u \rangle = \delta \langle f, \Omega \rangle > 0$, whence

$$\frac{\|(E - E_R)u\|}{\|Eu\|} \geq \frac{\delta \langle f, \Omega \rangle}{\|u\|} e^{(3-\mu)\sigma} - 1 \longrightarrow \infty. \quad ** \quad (93)$$

This is a fixed actual retained vector for each fixed $\lambda > 0$, although preparing its exact ground-orthogonal coefficient requires unknown ϵ . The computable $v = (A_{21} - \mu)\Omega$ is instead a bright Ritz eigenvector with true ground overlap $-\delta \langle \psi, \Omega \rangle \neq 0$; its true-relative error tends to one, while its Ritz-output-relative error diverges. At the cap forward certifies the first error above 6.9 at $\sigma = 6$, and the second within 10^{-4} of one at $\sigma = 8$. The mechanism is distinct ground overlap, not divergence of either correctly centered heat operator. At $\lambda = 0$ or $\sigma = 0$ the retained physical error is exactly zero. Sources: `research/round24/forward/z1/report.md`, `research/round24/reverse/z1/report.md`, `research/round24/skeptic/z1-review.md`.

In the remaining heat estimates, integration coordinates t, s and joining values T are dimensionless heat coordinates, equal in units to σ ; a duration t corresponds to physical time $(\hbar/\alpha)t$.

Z2 reconstructs the entire retained-to-omitted map, not just its ground column. With $\eta_{pq} = 4W_p W_q$, the actual orthogonal residual channels give

$$\begin{aligned} B\Omega = 0, \quad B \sum_p b_p \phi_p &= -\frac{\lambda}{2} \left[\sum_p b_p \chi_p + \sum_{p < q} (b_p + b_q) \eta_{pq} \right], \\ (B^* B)|_{\text{faces}} &= \frac{\lambda^2}{4} (19I + J_{20}), \quad \|B\| = \frac{\sqrt{39}}{2} \lambda. \quad ** \end{aligned} \quad (94)$$

This follows by expanding the squared norm on arbitrary complex coefficients. For the exact retained vacuum evolution,

$$BE_R(s)\Omega = -\frac{\lambda^2}{D} (1 - e^{-Ds})F, \quad \|BE_R(s)\Omega\| = \beta(1 - e^{-Ds}), \quad \beta = \frac{\lambda^2 \sqrt{195}}{2D}. \quad (95)$$

The zero initial omitted column therefore does not remain zero. In fact $Q(L - \epsilon)^2 \Omega = \lambda^2 F$. The exact centered Duhamel defect is $B + \delta P$, not B alone. Thus

$$V(t) = \beta[t - (1 - e^{-Dt})/D] + \delta_+ t \quad \text{bounds} \quad \|(E(t) - E_R(t))\Omega\|. \quad (96)$$

For $f_0 = \langle f, \Omega \rangle$ and $q_0 = \|(I - G_R)\Omega\|$, an independent spectral bound is $J(t) = p + (q_0 + p)e^{-(3-\mu)t} + q_0e^{-Dt}$, and the true denominator is at least $f_0 - p$. The increasing V and decreasing J can be joined:

$$\sup_{t \geq 0} \frac{\|(E(t) - E_R(t))\Omega\|}{\|E(t)\Omega\|} \leq \frac{\max\{V(T), J(T)\}}{f_0 - p}. \quad ** \quad (97)$$

For normalized $x \in P_{21}$ with $\|x - \Omega\| \leq \eta$, add to the early bound $\eta[(\rho + \delta_+)t + \|B\|(1 - e^{-3t})/3]$, and use late bound $p + (q_0 + \eta + p)e^{-(3-\mu)t} + (q_0 + \eta)e^{-3t}$ with denominator $f_0 - \eta - p$. This preserves the same prepared input in both evolutions. The continuous envelopes $\rho, \beta \leq \sqrt{195}\Lambda^2/6$, $g \geq 3 - 20\Lambda$, $q_0 \leq \sqrt{5}\Lambda/3$, and $f_0 \geq \sqrt{1 - 5\Lambda^2/9}$ cover every $0 \leq \lambda \leq \Lambda \leq 0.01$, rather than a parameter grid. They give

Input class	joining σ	all-time relative bound
Vacuum	3/2	0.00028
Normalized P_{21} ball, radius 0.01	8/5	0.00044
Normalized retained Ritz overlap ≥ 0.99	2	0.002

Reverse's ball without a normalization requirement has the distinct bound 0.000720 at join 29/10; its vacuum bound is 0.000600. These are not averaged with the sharper normalized results. Ritz overlap $a > p$ provides the checkable denominator floor $a - p$; uncertainty in overlap, coefficients, or preparation must be subtracted first. Comparing to an ideal vacuum instead of the same prepared input adds the preparation bias separately. Rational projectors and positive-gap scalar evaluation add negligible uniform numerical cost while preserving an exact ground coefficient. Z1's excluded ground-null inputs remain excluded. Sources: `research/round24/forward/z2/report.md`, `research/round24/reverse/z2/report.md`, `research/round24/skeptic/z2-review.md`.

10.2.5 ** AC1–AC2: a 293-state physical space and delayed new-channel loading

Adjoin every old residual channel to P_{21} and close under K . A shared-edge pair must be split into its spin-zero and spin-one common-edge parts. In exact rational coordinates the basis and metric are

Vector type	count	norm squared	electric energy
Ω	1	1	0
$\phi_p = 2W_p$	20	1	3
$\chi_p = 4W_p^2 - 1$	20	1	8
$d_{pq} = 4W_pW_q$, edge-disjoint	128	1	6
$s_{pq} = 8X_{pq,s}$, shared-edge singlet	62	1	9/2
$t_{pq} = 8X_{pq,t}$, shared-edge triplet	62	3	13/2

Thus P_{293} has rank $1 + 20 + 20 + 128 + 124 = 293$. Orthogonality follows from pair center parity, distinct spin-one supports, and orthogonal common-edge Casimir projections. This is a generated K -reducing subspace, not an entire electric shell; omitted intertwiners remain in Q_{293} .

The forward-origin complete magnetic matrix follows from center parity and the exact old columns

$$S\Omega = \frac{1}{2} \sum_p \phi_p, \quad S\phi_p = \frac{1}{2}\Omega + \frac{1}{2}\chi_p + \frac{1}{2} \sum_{q \text{ edge-disjoint}} d_{pq} + \frac{1}{4} \sum_{q \text{ sharing}} (s_{pq} + t_{pq}). \quad ** \quad (98)$$

All new vectors are even under simultaneous face-sign reversal and all old face vectors are odd, so same-parity magnetic entries vanish. Metric self-adjointness fixes the reverse entries, notably 3/4

from a triplet to each face, not $1/4$. The assembled matrix has 1088 directed nonzero magnetic entries and diagonal Gram with 231 weights one and 62 weights three. Its complete omitted map satisfies

$$B_+ = Q_{293}LP_{293} = -\lambda Q_{293}SP_{293}, \quad B_+P_{21} = 0, \quad \|B_+\| \leq 20\lambda. \quad (99)$$

The bound covers all 272 new columns together, including unexpanded omitted intertwiners; it is not an evaluation of their full Gram.

Because the old Ritz residual is now retained, its angle to the new Ritz ground is at most $p_{21} = \rho_{21}/(3 - \mu_{21})$. Since $B_+f_{21} = 0$,

$$\rho_+ \leq 20\lambda p_{21}, \quad \|G - G_+\| \leq \frac{\rho_+}{3 - \mu_+}, \quad 0 \leq \mu_+ - \epsilon \leq \frac{\rho_+^2}{3 - \mu_+}. \quad (100)$$

At the coupling cap, conservative exact envelopes are $p_{21} \leq 1/12000$, $\rho_+ \leq 1/60000$, $p_+ \leq 1/168000$, and $\delta_+ \leq 1/1008000000$. AC1 improves ground certification but its coarse all-time ball bound 0.0014 is worse than Z2's 0.00044; increasing dimension alone does not guarantee a better error certificate. Its parent accuracy gate was therefore limited. Sources: `research/round26/forward/ac1/report.md`, `research/round26/reverse/ac1/report.md`, `research/round26/skeptic/ac1.md`.

AC2 exploits the actual loading sequence. Let $N = P_{293} - P_{21}$, $P_f = P_{21} - |\Omega\rangle\langle\Omega|$, $A_+ = P_{293}LP_{293}$, and $C = NA_+P_{21}$. Exact matrix identities give $NA_+N = D_N + cN$, $D_N \geq 9/2$, $C\Omega = 0$, and $\|C\| = \sqrt{39}\lambda/2$ from (94). For original inputs $x \in P_{21}$, the new retained component $y(t) = Ne^{-t(A_+ - \mu_+)}x$ obeys the full equation

$$y' = -(D_N + c - \mu_+)y - CP_f e^{-t(A_+ - \mu_+)}x, \quad y(0) = 0. \quad (101)$$

Its forcing uses the entire retained trajectory and therefore includes feedback. For $\|x\| = 1$, $\|x - \Omega\| \leq \eta = 0.01$ and $0 \leq \lambda \leq \Lambda \leq 0.01$, put

$$g = 3 - 20\Lambda, \quad p_{21} = \frac{7\Lambda^2}{3g}, \quad q_+ = \frac{3\Lambda}{4} + \frac{3p_{21}}{2}, \quad b = q_+ + \eta, \quad \bar{C} = \frac{25\Lambda}{8}, \quad d = 9/2. \quad (102)$$

The new ground/excited decomposition bounds the forcing by $q_+ + be^{-gt}$. Convolution yields

$$\|y(t)\| \leq \bar{C} \left[q_+ \frac{1 - e^{-dt}}{d} + b \frac{e^{-gt} - e^{-dt}}{d - g} \right]. \quad (103)$$

Duhamel with the complete B_+ and its separate centering defect consequently has increasing envelope $\bar{V}(t) = \delta_+t + 20\Lambda\bar{C}[q_+t/d + b/(gd)]$. The independent late envelope is $J(t) = p_+ + (2b + p_+)e^{-gt}$. The true full output is bounded below using the old full-ground result, $D_\eta = 1 - 5\Lambda^2/9 - \eta - p_{21} > 0$. At $\Lambda = 0.01$, the join $t_0 = 13/5$ and $e^{-182/25} < 1/1400$ give

$$D_\eta = \frac{7127}{7200}, \quad \bar{V}(t_0) = \frac{457097}{12600000000}, \quad J(t_0) \leq \frac{2441}{78400000}, \quad (104)$$

$$\sup_{\sigma \geq 0} \frac{\|(E - E_+)x\|}{\|Ex\|} \leq \frac{457097}{12472250000} < 0.000037. \quad **$$

All envelopes cover the continuous coupling interval. Reverse independently uses face-block norms $\|NSP_f\| = \sqrt{39}/2$ and $\|P_fS(I - P_f)P_{293}\| = \sqrt{59}/2$ to obtain a delayed-leakage integral

$$\int_0^t \|B_+E_+(s)x\| ds \leq \frac{130}{9}\lambda^2 \left[\eta \frac{1 - e^{-3t}}{3} + \frac{4\lambda}{3} \left(t - \frac{1 - e^{-3t}}{3} \right) \right], \quad (105)$$

leading to 0.000048 for the ball and 0.000043 for the vacuum. The sharper result is confined to original P_{21} inputs: new retained preparations would not have $y(0) = 0$. At $\sigma = 0$ and at $\lambda = 0$ the physical retained error is exactly zero. AC2 certifies the exact retained semigroup; numerical evaluation was still a separate task. Sources: `research/round26/forward/ac2/report.md`, `research/round26/reverse/ac2/report.md`, `research/round26/skeptic/ac2.md`.

10.2.6 ** AF1–AF2: execute the heat and prevent secular centering error

AF1 fixes $\lambda = 0.01$, $\sigma = 1$, and input Ω , and exports all 293 coordinates. Every norm uses the actual diagonal Gram Γ , with $\text{Tr } \Gamma = 417$. A trial residual identifies the minimum only after the spectral separation $\lambda_2(A_+) \geq 3$ and $\mu_+ \leq 1/5$ are proved. For a rational trial vector v , define its Rayleigh quotient a and residual $\rho_v^2 = \|A_+ v\|_\Gamma^2 / \|v\|_\Gamma^2 - a^2$. Then

$$a - \frac{\rho_v^2}{3 - a} \leq \mu_+ \leq a, \quad \|P_v - G_+\| \leq \frac{\rho_v}{3 - a}, \quad P_v x = v \frac{\langle v, x \rangle_\Gamma}{\langle v, v \rangle_\Gamma}. \quad (106)$$

The first inequality is again the full retained spectral-measure argument, not a small residual near an unidentified eigenvalue.

Forward uses an explicit one-step electric residual correction, a center radius below 10^{-11} , and the degree-100 polynomial in $A_+ - \hat{\mu}$; reverse uses the old Ritz energy interval, center radius below 10^{-8} , and degree 80. Exact integer or rational Horner arithmetic has no accumulated floating roundoff. Since $\|A_+ - \hat{\mu}\| \leq 9$, their respective polynomial bounds are $2^{14}9^{101}/101!$ and $2 \cdot 9^{81}/81!$. Coordinate grids 10^{-50} and 10^{-30} cost at most $21/(2 \cdot 10^{50})$ and $11 \cdot 10^{-30}$ in the physical metric. The separate center cost is $d_\mu/(1 - d_\mu)$ at this time. Adding the independently retained physical omission and dividing by the true floor $7199/7200$ gives full relative point bounds 0.000015 forward and 0.000014 reverse. Exact Gram comparisons put their exported-vector difference inside the sum of their independent numerical radii. Sources: `research/round26/forward/af1/report.md`, `research/round26/reverse/af1/report.md`, `research/round26/skeptic/af1.md`.

AF2 prevents the uncertain center from being exponentiated indefinitely. It fixes the same coupling and original normalized radius-0.01 P_{21} class, and defines a piecewise approximation:

$$\hat{E}_+(\sigma)x = \begin{cases} P_n(\sigma(A_+ - \hat{\mu}))x, & 0 \leq \sigma \leq 8, \\ P_v x, & \sigma > 8, \end{cases} \quad P_n(H) = \sum_{j=0}^n \frac{(-H)^j}{j!}. \quad (107)$$

Forward uses degree 256 with uniform Taylor error $2 \cdot 72^{257}/257! < 10^{-25}$, center cost $8d_\mu/(1 - 8d_\mu)$, and complex export allowance $15 \cdot 10^{-40}$. Its corrected rational trial is certified by (106), independently of the heuristic correction's convergence. Reverse constructs $v = (I - A_+/9)^{64}\Omega$ exactly. The excited-to-ground power ratio is at most $15/22$ and vacuum ground overlap squared is at least $14/15$; the final Rayleigh residual certifies the actual resulting vector. It uses degree 240, Taylor bound $16 \cdot 72^{241}/241!$, and real export allowance $11 \cdot 10^{-30}$.

The late branch has no uncertain ground exponent. For every $\sigma > 8$,

$$\|e^{-\sigma(A_+ - \mu_+)} - P_v\| \leq \frac{\rho_v}{3 - a} + e^{-(14/5)\sigma}. \quad (108)$$

The full retained excited tail is below $1/(5 \cdot 10^9)$ in forward's rational certificate and below 10^{-9} in reverse's. Each complete numerical allowance e_{num} is below 10^{-8} . Therefore

$$\frac{\|\hat{E}_+(\sigma)x - E(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{457097}{12472250000} + \frac{e_{\text{num}}}{7127/7200} < 0.000037 \quad (\sigma \geq 0). \quad ** \quad (109)$$

The two terms are physical compression and retained computation errors. The sample exports include the exact normalized preparations $(159999\Omega + 800\phi_0)/160001$ and, forward only, $(159999\Omega + 800i\phi_{19})/160001$, whose squared vacuum distance is $4/160001 < 10^{-4}$. Forward exports eighteen full complex vectors at $\sigma = 0, 1, 13/5, 8, 9, 1000$; reverse exports twelve real vectors, replacing the last time by 10^6 . Continuous coverage comes from the two analytic branch bounds, not interpolation between these exports. Outputs are not renormalized. The reverse command interface accepts exact rational times and 293 real rational coordinates, enforcing original support, normalization, and radius; the operator estimate covers complex vectors, but that does not make

the real interface a complex interface. Fixed-coupling computation is not continuous-coupling execution. Time, coefficient, or preparation measurement uncertainty requires its own budget.

The remaining finite-graph tasks are growing-graph control and additional observable/input classes with complete new omission certificates. Neither the all-time heat result nor its small error implies real-time accuracy, a homogeneous numerical gap, physical scale matching, or a continuum construction. Sources: `research/round26/forward/af2/report.md`, `research/round26/reverse/af2/report.md`, `research/round26/skeptic/af2.md`.

10.3 Canonical endpoint observables and an exact reached spectrum

Use the canonical summable full-link model, with complete selected-strip and free-link reference factors, unique reference ground Ω , and

$$\begin{aligned} H_q &= H_0 + V_q, & V_q &= -\frac{\alpha\tau_q}{24} \sum_{f \text{ omitted}} q^{|a_f|1} x_f, & x_f &= \frac{1}{2} \text{Tr } U_f, \\ b(q) &= \frac{p(q)}{24(1-q)^3(1+q)^2(1+q^2)}, & p(q) &= 2 + 5q + 5q^2 + 6q^3 + 3q^4, \\ \tau_q &= \frac{\eta}{8b(q)}, & 0 < q, \eta < 1. \end{aligned} \quad (110)$$

The perturbation is norm convergent with $\|V_q\| = \alpha\eta/8$ and $D(H_q) = D(H_0)$. The admitted stationary ground projection satisfies

$$d_q = \|P_q - P_0\| \leq \hat{d}_q := \frac{\sigma_q}{\bar{g}}, \quad \sigma_q^2 = \frac{\alpha^2 \tau_q^2 b(q^2)}{96}, \quad \bar{g} = \frac{\alpha(1-\eta)}{8}, \quad \hat{d}_q = O((1-q)^{3/2}). \quad (111)$$

For the convention $\beta_q^t(B) = e^{-itH_q/\hbar} B e^{itH_q/\hbar}$, define $C_q^B(t) = \omega_q(B\beta_q^t(B)) - \omega_q(B)^2$. All endpoint conclusions in this subsection are restricted to

$$T_q = \frac{z}{\eta} \frac{\hbar}{\alpha} (1-q)^{-3}, \quad 0 < z \leq 10^{-6}, \quad s_q = \alpha\tau_q T_q / \hbar \longrightarrow \frac{8z}{7}, \quad s_q \leq \frac{3z}{2}. \quad (112)$$

The variable s_q is a proof coordinate at the original physical time T_q . The trigonometric formulas below have algebraic extensions, but no full stationary endpoint theorem beyond the displayed z interval is asserted.

10.3.1 ** U1–U2: a first-order resonance becomes a full-system witness

Let $O = (3, 1, 0)$ and $D = (3, 2, 1)$, and take paths $X : O \rightarrow (3, 2, 0) \rightarrow D$, $Y : O \rightarrow (3, 1, 1) \rightarrow D$, and $Z : O \rightarrow (4, 1, 0) \rightarrow (4, 2, 0) \rightarrow (4, 2, 1) \rightarrow D$. Their eight links are free singleton factors. The two real characters $\phi_X = \text{Tr}(XZ^{-1})$, $\phi_Y = \text{Tr}(YZ^{-1})$ are orthonormal and have energy $E = 6(3\alpha/4) = 9\alpha/2$. Set $\psi_+ = (\phi_X + \phi_Y)/\sqrt{2}$ and $W = (|\psi_+\rangle\langle\Omega_F| + \text{adjoint}) \otimes I_{F^c}$. It is a fixed gauge-invariant bounded local operator, with $W^3 = W$, $\|W\| = 1$, and reference variance one. It is not a Wilson multiplier.

The yz face at O has $x_{f_0} = \text{Tr}(XY^{-1})/2$. Haar character convolution gives $\langle\phi_Y, x_{f_0}\phi_X\rangle = 1/4$. Every other face has an unmatched free-link center parity, and all diagonal terms vanish by the same selection rule. The actual full-model matrix element is therefore

$$\langle\psi_+, V_q\psi_+\rangle = -\frac{\alpha\tau_q q^4}{96}. \quad ** \quad (113)$$

It gives first-order phase-removed response $is_q q^4/96 \rightarrow iz/84$. The vacuum subtraction uses $\langle\Omega, V_q\Omega\rangle = 0$, not $V_q\Omega = 0$. A derivative alone did not establish an endpoint; U1's gate is limited and its two-state plane is not assumed invariant.

U2 supplies the complete remainder. Each complete reference factor owns at most ten links and meets at most forty omitted faces; one face reaches at most four factors. Starting from eight factors, the ordered connected word count at depth n is bounded by $\prod_{k=0}^{n-1} 40(8+3k)$. Combining coupling $\alpha\tau_q/24$, commutator factor two and simplex $|t|^n/n!$ gives

$$\|\text{order } n\| \leq \frac{(8/3)^n}{n!} (10\alpha\tau_q|t|/\hbar)^n. \quad (114)$$

Strong bounded-generator Dyson evolution first justifies the expansion for each $q < 1$; connected counting then supplies its useful uniform bound without assuming operator-norm continuity through the unbounded H_0 . For $x = 10\alpha\tau_q|t|/\hbar < 1$,

$$R(x) = (1-x)^{-8/3} - 1 - \frac{8}{3}x \leq \hat{R}(x) := \frac{44x^2}{9(1-x)^5}. \quad (115)$$

Taylor's integral remainder proves this using the second derivative $(88/9)(1-x)^{-14/3}$; reverse's independent algebraic bound within the same author's reconstruction is $x^2(6-8x+3x^2)/(1-x)^3$. State replacement in the bounded connected expression costs $6\hat{d}_q$, so

$$\begin{aligned} |e^{iEt/\hbar}C_q^W(t) - 1 - is_qq^4/96| &\leq 6\hat{d}_q + \hat{R}(10s_q), \\ \liminf_{q \uparrow 1} |C_q^W(T_q) - C_0^W(T_q)| &\geq \frac{z}{84} - \frac{44}{9} \frac{(80z/7)^2}{(1-80z/7)^5} > \frac{z}{168} > 0. \end{aligned} \quad ** \quad (116)$$

The final strict inequality follows by monotonicity of $\hat{R}(80z/7)/z$ and exact arithmetic at $z = 10^{-6}$. This makes the earlier sufficient $\gamma < 3$ time-window exponent sharp in the existence-of-one-observable sense; it does not decide every observable or compute the limiting trajectory. Both U loops use same-author self-review. Sources: [research/round23/forward/u1/report.md](#), [research/round23/reverse/u1/report.md](#), [research/round23/forward/u2/report.md](#), [research/round23/reverse/u2/report.md](#), [research/round23/skeptic/u1.md](#), [research/round23/skeptic/u2.md](#).

10.3.2 ** V1–V2: finite outcomes are not a finite implementation

The exact spectral projections $P_{\pm} = (W^2 \pm W)/2$, $P_0 = I - W^2$ give outcomes $-1, 0, +1$, with an infinite-dimensional local zero eigenspace. For self-adjoint $\|A - W\| \leq \delta$, $\|A\| \leq 1 + \delta$, two applications of the product difference inequality, including the connected means, prove

$$|C_q^A - C_q^W| \leq 4\delta + 2\delta^2, \quad |(C_q^A - C_0^A) - (C_q^W - C_0^W)| \leq 8\delta + 4\delta^2. \quad (117)$$

This is uniform in time. Choosing fixed $\delta = z/100000$ retains endpoint margin greater than $z/200$; for contractions the two-model cost improves to 8δ . A binary first-moment POVM instead has outcome-square one and does not identify $\omega(W^2)$ with its outcome variance.

Ordinary sequential projective measurements cannot generally recover the complex unordered correlation. Set the self-adjoint unitary completions $R_{\pm} = W \pm P_0$. Then

$$\omega(W\beta^t(W)) = \frac{1}{4} \sum_{s,r \in \{+, -\}} \omega(R_s\beta^t(R_r)). \quad ** \quad (118)$$

Four controlled-unitary products and two ancilla quadratures require eight binary settings, plus a separate mean measurement. This applies the established interferometric primitive; it assumes coherent access to the actual full both-sign evolution and the intended stationary states. The exact PVM and instrument identity do not construct those resources.

A separate V1 obstruction concerns the full local Haar operator norm. For every multiplication operator M_g , a weakly-null orthonormal sequence in a positive-measure level set of $|g|$ gives

$\|W - M_g\| \geq \|g\|_\infty$ because W_F has finite rank. The triangle inequality also gives $\|W - M_g\| \geq 1 - \|g\|_\infty$, hence

$$\inf_g \|W_F - M_g\| \geq \frac{1}{2}. \quad ** \quad (119)$$

The gate admits this in the full local Haar norm, not an unconstructed physical-quotient norm. It blocks this norm-transfer route to Wilson observables, not the existence of a different Wilson witness.

For V2, the actual eigenvectors defining W prove domain preservation and

$$\|[H_0, W]\| = 9\alpha/2, \quad \|[H_q, W]\| \leq \alpha(9/2 + \eta/4). \quad (120)$$

Graph closure extends the finite-factor commutator, and strong integration then gives a norm-Lipschitz physical-time bound for this particular observable. For one predeclared endpoint, eighteen independent bounded-output settings, N trials per setting, and tolerance r , the exponential-moment argument and union bound give failure probability at most $36e^{-Nr^2/2}$. In forward's convention, full trace-norm preparation error s , complete-setting operator error g , and a common per-model time offset $|\Delta t| \leq (\hbar/\alpha)\theta$ yield

$$N \geq \frac{2}{r^2} \log \frac{36}{\kappa}, \quad |\hat{D} - (C_q^W(T_q) - C_0^W(T_q))| \leq 8(r + s + g) + (9 + \eta/4)\theta \quad (121)$$

with confidence at least $1 - \kappa$. Reverse permits per-shot timing jitter, uses half trace distance ε_p , and obtains instead $8(r + 2\varepsilon_p + \varepsilon_i) + 2(9 + \eta/4)\alpha\hbar/\hbar$; the timing coefficients are not interchangeable.

At $\eta = 1/2, z = 10^{-6}, q = 1 - 10^{-6}$, forward takes $r = s = g = \theta = 10^{-10}$ and $N = 1.8 \cdot 10^{21}$, giving $3.24 \cdot 10^{22}$ preparations, a finite- q true margin above 10^{-8} , and confidence radius $3.3125 \cdot 10^{-9}$ at physical time $2 \cdot 10^{12}\hbar/\alpha$. Reverse's different example, $q = 1 - 10^{-8}$, has radius $8.4225 \cdot 10^{-10}$ and time $2 \cdot 10^{18}\hbar/\alpha$. Neither is an implemented or efficient experiment, an optimal resource lower bound, or confidence over a continuum of times. Sources: `research/round24/forward/v1/report.md`, `research/round24/reverse/v1/report.md`, `research/round24/skeptic/v1-review.md`, `research/round24/forward/v2/report.md`, `research/round24/reverse/v2/report.md`, `research/round24/skeptic/v2-review.md`.

10.3.3 ** Y1–Y2: spatial collars and an actual scalar limit

Starting from the eight free factors, recursively include every complete owner of every omitted face meeting the current factors. Complete selected strips must be included because reference evolution spreads an operator within its whole factor. Forward retains all omitted faces internally supported on the new factor set; reverse retains those meeting the preceding set. The conventions therefore agree on factors and links but not on their finite Hamiltonians:

Depth	factors	links	forward faces	reverse faces
1	35	98	20	19
2	135	297	129	125

Both are unrenormalized subsets of the actual q profile, retain all Gauss actions, and still have infinite-dimensional onsite Haar spaces. Every connected word of length at most k lies in collar k ; longer retained words cancel exactly when comparing the two expansions. One full tail, not two, therefore suffices:

$$\sup_{|t| \leq T_q} \left\| \beta_q^t(W) - \beta_{q,k}^t(W) \right\| \leq D_k(15z), \quad D_k(x) = \frac{(8/3)_{k+1} x^{k+1}}{(k+1)!(1-x)^{k+4}}. \quad ** \quad (122)$$

The bound includes every boundary excursion and return. Reverse independently uses the coefficientwise larger $(3)_n/n!$ and exact rational tail

$$B_k(x) = \frac{x^{k+1}[(k+2)(k+3) - 2(k+1)(k+3)x + (k+1)(k+2)x^2]}{2(1-x)^3}. \quad (123)$$

Each regional Hamiltonian has its own unique stationary ground. Comparing both states and both connected means costs $6(d_q + d_{q,k})$, bounded by $12\hat{d}_q$ in forward's route. Thus the forward 98-link region retains limiting difference greater than $z/250$; at $z = 10^{-6}$ its spatial error is below $1.101 \cdot 10^{-9}$ and depth two below $2.568 \cdot 10^{-14}$. Finite spatial support does not make these evaluated finite matrices. Sources: `research/round24/forward/y1/report.md`, `research/round24/reverse/y1/report.md`, `research/round24/skeptic/y1-review.md`.

For a fixed collar, its reference $K_k = H_{0,k}/\alpha$ is a finite sum of link Casimirs plus bounded strip potentials and scalar ground shifts. Peter–Weyl projections below finite energy have finite rank; hence K_k has compact resolvent even before gauge reduction. Let $A_k = -\sum_{f \in O_k} x_f/24$ and retain the entire degenerate energy projections in $\bar{A}_k = \sum_E P_E A_k P_E$. This is a bounded self-adjoint strong direct sum preserving $D(K_k)$; its vacuum block vanishes.

The needed averaging argument can be proved directly. For a self-adjoint pure-point K , bounded A , and $\ell > 0$, put $Z_\ell(s) = e^{isK/\ell} e^{-is(K/\ell+A)}$ and $R_\ell(s) = \int_0^s (e^{iuK/\ell} A e^{-iuK/\ell} - \bar{A}) du$. On a fixed energy block the nonresonant coefficient is

$$P_F R_\ell(s) P_E = \frac{\ell(e^{is(F-E)/\ell} - 1)}{i(F-E)} P_F A P_E \quad (F \neq E). \quad (124)$$

Orthogonal output energies and density imply strong convergence of this primitive, uniformly on compact s intervals and compact input sets. Integration by parts in Duhamel makes it act only on the compact averaged orbit and its bounded derivative, giving strong convergence of Z_ℓ and its adjoint to $e^{-is\bar{A}}$. If $\Delta_E = \text{dist}(E, \text{spec } K \setminus \{E\}) > 0$, the same proof yields the fixed-block estimate

$$\sup_{|s| \leq S} \|(Z_\ell(s) - e^{-is\bar{A}}) P_E\| \leq \frac{2\ell M}{\Delta_E} (1 + 2SM), \quad \|A\| \leq M. \quad (125)$$

This is not a bound over all excited energies. Compact-resolvent doublets with vanishing splittings provide explicit counterexamples to a general operator-norm averaging inference.

In the actual collar, $M_k = N_k/24$ and $\|A_{q,k} - A_k\| \leq (1-q) \sum_{f \in O_k} |a_f|_1/24$. At $\ell = \tau_q$ and the original T_q , the exact ordered scalar is

$$e^{iEt/\hbar} \langle \Omega, W_{q,k}^\beta(W) \Omega \rangle = \langle \psi_+, Z_{q,k} W Z_{q,k}^* \Omega \rangle. \quad (126)$$

Both strong propagator limits are needed. The averaged vacuum is fixed, so for the entire energy-9/2 block $B_k = P_{k,9/2} A_k P_{k,9/2}$ its regional limit is $f_k(z) = \langle \psi_k, e^{-i(8z/7)B_k} \psi_k \rangle$. The uniform spatial bound makes f_k Cauchy, and choosing k first and then $q \uparrow 1$ proves

$$e^{iET_q/\hbar} C_q^W(T_q) \longrightarrow F_\infty(z), \quad F_\infty(z) = \lim_{k \rightarrow \infty} f_k(z), \quad |F_\infty - f_k| \leq D_k(15z). \quad ** \quad (127)$$

No global $q = 1$ generator or compact resolvent of the infinite reference is used. At this gate the actual matrices and their excited separations remained unevaluated. The forward-origin disk

$$|F_\infty(z) - 1 - iz/84| \leq D_2(15z) + \frac{1}{2}(8z/7)^2(129/24)^2 \quad (128)$$

has radius below $1.89 \cdot 10^{-11}$ at $z = 10^{-6}$, by spectral Taylor remainder and the known first moment $-1/96$. It encloses an actual scalar, not a computed trajectory. This proof exceeds the finite-spectral bounded reference hypotheses of the related norm-averaging theorem, so its pure-point strong argument is essential. Sources: `research/round24/forward/y2/report.md`, `research/round24/reverse/y2/report.md`, `research/round24/skeptic/y2-review.md`.

10.3.4 ^{**} AA1–AA2: compute a reducing component, not the entire shell

The unit cube translated by $O = (3, 1, 0)$ has twelve free links. On its physical energy-9/2 shell, quarter-energy arithmetic gives $3n_{1/2} + 8n_1 + 15n_{3/2} = 18$. Besides six half-spin links, the only solution uses one half-spin and one spin-3/2 edge and is excluded by a gauge leaf. At each cube vertex zero or two half-spin edges are required, so six occupied edges form one simple six-cycle with a unique bivalent invariant intertwiner. The sixteen six-cycles therefore form a complete orthonormal physical cube shell. Orthogonality follows from an unmatched edge-center sign.

A face can connect two such cycles only when their symmetric difference is its four-edge boundary, replacing two adjacent edges by the other two. The same Haar convolution as U1 makes every allowed matrix element 1/4; thus the compressed potential is $-A/96$, with A the exact sixteen-node flip adjacency. Enumeration and exact polynomial multiplication give

$$\begin{aligned} A(A^2 - 4I)(A^2 - 12I) &= 0, \\ \text{spec } A &= \{0^{(8)}, 2^{(3)}, (-2)^{(3)}, (2\sqrt{3})^{(1)}, (-2\sqrt{3})^{(1)}\}. \end{aligned} \quad ** \quad (129)$$

Twelve vertices have degree two and four degree six. The original coherent source, reconstructed from its paths, has adjacency moments 1, 1, 4, 8, 32, 80, 320 through order six. Polynomial spectral projectors give weights 1/3 at zero, 3/8 and 1/8 at $+2, -2$, and $1/12 \pm 1/(8\sqrt{3})$ at $\pm 2\sqrt{3}$. AA1 correctly remained limited because cube-shell completeness alone did not exclude resonant exterior loading.

AA2 proves that missing inclusion. A face external to the cube but sharing cube edges shares exactly one, with twenty such actual faces. Its opposite edge is free and the remaining two sides have distinct exterior reference owners. The minimum $-3/4$ change of an occupied common edge is canceled by $+3/4$ on the opposite free edge. A remaining free side costs 3/4; a strip side creates an endpoint-charged vector orthogonal to its gauge-invariant strip ground and costs at least 1/8. Distinct ownership therefore gives a spectral support increment at least 1/4, not merely a mean-energy inequality. Cube-disjoint faces excite the exterior by at least 1/8. For internal faces with r occupied edges and m spin-one fusion branches, the change is

$$\Delta E = 3 - \frac{3}{2}r + 2m, \quad (130)$$

whose nonzero values have magnitude at least 1/2. Consequently the sixteen-state space $\mathcal{L}$ times the unchanged exterior ground reduces every sufficiently large regional pinched operator. It need not exhaust the whole energy shell, and the instantaneous interaction still leaks. All participating nonzero frequencies from $\mathcal{L}$ or the vacuum are at least 1/8 in magnitude, giving the evaluated primitive bound $16\tau_q M_k$ and averaging error $b_k = 16\tau_q M_k(1 + 3zM_k)$.

These facts promote the AA1 cube scalar to the actual full canonical endpoint:

$$\begin{aligned} F_\infty(z) &= \frac{1}{3} + \frac{1}{2} \cos(2\theta) + \frac{1}{6} \cos(2\sqrt{3}\theta) + i \left[\frac{1}{4} \sin(2\theta) + \frac{1}{4\sqrt{3}} \sin(2\sqrt{3}\theta) \right], \\ \theta &= z/84, \quad 0 < z \leq 10^{-6}, \quad |F_\infty - 1 - i\theta| \leq 2\theta^2 = \frac{z^2}{3528}. \end{aligned} \quad ** \quad (131)$$

The complete second moment is four; it also gives $|F_\infty - e^{i\theta}| \leq 3\theta^2/2$ by centering the spectral variable at its mean one. A two-state fit would lose the actual second and higher moments.

For finite q , keep the exact weighted flip matrix $Q(q)$, whose entries are q^4 or q^5 and whose symmetric row norm is at most six. Combining direct full stationary-state replacement, the collar tail, and the two averaging columns gives

$$\left| e^{iET_q/\hbar} C_q^W(T_q) - \langle \psi_+, e^{is_q Q(q)/96} \psi_+ \rangle \right| \leq 6\hat{d}_q + D_k(15z) + 32\tau_q M_k(1 + 3zM_k). \quad (132)$$

A degree-eight rational scalar Taylor center adds at most $(s_q/16)^9/9!$; comparison directly to the limiting formula separately adds $s_q(1 - q^5)/16 + |s_q - 8z/7|/16$. At $\eta = 1/2, z = 10^{-6}, q = 1 - 10^{-12}$

and depth three with $N_3 = 332$, the state, spatial, averaging, and scalar errors are respectively below $6.547 \cdot 10^{-19}$, $5.455 \cdot 10^{-19}$, $2.530 \cdot 10^{-34}$, and $1.334 \cdot 10^{-70}$, totaling less than $1.201 \cdot 10^{-18}$ about an exact rational center. The physical time is $2 \cdot 10^{30} \hbar/\alpha$. This is a mathematical certificate, not physical feasibility or clock calibration. AA1/AA2 are same-author self-review. Sources: `research/round25/solo/aa1/report.md`, `research/round25/solo/aa2/report.md`, `research/round25/advisor/aa1-gate.json`, `research/round25/advisor/aa2-gate.json`.

10.3.5 ** AD1–AD2: null and informative Wilson multipliers

AD1 tests actual elementary Wilson multiplication. Forward chooses the yz face at O and reverse the xy face there; their finite- q collars and constants remain distinct. Each $B = \text{Tr } U_f$ has norm two, reference variance one, and created vector $\psi = B\Omega$ of energy 3α . Its square creates an extra spin-one state: $B\psi = \Omega + \chi_{1,f}$. For internal cube faces the overlap counts are $r = 0, 1, 4$, so (130) has no zero branch. The complete exterior charge argument excludes every external equal-energy image. Therefore $\bar{A}_{q,k}\psi = 0 = \bar{A}_{q,k}\Omega$, with participating nonzero frequencies bounded below by $1/8$. Applying the ordered scalar identity with the actual multiplier norm gives, in forward's eight-factor convention,

$$\begin{aligned} \left| e^{i3\alpha T_q/\hbar} C_q^B(T_q) - 1 \right| &\leq 24\hat{d}_q + 4D_k(15z) + 48\tau_q M_k(1 + 3zM_k), \\ \lim_{q \uparrow 1} e^{i3\alpha T_q/\hbar} C_q^B(T_q) &= 1. \end{aligned} \quad ** \quad (133)$$

The right-column error costs $2b_k$ and the remaining created-vector error b_k ; variance one does not make this multiplier norm one. Reverse's twelve-factor seed uses $(4)_n/n!$ and its own tail $(4)_{k+1}x^{k+1}/[(k+1)!(1-x)^{k+5}]$, with conservative averaging cost $4b_k$. The null phase-removed scalar carries no slow-rate information. Nonzero variance therefore does not establish an informative probe, and AD1's scale bridge remains limited. Its selected forward finite- q example has radius below $5 \cdot 10^{-18}$ at the same impractical clock as AA2. Sources: `research/round26/forward/ad1/report.md`, `research/round26/reverse/ad1/report.md`, `research/round26/skeptic/ad1.md`.

AD2 freezes the actual original six-cycle multipliers $B_X = \chi_X = \text{Tr}(XZ^{-1})$ and $B_\Sigma = (\chi_X + \chi_Y)/\sqrt{2}$. Their exact multiplication norms are two and $2\sqrt{2}$; both have mean zero and variance one. Their created vectors are respectively ϕ_X and ψ_+ , but their fourth moments are two and $5/2$, versus one for the rank observable. In particular the extra squared norms of $(B_X^2 - I)\Omega$ and $(B_\Sigma^2 - I)\Omega$ are one and $3/2$. Equality of a created vector therefore transfers only the scalar mechanism proved with the actual multiplier, not the entire operator or its error budget.

The reducing component from AA2 applies to both sources. Exact polynomial spectral projectors give for the single X source weights $2/3$ at zero, $1/8$ at each of ± 2 , and $1/24$ at each of $\pm 2\sqrt{3}$. Its moments are 1, 0, 2, 0, 16, 0, 160. Consequently, for the full stationary correlations on the original clock,

$$\begin{aligned} F_X(z) &= \frac{2}{3} + \frac{1}{4} \cos(2\theta) + \frac{1}{12} \cos(2\sqrt{3}\theta), \\ F_\Sigma(z) &= F_\infty(z) \text{ of (131),} \quad \theta = z/84, \quad 0 < z \leq 10^{-6}, \\ \theta^2 - \frac{2}{3}\theta^4 &\leq 1 - F_X(z) \leq \theta^2, \quad |F_\Sigma(z) - 1 - i\theta| \leq 2\theta^2. \end{aligned} \quad ** \quad (134)$$

The positive lower quadratic witness follows from $1 - \cos x \geq x^2/2 - x^4/24$ and the complete second/fourth moments. The coherent sum has a linear imaginary witness. A constant fitted to X's zero slope fails its curvature; a single cosine fitted to second moment two predicts fourth moment four, whereas the actual fourth moment is sixteen. A single phase fitted to the coherent mean one predicts second moment one, whereas the actual value is four. These are discriminators of named reduced descriptions, not universal model-identification claims.

For a source $\psi_j = B_j \Omega$ of norm one and $J_j = \|B_j\|$, the complete finite- q budget is

$$\left| e^{iET_q/\hbar} C_q^{B_j}(T_q) - \langle \psi_j, e^{is_q Q(q)/96} \psi_j \rangle \right| \leq 6J_j^2 \hat{d}_q + J_j^2 D_k(15z) + (1 + J_j)b_k. \quad (135)$$

The factor $1 + J_j$ follows by first replacing the vacuum column and then using the normalized created source on the remaining column. Use $J_X^2 = 4$, $J_\Sigma^2 = 8$, and $1 + 2\sqrt{2} < 4$; rank-norm budgets would undercount state and spatial errors. At the AA2 sample parameters, degree-eight exact rational centers have total radii below $5 \cdot 10^{-18}$ for X and 10^{-17} for the sum. Their certified displacements exceed these radii, giving actual finite- q as well as limiting witnesses. If comparing to the limiting trigonometric formulas, the coefficient/time correction after (132) remains separate.

The Wilson obstruction in V1 is thus resolved by new source-specific observables, not by falsely approximating the rank operator in the forbidden norm. The remaining bridge is empirical or cross-model scale identification with frozen observables, generators, preparations, and held-out data. The elementary null probe, multiple-frequency witnesses, and physical-time budgets establish none of that matching by themselves. They also supply no global $q = 1$ dynamics, homogeneous stability theorem, or four-dimensional continuum mass gap. Sources: `research/round26/forward/ad2/report.md`, `research/round26/reverse/ad2/report.md`, `research/round26/skeptic/ad2.md`.

11 What changed, and what the results can be used for

The most consequential changes are changes in what the evidence licenses. Early numerical consistency became a demand for an explicit regulator and common current/stress definition. A finite matrix gap became a full-space statement only after controlling the representation tail. An appealing local inverse became a commutator identity with a retained boundary defect. A nonzero-variance Wilson probe became an identification problem after its slow response proved null. A formal retained semigroup became an implemented approximation only after the arithmetic error and the late-time ground component were controlled. These changes are useful even when the original parent objective remains open.

11.1 From closed retained dynamics to complete channel loading

For a self-adjoint generator L and an orthogonal retained projection P , let

$$A = PLP, \quad B = QLP, \quad D = QLQ, \quad Q = I - P. \quad (136)$$

Assume the domain and boundedness conditions used in the relevant projected problem. With $x = Pu$ and $y = Qu$, the exact heat equations are $x' = -Ax - B^*y$, $y' = -Bx - Dy$. Solving the second equation gives

$$x'(t) = -Ax(t) - B^*e^{-tD}y(0) + \int_0^t B^*e^{-(t-s)D}Bx(s) ds. \quad (137)$$

This is the familiar projected-memory construction. The workbench contribution is its physical channel identification and controlled specialization, rather than invention of the formula. If $BP_{\text{old}} = 0$, initial leakage on an old subspace vanishes. It does not follow that $Be^{-tA}P_{\text{old}} = 0$: A can load a new retained channel on which B acts. AC2 exploits this delay quantitatively instead of deleting that channel.

Possible applications include certified reduced models for finite gauge systems, testing truncation designs, and preparing benchmarks for quantum-simulation algorithms. The practical advance is a bound on the full output, rather than convergence between two arbitrary finite cutoffs. A useful larger-graph application still requires a new complete basis, residual, and denominator calculation.

11.2 From a useful heat bound to an executable all-time result

The exact physical approximation and the finite-arithmetic approximation contribute different errors. On the declared normalized preparation class, suppose

$$\|E(\sigma)x - E_P(\sigma)x\| \leq \varepsilon_{\text{phys}}, \quad \|E_P(\sigma)x - \hat{E}_P(\sigma)x\| \leq \varepsilon_{\text{num}}, \quad \|E(\sigma)x\| \geq d_0 > 0. \quad (138)$$

The triangle inequality yields

$$\frac{\|E(\sigma)x - \hat{E}_P(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{\varepsilon_{\text{phys}} + \varepsilon_{\text{num}}}{d_0}. \quad (139)$$

All three quantities need proof on the same input and time class. In particular, a relative theorem may fail when the true output decays to zero, even if an absolute theorem holds.

The all-time computational design addresses a second issue: a small error in a ground-energy shift causes secular drift. If an exact ground vector has energy μ but the program shifts by $\hat{\mu} = \mu + \delta$, its ground component is multiplied by $e^{\delta\sigma}$. A fixed nonzero δ cannot support a uniform all-time guarantee. AF2 instead joins a certified finite-time polynomial evaluation to a certified ground projector with an exponentially bounded excited tail. The join is an algorithmic choice, not a new physical clock. Possible uses include long-heat-time state preparation, reproducible benchmark data, and testing alternative polynomial or rational approximations. None of these uses licenses real-time evolution without a separate theorem.

11.3 From a source correction to a nonlinear stability problem

An identity $[J, G] = -A + R$ with $\|R\| < \|A\|$ is useful evidence about the specified source A and generator G . The next transformation generally changes both the source and the diagonal operator. A full iteration needs, for example, bounds on a declared map

$$(G_n, A_n) \mapsto (G_{n+1}, A_{n+1}, J_n) \quad (140)$$

in one complete interaction norm, along with control of scalar and diagonal updates and the summability of the corrections. This display states a future obligation, not a derived recurrence. AE2 proves finite original weighted norms and per-source contraction in its smaller coupling region; it does not prove contraction of the collective weighted interaction. The distinction prevents an unsupported homogeneous numerical mass-gap claim.

The current construction may serve as a local preconditioner, an explicit approximate homological solver, or a starting point for a future stability iteration. Such applications are proposed mathematical uses; no faster numerical algorithm or new many-body phase has been measured here.

11.4 From a resonant state to an informative observable

An operator B can prepare a state with nonzero variance while the selected averaged interaction annihilates that state. Then the selected slow readout is constant and cannot identify the slow parameter. AD1 supplies this control. AD2 uses actual six-cycle Wilson multiplication observables with nontrivial endpoints, retaining their own norms and higher moments. The distinction matters because a rank-one preparation surrogate and a multiplication operator may agree on Ω but differ elsewhere.

These results can inform the selection of mathematical test observables and simulation readouts. They do not demonstrate laboratory preparation, finite-resource feasibility at the sample physical clock, or a calibration connecting the summable model to the homogeneous model. The formula depends on the parameter combinations actually present; naming a frequency is not a measurement of an independent physical scale.

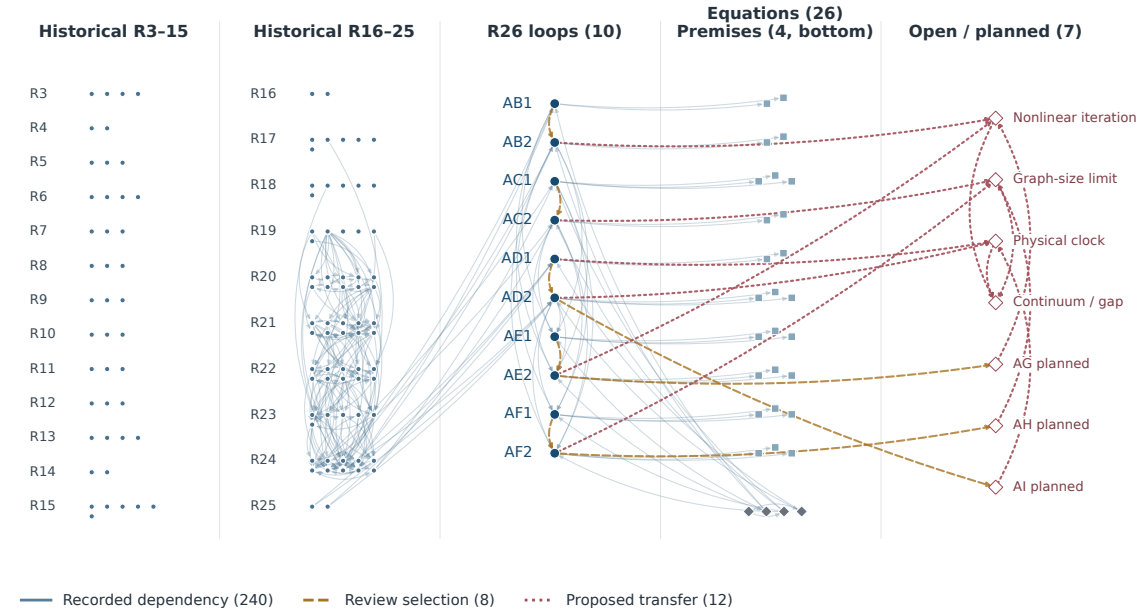
11.5 Evidence networks and nonphysical geometry

The released evidence network has 158 nodes and 260 directed links: 111 historical entries, 26 displayed equations, ten current loops, four premise nodes, and seven open targets. Its three edge types are source dependency, proposed transfer, and review selection. A source dependency records the declared use of a file or result; it is not necessarily a minimal mathematical dependency. Review-selection edges record chronology rather than logical implication. In particular, the drawing cannot close an edge labelled proposed transfer.

The grouped appendix and machine-readable ledgers make this network useful as a research-navigation and audit artifact. Figure 1 shows its organization; node coordinates have no physical interpretation. A new calculation can be attached to the exact obstruction it addresses, while failed or limited outcomes remain visible.

Recorded evidence relations: all 158 nodes and 260 directed edges

Schematic organization by historical round and current node kind; position and distance have no physical meaning.



Source: research/round26/network.json at commit 40960f39a3dc. Every structural node and edge is included; labels are selective. A recorded dependency is provenance, not permission to transfer a theorem between models. Planned nodes are not completed investigations.

Figure 1: The recorded evidence network, grouped for readability. Edges retain their audit meaning; geometric placement is a presentation choice. Open targets are not established conclusions.

12 Verification, stress tests, and interpretation of the figures

Verification has three distinct layers. First, the pinned workbench contains written model-specific derivations and source-bound historical admission records. Second, the Round26 release was replayed from a clean detached tree: 40 current producer executions and four inherited AA executions covered normal and optimized Python, 171 admission mutations were rejected in each mode, 156 generated files were compared, and seven interface suites passed. These are evidence of reproducibility and admission integrity, not a proof-assistant verification of the mathematics. Third, this manuscript adds a separately authored mathematical stress suite and parameterized calculators. Repeated execution in another interpreter mode is not counted as an additional independent derivation.

12.1 What the new checks challenge

The fresh calculator suite passes 194 checks; the separately authored skeptic suites pass 158 checks (90 general, 12 early, 33 middle, and 23 recent). Each corresponding normal and optimized Python run has identical result bytes. The standalone HTML formulas pass 15 additional checks. These are distinct programmed checks, not counts of independently proved theorems or measures of statistical confidence.

The independent skeptic suite derives selected identities without importing the producer implementations. It checks normalized Haar moments and their differential recurrence, tests a false moment closure, distinguishes two-trace from three-trace information, reconstructs the shared-link mixed derivative, enumerates the two-square shell minima, and recertifies the one-square continuous gap through a separate rational Sturm and full-tail argument. Further tests expose delayed omitted-channel loading, distinguish a source identity from an operator identity, count translated boundary crossings, check Wilson spectral moments and phase signs, and verify the rational preparation and combined AF2 error budgets. The audit ledger records which manuscript contributions were reviewed and which historical production runs were not rerun.

The calculator tests compare independent numerical representations where available: Bessel ratios against normalized Haar quadrature, matrix dimensions against one another as diagnostics, exact rational enclosures against the recorded coefficients, and boundary or invalid inputs against the declared domains. These tests deliberately include wrong models. For example, an approximation that drops variance can pass a selected equation and still fail the next moment identity. A retained initial column equal to zero can coexist with nonzero later leakage. A ground-energy rounding error can pass short-time tests and fail at sufficiently long times.

12.2 Arithmetic, analytic scope, and sampling

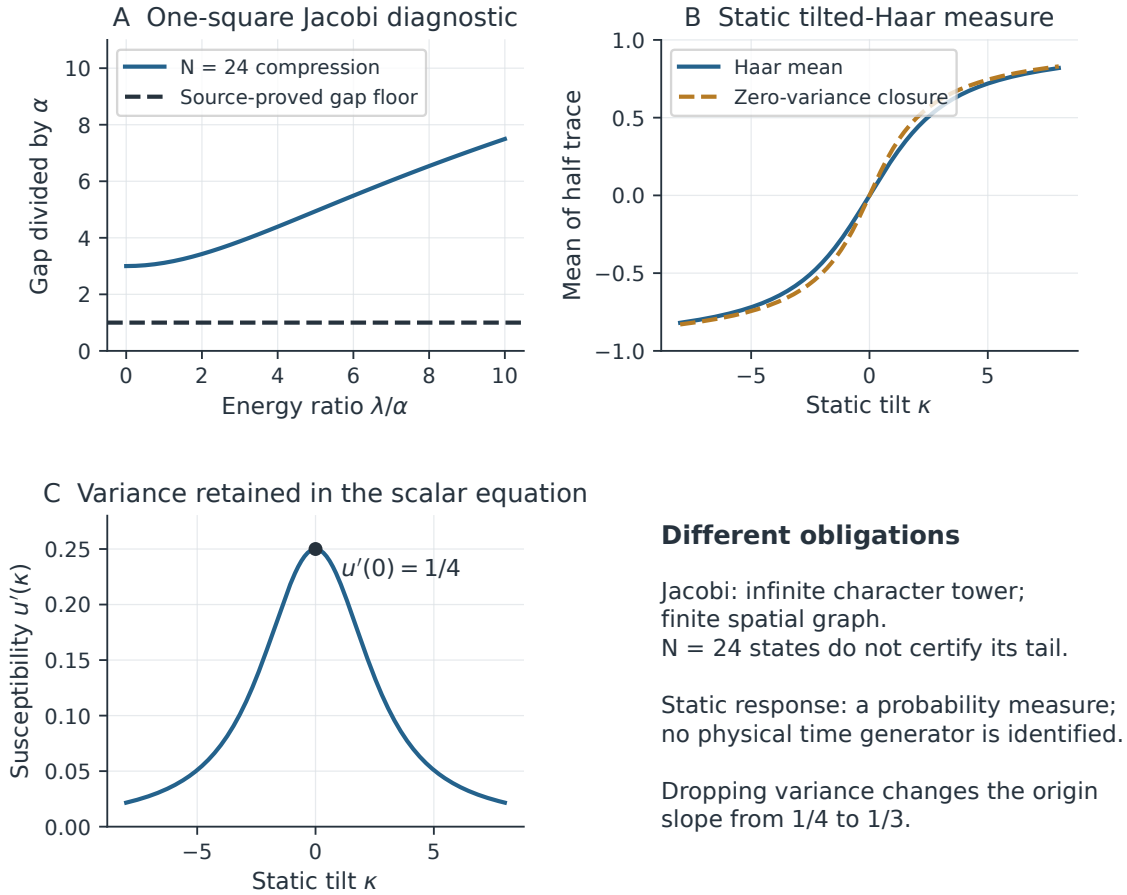
Exact rational arithmetic certifies the arithmetic statement it evaluates. It does not make an unevaluated transcendental quantity exact; such a quantity needs a proved enclosure. The companion distinguishes positive Taylor partial sums used as rigorous exponential bounds from high-precision floating diagnostics. Jacobi truncation comparisons are labelled diagnostics unless a separate omitted-tail and interval argument is invoked. The HTML calculators use ordinary double precision and display rounded values; they are explanatory interfaces to the formulas, not interval-proof engines.

A plot samples a formula. Its visual smoothness supplies no all-time, all-coupling, or infinite-volume proof. Uniform bounds in the paper come from the stated analytic arguments: complete interval coverage with a continuity estimate, a monotone early/late envelope, a convergent majorant, or a complete spectral/channel identity. Each plot states its parameter region and whether it shows an exact function, a sufficient upper budget, or numerical diagnostic data.

12.3 Known limits of this audit

Neither this manuscript audit nor a finite set of stress tests can establish that the work has no limitations or that every possible error has been excluded. The complete historical numerical production archive was not regenerated for this paper. The source checkpoint, original reports, fresh calculation scripts, source hashes, and per-contribution dispositions make that boundary inspectable. No unrestricted continuum conclusion is admitted by inference from the calculations. No experimental application, literature priority, or external peer-review status is claimed.

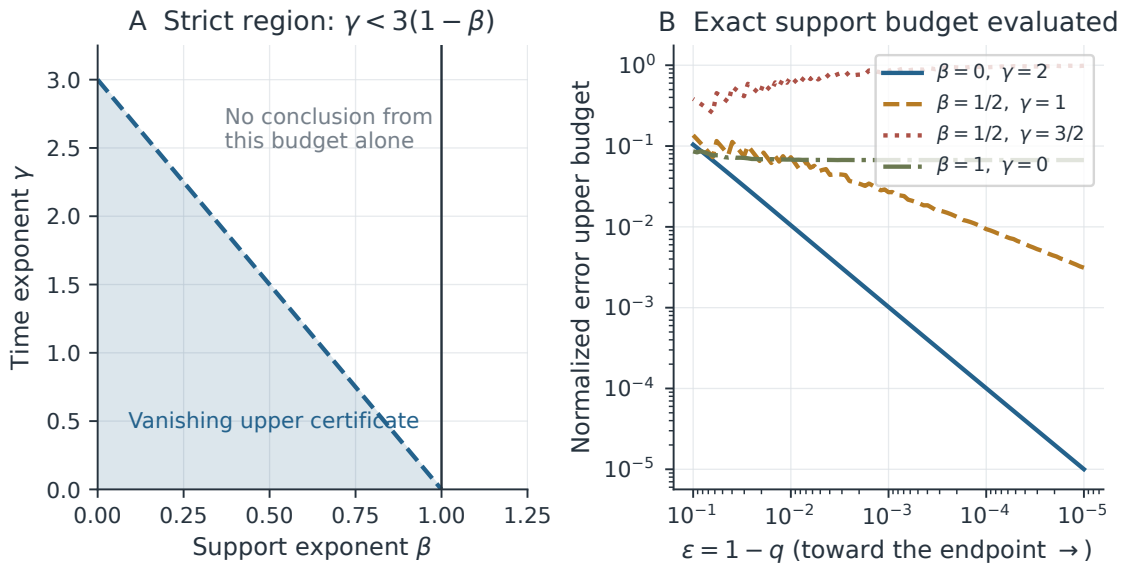
Distinct SU(2) reductions: spectrum and static response



Gap floor: inherited exact Round10 certificate on $0 \leq \lambda/\alpha \leq 10$. Static curves: exact Bessel formula evaluated numerically. Numerical agreement and truncation refinement are diagnostics; neither proves a continuum theorem.

Figure 2: Two distinct benchmarks. The one-square Jacobi eigenvalue curves are finite-matrix numerical diagnostics; the recorded continuous full-space certificate has its own tail and interval proof. The tilted-Haar mean and variance are static quantities, not a physical trajectory. Plot data and parameters are supplied with the calculators.

Complete support controls the available time window

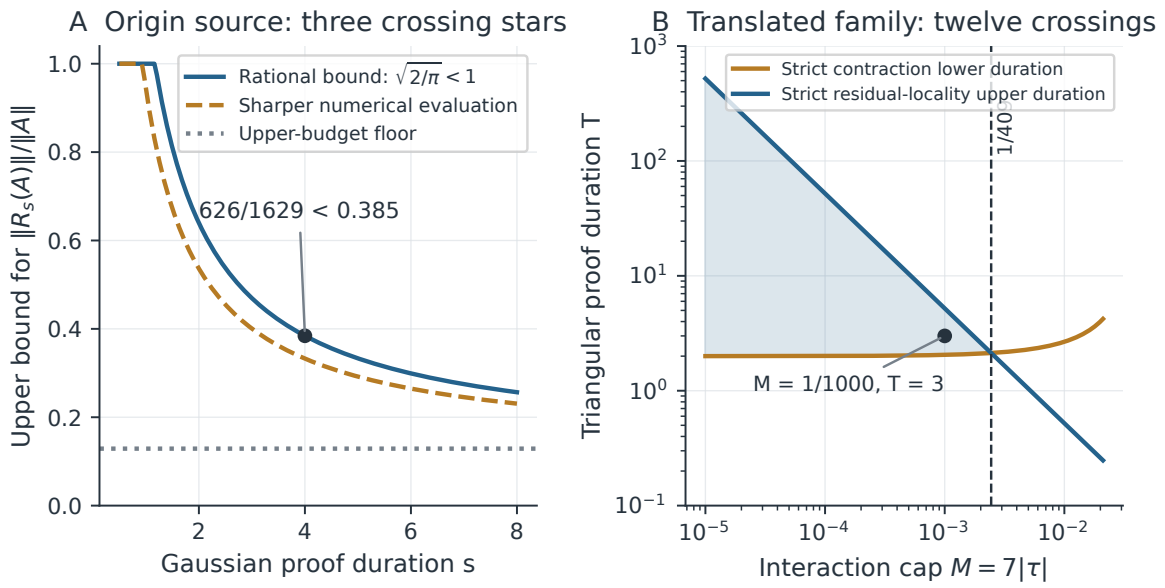


Canonical summable family, $\eta = 1/2$, $C = \ell = 1$; $L = \max(1, \text{floor}(\varepsilon^{-\beta}))$.

Boundary curves are excluded from the vanishing certificate. A positive or divergent upper bound is not a lower bound on actual correlation error. The $q = 1$ Hamiltonian is not inserted into the summable family.

Figure 3: Sufficient support/time windows and evaluated upper budgets. The strict exponent region follows from the stated complete-support asymptotics. Values outside it indicate failure of this sufficient certificate, not proof that the actual correlation error diverges.

Filtered correction: contraction and locality are separate



Panel A: AB2 original cap $M = 35/1664$. Panel B: AE2 requires $2/(1-25M) < T < 1/(192M)$. Its joint open interval exists exactly when $409M < 1$. The zero-frequency residual is one for both filters. These are sufficient upper certificates; neither a budget floor nor a failed locality estimate proves actual divergence.

Figure 4: Full-source Gaussian contraction and compact-filter certificate compatibility. Gaussian duration is a proof regulator. The compact construction has a smaller admitted coupling range and counts all translated crossings. The original-cap incompatibility concerns these sufficient bounds, not every possible filter or a lower bound on an actual error.

Wilson endpoints: one clock, distinguishable probes

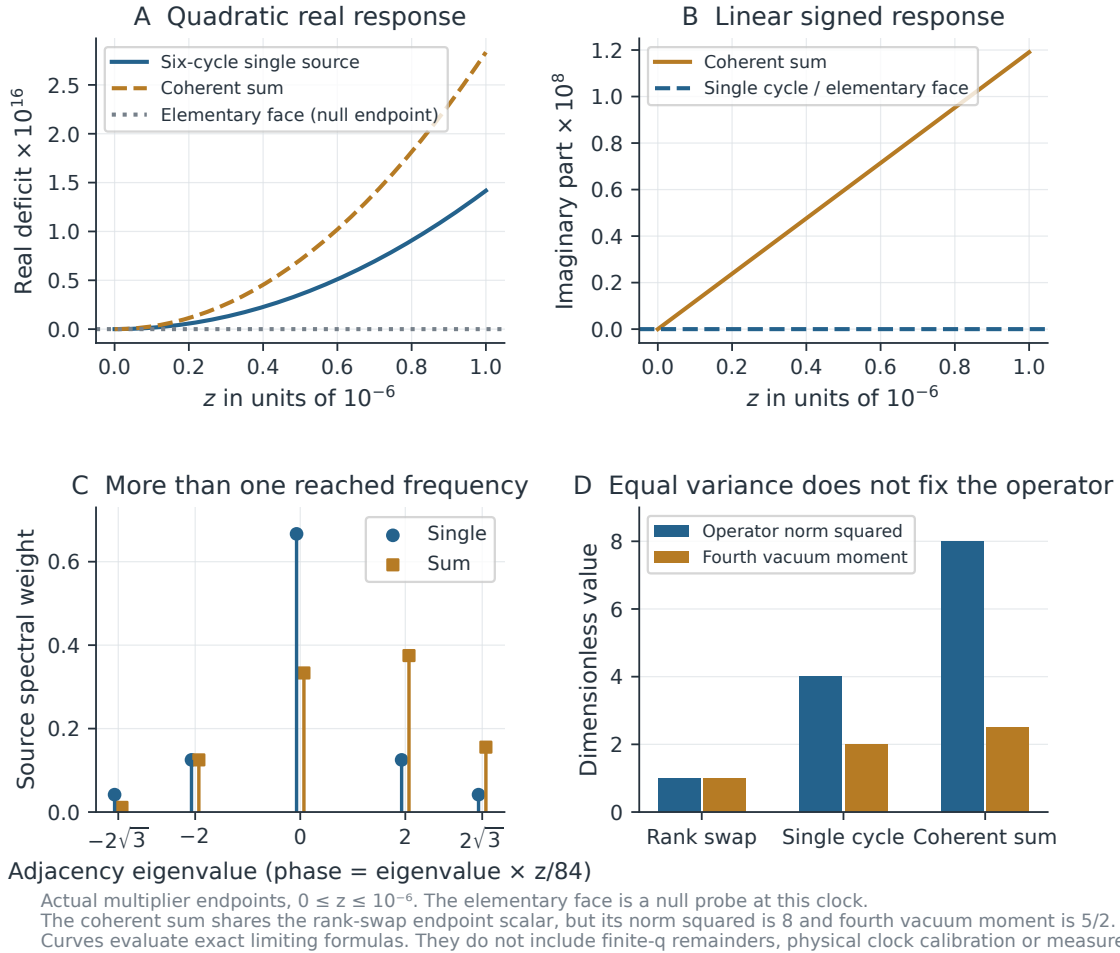
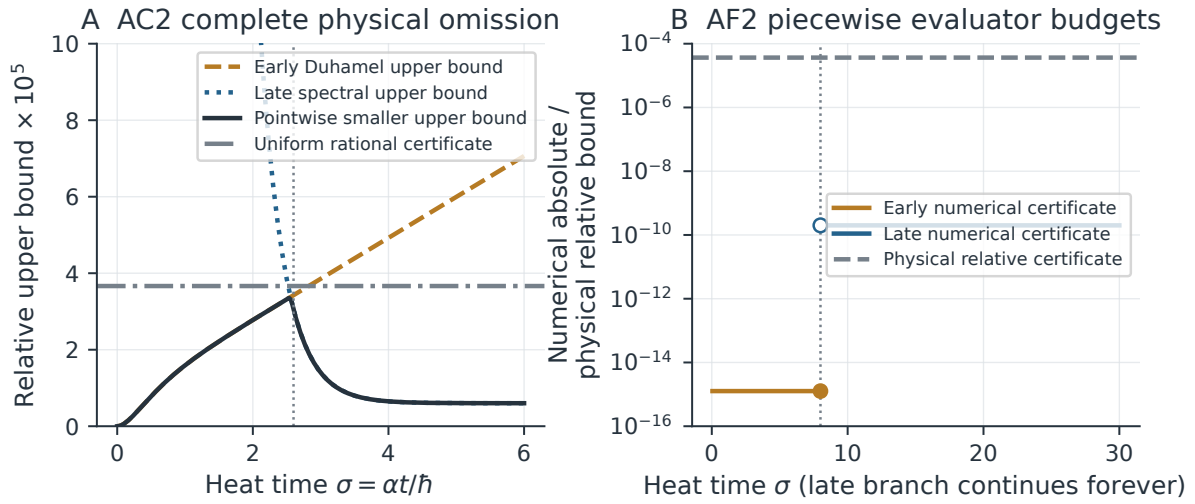


Figure 5: Canonical endpoint functions for the specified actual Wilson observables, their reached spectral weights, and distinctions between operator norms and moments. The elementary probe supplies a null slow-response control. These are model readouts; finite- q deviations and physical resource costs require their separate bounds.

Heat approximation: physical omission and numerical error



$\lambda \leq 0.01$ for AC2; $\lambda = 0.01$ for AF2. Inputs: normalized original P21 vectors within 0.01 of the vacuum.
 Physical bound = 457097/12472250000; denominator $\geq 7127/7200$. The 293-dimensional space is an enrichment.
 This companion recomputes scalar certificates. It does not execute the original 293-coordinate heat solver.

Figure 6: The physical AC2 early/late envelope and the separately charged AF2 numerical bound. The numerical branch changes at $\sigma = 8$ to prevent secular center error. The denominator applies to the original normalized P_{21} preparation ball; no relative claim for arbitrary ground-null inputs is shown. The late branch extends to all later times even though the horizontal display ends.

13 Next planned goals and remaining proof obligations

Round26 completes AB1/AB2, AC1/AC2, AD1/AD2, AE1/AE2, and AF1/AF2. The first three goal pairs were inherited from the earlier roadmap. AE and AF were selected only after those six reviews. This paper does not execute further physics loops. The next three goals below were planned from the final evidence. Their second loops must be chosen after their first-loop reviews; two later goals remain unselected.

1. **AG: the first actual nonlinear homogeneous update.** Start within AE2’s restricted coupling region. Derive the generated source, scalar term, and diagonal update from one actual transformation. Bound every translated crossing and the relevant unbounded domains in the original rooted support-cardinality norm. The decisive test is whether per-source improvement also improves the collective weighted budget. A failure would select a revised construction, not an assertion of all-stage induction.
2. **AH: larger physical graphs with complete channel control.** Freeze the next graph and build its actual Gauss-invariant retained basis. Establish all Gram, coupling, and omitted-channel identities before comparing accuracy. Compute the graph dependence of residual, ground-projection, and true-output denominator constants. A favorable graph-size trend remains weaker than a volume-uniform theorem.
3. **AI: observable and physical-scale identification.** Freeze actual Wilson observables, independent parameters, and physical observation times. Determine the parameter combinations identified by their complete readouts, using null probes and indistinguishable parameter families as controls. Add a genuinely discriminating observable or a state/domain/generator map with a bounded defect. A fitted clock that forces agreement is not an admissible substitute.

Beyond these near-term goals, a continuum implication requires an explicitly matched scale trajectory, control uniform in graph volume and cutoff, a nontrivial limiting theory, the required reconstruction/positivity properties, and a positive gap in physical units. These are separate obligations, not a consequence of the finite-graph error threshold. The present results provide controlled subproblems and counterexamples that organize this work; they do not settle it.

Authorship and assistance statement

This draft was prepared from the source-bound workbench with AI-assisted writing, calculation, and separate model-agent review. Human author names, affiliations, contributions, and submission responsibility have not been supplied. The listed historical roles describe the research record, not named human coauthors. Before public submission, the responsible authors should approve the claims and attribution, complete an independent expert mathematical review, and resolve any priority or proof-depth issues identified by the audit. The manuscript contains no invented authors, affiliations, experimental measurements, or arXiv identifier.

References

- [1] Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. Clay Mathematics Institute, Millennium Problem description. URL <https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>. Official problem statement; accessed 22 September 2026.
- [2] Isaac Newton. *Opticks: or, A Treatise of the Reflexions, Refractions, Inflexions and Colours of Light*. William Innys, London, 4 edition, 1730. URL <https://www.gutenberg.org/files/33504/33504-h/33504-h.htm>. Primary text in Project Gutenberg transcription.
- [3] Nikola Tesla. Art of transmitting electrical energy through the natural mediums. United States Patent US787412A, 1905. URL <https://patents.google.com/patent/US787412A/en>. Issued 18 April 1905; filed 16 May 1900.
- [4] Simon Wolfgang Mages, Matthias Aicher, and Andreas Schäfer. Euler–heisenberg lagrangian to all orders in the magnetic field and the chiral magnetic effect, 2010. URL <https://arxiv.org/abs/1009.1495>. Version 1 inspected.
- [5] Y. Kluger, J. M. Eisenberg, and B. Svetitsky. Pair production in a strong electric field: an initial value problem in quantum field theory. *International Journal of Modern Physics E*, 2:333–380, 1993. doi: 10.1142/S0218301393000121. URL <https://arxiv.org/abs/hep-ph/0311293>. Primary-author technical review; arXiv version 1 posted in 2003.
- [6] Jochen Zahn. The renormalized locally covariant dirac field. *Reviews in Mathematical Physics*, 26:1330012, 2014. doi: 10.1142/S0129055X13300124. URL <https://arxiv.org/abs/1210.4031>. Version 3, dated 18 October 2013, inspected.
- [7] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10:2445–2459, 1974. doi: 10.1103/PhysRevD.10.2445. URL <https://journals.aps.org/prd/abstract/10.1103/PhysRevD.10.2445>.
- [8] John Kogut and Leonard Susskind. Hamiltonian formulation of Wilson’s lattice gauge theories. *Physical Review D*, 11:395–408, 1975. doi: 10.1103/PhysRevD.11.395. URL <https://journals.aps.org/prd/abstract/10.1103/PhysRevD.11.395>.
- [9] Irian D’Andrea, Christian W. Bauer, Dorota M. Grabowska, and Marat Freytsis. New basis for hamiltonian SU(2) simulations. *Physical Review D*, 109:074501, 2024. doi: 10.1103/PhysRevD.109.074501. URL <https://journals.aps.org/prd/abstract/10.1103/PhysRevD.109.074501>. Equation locations inspected in arXiv v1; published author order and title used.
- [10] Henry Froland, Dorota M. Grabowska, and Zhiyao Li. Simulating fully gauge-fixed SU(2) hamiltonian dynamics on digital quantum computers, 2025. URL <https://arxiv.org/abs/2512.22782>. Version 1; abstract submission date 28 December 2025; HTML displays manuscript date 24 August 2026.
- [11] I. M. Burbano and Christian W. Bauer. Gauge loop-string-hadron formulation on general graphs and applications to fully gauge fixed hamiltonian lattice gauge theory, 2024. URL <https://arxiv.org/abs/2409.13812>. Version 2 inspected.
- [12] National Institute of Standards and Technology. DLMF, chapter 18, §18.5(iii), equation 18.5.10: Ultraspherical (gegenbauer) polynomials. NIST Digital Library of Mathematical Functions. URL <https://dlmf.nist.gov/18.5.E10>. Equation inspected; accessed 22 September 2026.

- [13] Gerald Teschl. *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, volume 157 of *Graduate Studies in Mathematics*. American Mathematical Society, 2 edition, 2014. URL <https://www.mat.univie.ac.at/~gerald/ftp/book-schroe/schroe2.pdf>.
- [14] D. A. Yarotsky. Quasi-particles in weak perturbations of non-interacting quantum lattice systems, 2004. URL <https://arxiv.org/abs/math-ph/0411042>. Version 1 inspected.
- [15] Joscha Henheik, Stefan Teufel, and Tom Wessel. Local stability of ground states in locally gapped and weakly interacting quantum spin systems. *Letters in Mathematical Physics*, 112, 2022. doi: 10.1007/s11005-021-01494-y. URL <https://arxiv.org/abs/2106.13780>. Version 3 inspected.
- [16] Sergey Bravyi, David DiVincenzo, and Daniel Loss. Polynomial-time algorithm for simulation of weakly interacting quantum spin systems. *Communications in Mathematical Physics*, 284: 481–507, 2008. doi: 10.1007/s00220-008-0574-6. URL <https://arxiv.org/abs/0707.1894>.
- [17] Simone Del Vecchio, Jürg Fröhlich, Alessandro Pizzo, and Stefano Rossi. Local iterative block-diagonalization of gapped hamiltonians: a new tool in singular perturbation theory. *Journal of Mathematical Physics*, 63:073503, 2022. doi: 10.1063/5.0084552. URL <https://arxiv.org/abs/2007.07667>.
- [18] Simone Del Vecchio, Jürg Fröhlich, and Alessandro Pizzo. Block-diagonalization of infinite-volume lattice hamiltonians with unbounded interactions, 2021. URL <https://arxiv.org/abs/2108.13907>. Version 1 inspected.
- [19] Bruno Nachtergaele and Robert Sims. On the dynamics of lattice systems with unbounded on-site terms in the hamiltonian, 2014. URL <https://arxiv.org/abs/1410.8174>. Version 1 inspected.
- [20] Ben T. McDonough, Chao Yin, Andrew Lucas, and Carolyn Zhang. Lieb–robinson bounds with exponential-in-volume tails. *PRX Quantum*, 6:040322, 2025. doi: 10.1103/n242-m1l2. URL <https://arxiv.org/abs/2502.02652>. ArXiv version 2, dated 2026, inspected.
- [21] Yu Tong, Victor V. Albert, Jarrod R. McClean, John Preskill, and Yuan Su. Provably accurate simulation of gauge theories and bosonic systems. *Quantum*, 6:816, 2022. doi: 10.22331/q-2022-09-22-816. URL <https://arxiv.org/abs/2110.06942>.
- [22] Hao Shen, Rongchan Zhu, and Xiangchan Zhu. A stochastic analysis approach to lattice Yang–Mills at strong coupling, 2022. URL <https://arxiv.org/abs/2204.12737>.
- [23] Konrad Osterwalder and Robert Schrader. Axioms for euclidean green’s functions. II. *Communications in Mathematical Physics*, 42:281–305, 1975. doi: 10.1007/BF01608978. URL <https://projecteuclid.org/journals/communications-in-mathematical-physics/volume-42/issue-3/Axioms-for-Euclidean-Greens-functions-II-with-a-n-Appendix-by/cmp/1103899050.pdf>. With an appendix by Stephen Summers; original article inspected through an open PDF mirror.
- [24] Arpan Chatterjee, Marco Frasca, Anish Ghoshal, and Stefan Groote. Exact lattice identities and continuum-limit Dyson–Schwinger equations for Yang–Mills theory, 2026. URL <https://arxiv.org/abs/2608.05415>. Version 1 inspected.

A Contribution catalog, applications, and remaining obligations

The double star marks workbench derivation, computation, repair, or specialization; it does not assert literature priority. Related early studies are consolidated; later loop increments remain separately indexed and grouped by subject. Every row points to a derivation or source and states a possible application, the change from earlier understanding, and a remaining obligation. Application means a supported mathematical use or proposed use within the stated model, not a measured technological benefit. The separate skeptical ledger records the level of fresh verification. Additional claim details and source paths are retained in `audit/contribution-catalog.json`.

Table 2: Organized contributions and their limits.

Contribution and derivation	Application and change	Limitation and next obligation
Origins and finite gauge benchmarks		
early01** Matched finite Maxwell–Dirac closure and continuation Eqs. (1), (2), (3), (4), (5). <i>Scope:</i> model-specific derivation and implementation using established finite ODE theory <code>evidence/qeg-research/round3/theory.md</code>	<i>Use:</i> Finite homogeneous backreaction and a global finite-dimensional existence theorem under $Z \geq z^* > 0$. <i>Change:</i> A source-off electric reversal became a reproducible finite-model result; energy conservation alone does not determine whether matching is physically correct.	Static B, flat geometry, finite positive quadrature, finite Landau range; adjacent refinements are not rigorous trajectory enclosures. The 1e-18 advisor benchmark remains failed. <i>Next:</i> Common covariant current/stress renormalization, regulator convergence and quantum fluctuation control.
early02** Complete causal tangent and stationary second variation Eqs. (6), (7). <i>Scope:</i> model-specific derivative, positive quadratic form and explicit falsifier <code>evidence/qeg-research/round4/response_contract.md</code>	<i>Use:</i> Causal finite-source response; stationary electric/transformed tangent control. <i>Change:</i> The signed first variation is not a norm; even a conserved positive second variation does not control raw potential/spin tangents.	Fixed grid and matching; stationary pure-vacuum base and norm-preserving tangents $h_i \cdot \eta_i = 0$; no time-uniform driven stability theorem. <i>Next:</i> Construct and test an appropriate coercive response energy for a declared driven base solution.
early03** Finite-window coefficient certificate and cofinal matching obstruction Eqs. (8), (9), (10). <i>Scope:</i> exact integral derivation, rational certificate and scoped obstruction <code>evidence/qeg-research/round6/extensions.md</code>	<i>Use:</i> $Z > 3/4$ for fixed $b=10, K=20$, $\alpha_{em} < 1/137$, every finite Landau cutoff and positive constant-exact quadrature; obstruction to unchanged matching in joint cutoff removal. <i>Change:</i> The continuous integral maximizer does not transfer to a quadrature rule, and a positive finite coefficient is not regulator-uniform.	Continuous-window max theorem differs from discrete certificate; tail bound controls only C; bounded-addition obstruction requires uniform field-domain bounds and fixed/bounded potential. <i>Next:</i> A different controlled matching/renormalization path and full dynamical error estimates.
early04** Anisotropic gravitational and regulated Dirac interface Eqs. (11), (12). <i>Scope:</i> model-specific consistency derivations and implementation controls <code>evidence/qeg-research/round3/gravity.md</code>	<i>Use:</i> Bianchi-I constraint residual, shear diagnostics, regulated mode current/stress and counterterm Ward checks. <i>Change:</i> A fixed-FLRW occupation calculation is not Einstein–Maxwell–Dirac closure. Energy conservation leaves a pressure nullspace; the correct off-constraint coefficient here is $-\theta$.	Bare finite comoving mode quantities; Ohmic and scalar fixtures have their own classical matter; no renormalized quantum pressure or self-consistent strong-curvature solution. <i>Next:</i> Common covariant subtractions and a specified suitable state, then compatible constraints and metric/current/stress response.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
early05** Action-derived scalar coupling and reference-subtraction taxonomy Eqs. (14). <i>Scope:</i> reduced-model derivation and separate classical implementation evidence/qeg-research/round7/research_report.md	<i>Use:</i> Finite scalar-Maxwell-Dirac continuation and independently conserved classical scalar-gravity fixtures. <i>Change:</i> A dynamical scalar, a renormalization scale, a subtraction reference and a numerical cutoff are different objects.	New scalar matter changes the theory; the classical gravity model and quantum finite model have no shared renormalized stress. <i>Next:</i> Prove any proposed identifying/decoupling map; retain scalar exchange and stress in new models.
early06** Positive spectral-measure diagnostics and conditional gap bridges Eqs. (15), (16). <i>Scope:</i> known spectral theorems rederived; workbench counter-benchmark implementation research/round8/spectral-proof.md	<i>Use:</i> False-plateau, hidden-channel and vanishing finite-gap controls; conditional common-gap survival and dense-channel decay theorem. <i>Change:</i> Effective mass approaches the lowest overlapping threshold from above and cannot certify a lower bound from a short plateau.	No reconstructed continuum QFT or regulator-uniform physical rate; periodic massless plot excludes its zero mode. <i>Next:</i> Construct the physical limit and a common all-time rate on a dense centered observable family.
early07** Finite Haar hierarchy characterization and Wilson identities Eqs. (17). <i>Scope:</i> elementary finite-measure theorem rederived and authentic lattice implementation research/round9/advisor/advisor.md	<i>Use:</i> Full compact connected-group Gibbs-law characterization; SU(2) staple residual and non-Gibbs atomic falsifier. <i>Change:</i> One Ward identity, especially the identically zero beta=0 residual, cannot identify the probability law.	Finite graph, smooth action, all generators/insertions for converse; independent-link staple premise; beta=2.2 autocorrelation insufficient. <i>Next:</i> Controlled physical transfer/observable estimates and adequately sampled finite-volume data.
early08** Exact central-convolution spectrum and continuum-time rotor limit Eqs. (18). <i>Scope:</i> known harmonic analysis rederived and numerically implemented research/round9/transfer/PROOF.md	<i>Use:</i> All-spin Bessel spectrum, exact gap, negative-beta positivity falsifier and strong rotor semigroup limit. <i>Change:</i> Pointwise positive kernels need not be positive operators, and a rotor spectrum is not the locally gauge-projected interacting transfer spectrum.	Auxiliary SU(2) operator; fixed-spin asymptotics; continuum time only, no continuum space; strong convergence, not norm convergence. <i>Next:</i> Actual many-link gauge transfer construction, interactions and matched physical clock.
early09** Product-L1 operator-norm counterexample and transfer stability Eqs. (19). <i>Scope:</i> exact family of counterexamples and standard conditional norm lemma research/round9/advisor/advisor.md	<i>Use:</i> Positive symmetric Markov family with product L1 tending to zero and fixed operator discrepancy; common-vacuum gap and finite-time comparison. <i>Change:</i> Kernel product-L1 does not bound essential row/column norms; $o(a)$, not merely $o(1)$, is sufficient for telescoped fixed-time convergence.	Local objection to a displayed estimate, not a verdict on every claim of a cited preprint; shared vacuum and common Hilbert space required. <i>Next:</i> Uniform Schur or Hilbert-Schmidt bounds on the actual model and a justified vacuum/space identification.
early10** One-square full-space spectral enclosure and coupling cover Eqs. (20), (21). <i>Scope:</i> certified implementation and explicit error contract for known one-square reduction research/round10/advisor/advisor.md	<i>Use:</i> Full-representation gap ≥ 0.999999 alpha on λ/α in $[0,10]$; correct radial and character normalizations. <i>Change:</i> Finite graph does not mean finite-dimensional Hilbert space; sampled coupling points do not cover an interval.	$\alpha > 0$, $\lambda \geq 0$ for displayed tail; finite graph; all-coupling monotonicity unproved; $\alpha > 0$ counterexample is an unmatched path. <i>Next:</i> More plaquettes with shared-link Gauss constraints and coupled tails.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
early11** Three-trace quotient, induced operator and complete electric shells Eqs. (22), (23), (24), (26). <i>Scope:</i> graph-specific derivation and exact polynomial implementation research/round10/skeptic/coordinate-note.md	<i>Use:</i> Complete physical quotient measure/domain, shared-link kinetic operator, spin-network spectrum and exact degree tail threshold. <i>Change:</i> Two separate traces miss a physical variable; two independent rotors reproduce some low tests but fail Kz and K(xy).	Two adjacent open squares, seven links, all six vertex Gauss constraints; no arbitrary boundary condition on quotient singularities. <i>Next:</i> Extend exact induced operators/degree bounds to larger connected graphs.
early12** Interacting two-square certificates and global heat comparison Eqs. (25), (27). <i>Scope:</i> graph-specific exact boundary algebra, gap certificate implementation and explicit conservative estimate research/round11/advisor/advisor.md	<i>Use:</i> Coupling-square floor >0.96 α ; specific full gap interval at (1,2,3,1); explicit positive bound for every finite nonnegative pair. <i>Change:</i> A monomial form matrix is not an ordinary Euclidean Hamiltonian, and positive pointwise gaps need quantified tails and coverage.	$\alpha > 0$, nonnegative magnetic coefficients, $k=0,1$ when $\mu_1 < R < \tau$; finite graph only; global heat bound deteriorates. <i>Next:</i> Connected-volume control retaining locality; sharper actual boundary norm and matched observables.
early13** Volume, measure, clock and non-triviality obstructions Eqs. (28). <i>Scope:</i> exact scoped counterbenchmarks and standard conditional limit criteria research/round12/volume/volume-bridge.md	<i>Use:</i> Free graph gap 3α ; explicit extensive bound; exact tensor counterbenchmark; Wilson Gibbs/ground-state inequivalence; conditional true-ground-state curvature route. <i>Change:</i> Decay of a lower certificate need not mean a vanishing actual gap; stochastic, Euclidean and physical Hamiltonian gaps are distinct.	Tensor copies are disconnected; no uniform true-ground-state curvature bound; no physical continuum construction or vacuum convergence. <i>Next:</i> Actual connected Hamiltonian estimates, embeddings, physical scale, strongly continuous nontrivial limit and full spectral exclusion.
early14** Factorial representation error and certified finite numerical state Eqs. (29), (30). <i>Scope:</i> finite-graph theorem and exact rational computation; priority unverified research/round12/advisor/advisor.md	<i>Use:</i> Full-Hilbert-space error for finite-support preparation and fixed action; rigorous residual/midpoint/Taylor layers; exact two-step state certificate. <i>Change:</i> Historical work/norm/refinement diagnostics did not certify omitted representations; R12 supplies a new theorem and a separate computed-state certificate.	Fixed finite graph and action horizon, signed drivers require absolute values, phase must match; numerical state need not be normalized; energy unbounded. <i>Next:</i> Sharper trajectory boundary integrals and preparation certificates; connected graph and physical-time scaling.
early15** Compact hierarchy fluctuation and formula audits Eqs. (31). <i>Scope:</i> exact explicit closure counterbenchmark and version-specific algebra checks research/round12/closure/closure-audit.md	<i>Use:</i> Exact recurrence, nonzero variance lower bound, residual $\kappa \cdot v$ and rank/counting/phase/prefactor tests. <i>Change:</i> Adding a variance symbol exposes another obligation; it does not close an infinite hierarchy.	One-plaquette Euclidean measure differs from Hamiltonian dynamics; tested closure is point concentration; external display checks apply to 2608.05415v1. <i>Next:</i> Quantified hierarchy truncation or an independently constructed measure with controlled moments.
early16** Finite obligation planner and historical admission repairs <i>Scope:</i> workflow implementation, not a mathematical proof kernel evidence/qeg-research/round4/README.md	<i>Use:</i> Finite forward/backward rule search and replay; source binding and retained defect regressions. <i>Change:</i> Successful script exit, exact binary input rationalization and counts of passing checks do not establish the physical theorem or transcendental rounding.	Finite reviewed rule libraries; R8 symbolic checker unavailable; no comprehensive fresh execution in this manuscript. <i>Next:</i> Independent theorem-premise and evidence admission review with explicit complete dependencies.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
early17** Inherited context and logical gravity controls Eqs. (13). <i>Scope</i> : surviving context; workbench counterexamples, with missing earlier packages evidence/qeg-research/round3/chapters/01_master_action.md	<i>Use</i> : Common-functional target, rotating-mode connection, energy/noise counterexample, spurious truncated concurrence and horizon integrability screen. <i>Change</i> : The same means can have different noise; nonlinear witnesses at an omitted order can be artifacts; fixed geometry modes are not gravity closure.	No independently identifiable R1/R2 contracts or source packages. Imported JSON has provenance and numerical cases but is not a new replay. Contextual conjectures and black-hole/mixing endpoints remain open. <i>Next</i> : Recover missing source packages if available; otherwise keep provenance gap and separately formulate the next physical initial-value problem.
Static, spectral, and local-limit development / dynamics		
R13-locality** Local finite-time volume limit Eqs. (43). <i>Scope</i> : historical study — scope below research/round13/locality/locality.md	<i>Use</i> : Local finite-time boundary comparison <i>Change</i> : Removed the extensive global-volume factor for local dynamics	Fixed spacing and common envelope; no vacuum comparison or continuum-uniform velocity <i>Next</i> : Sharper physical scaling estimate and an appropriate observable algebra
R20-g1** Literal-box real-time dynamics converges in operator norm on each local observable at fixed physical time, with a strongly continuous Hilbert implementation. Section 7. <i>Scope</i> : accepted research/round20/forward/g1/report.md	<i>Use</i> : All-phase fixed-time literal-box dynamics limit <i>Change</i> : Bounded strong Duhamel avoids false operator-norm differentiation	Full bounded local algebra need not have point-norm time continuity <i>Next</i> : Limiting state's GNS generator
Static, spectral, and local-limit development / homogeneous		
R13-stability** Qualitative homogeneous stability Section 9.1. <i>Scope</i> : historical study — scope below research/round13/advisor/weak-coupling-stability.md	<i>Use</i> : Apply a known product-vacuum theorem to full rotor links <i>Change</i> : Matched infinite-dimensional sites, boundaries and gauge-invariant uniqueness	Numerical smallness constants unevaluated; weak bare coupling exits this regime <i>Next</i> : Extract a matched explicit local stability threshold
R17-a1** Exact local projector and incidence bounds; physical counterexample to the full pure relative-form shortcut. Dense uniform W threshold remains open. Section 9.1. <i>Scope</i> : accepted research/round17/forward/a1/report.md	<i>Use</i> : Diagnose diagonal control and vacuum mixing <i>Change</i> : Actual physical trial disproves pure relative bound for full W	Does not refute valid additive bounds or dressing <i>Next</i> : Restricted link-disjoint support theorem

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R21-i1** Complete 24-link blocking gives 21 omitted anchor faces and exact normalized local interaction norm $7 \tau $; qualitative fixed-spacing stability applies with unevaluated source constants. Section 9.1. <i>Scope:</i> limited research/round21/forward/i1/report.md	<i>Use:</i> Exact homogeneous theorem dictionary <i>Change:</i> Complete ownership, all crossing supports and boundary-aware padding	Yarotsky constants remain unevaluated; no declared numerical τ <i>Next:</i> Construct explicit local rotations and audit global iteration
R21-i2** Exact one-star vacuum rotation cancels first-order vacuum mixing and retains a bounded quadratic remainder; a bare pure relative bound for the original interaction fails. Section 9.1. <i>Scope:</i> limited research/round21/forward/i2/report.md	<i>Use:</i> Local domain-safe vacuum dressing <i>Change:</i> Exact cancellation sign and full quadratic remainder	One-star result; overlapping generated terms not yet controlled <i>Next:</i> Simultaneous finite-volume rotation
R22-o1** Complete simultaneous homogeneous rotation has explicit weighted remainder with preserved unbounded operator domain. Eqs. (52). <i>Scope:</i> limited research/round22/reverse/o1/report.md	<i>Use:</i> One simultaneous homogeneous rotation <i>Change:</i> All ordered connected supports, root counts and unbounded domain retained	One-step radius is not gap threshold or iteration closure <i>Next:</i> Generated-support recurrence including diagonal part
Static, spectral, and local-limit development / moments		
R13-moments** Compact moment certificates and uniqueness Eqs. (32). <i>Scope:</i> historical study — scope below research/round13/moments/README.md	<i>Use:</i> Exact scalar Haar moment enclosures <i>Change:</i> Retained variance and all recurrence/support constraints; proved complete-hierarchy uniqueness	No hierarchy rate; code only levels 1–6 and slack must vanish <i>Next:</i> Connected shared-link hierarchy with complete identities
R13-response** Exact scalar response Eqs. (33). <i>Scope:</i> historical study — scope below research/round13/response/response.md	<i>Use:</i> Coupling-response analysis and residual error transport <i>Change:</i> Separated regular Haar branch, singular branches and changed-prior countermodel	Static response; sampled floating error is not a global enclosure <i>Next:</i> Certify a full residual envelope between numerical nodes
R14-loop1** Exact conditional Haar reduction with removable zero field. Section 5.1. <i>Scope:</i> accepted within declared finite-model scope research/round14/forward/loop1.md	<i>Use:</i> Controlled two-holonomy interaction and local covariance sign <i>Change:</i> Exact conditional field and removable zero-field reduction replace factorization	Small- η bound insufficient at selected central fixture; floating comparisons diagonal-agnostic <i>Next:</i> Exact rational same-action covariance enclosures

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R14-loop2** 23 exact finite-measure covariance enclosures independently replayed using characters. Section 5.1. <i>Scope:</i> accepted within declared finite-model scope research/round14/forward/loop2.md	<i>Use:</i> Normalized rational covariance certification <i>Change:</i> 23 exact enclosures certify selected central target including full tails	Static finite action; cap48 distinct from uncapped convergence theorem <i>Next:</i> Continuous target-interval coverage
R15-A1** All8 positive exact midpoints, all8 coarse transported lower bounds insufficient. Section 5.1. <i>Scope:</i> accepted correctly diagnosed insufficiency research/round15/forward/A1/derivation.md	<i>Use:</i> Transport exact covariance point intervals <i>Change:</i> Eight positive midpoint results correctly fail all eight coarse cells	Positive point values do not prove complete interval positivity <i>Next:</i> Bisect failing cells using unchanged point accuracy
R15-A2** Refined failing cell radii rather than increasing already sufficient point precision. Section 5.1. <i>Scope:</i> accepted exact complete positive cover research/round15/forward/A2/result.md	<i>Use:</i> Complete positive cover of eta in $[1/8, 1/4]$ <i>Change:</i> Refined geometry rather than unnecessarily increasing point precision	Finite SU^2 action only; cap failures retained <i>Next:</i> Higher-dimensional or larger-graph certified covers
Static, spectral, and local-limit development / gluing		
R15-B1** Closed cube has six-way dependence while any five face class observables factorize; use actual character gluing and distinct face derivatives in B2. Eqs. (34). <i>Scope:</i> accepted exact finite cube graph and Haar identities research/round15/forward/B1/derivation.md	<i>Use:</i> Graph-faithful closed-cube Haar contraction <i>Change:</i> Proved six-way dependence alongside any-five class-function factorization	Does not assert six independent holonomies or Hamiltonian gap <i>Next:</i> Full normalized Wilson-weighted cube integral
R15-B2** Character dimension and separate face derivatives retained; coarse degree failures remain explicit. Section 5.2. <i>Scope:</i> accepted exact finite cube integral and rejected factorization research/round15/forward/B2/derivation.md	<i>Use:</i> Finite closed-cube Wilson expectation <i>Change:</i> Retained dimension factors and distinct-face derivatives with full Taylor tails	Product expectation is not a connected six-face cumulant; coarse degrees fail <i>Next:</i> Shared-face higher-incidence graph

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R16-loop1** Exact finite two-cube Haar/fusion identities and separate conditional physical energy-scale audit. Eqs. (35). <i>Scope:</i> accepted research/round16/forward/loop1/derivation.md	<i>Use:</i> Shared-face Haar fusion on 12-vertex, 20-link, 11-face graph <i>Change:</i> Replaced invalid single-surface label rule by two-disk fusion	Static graph; full 11-face and outer 10-face actions differ <i>Next:</i> Same observable comparison with shared action on/off
R16-loop2** Positive normalized all-eleven observable difference for outer 1/8 and shared 1/8 versus 0, exact width $\leq 1e-12$. Section 5.2. <i>Scope:</i> accepted research/round16/forward/loop2/derivation.md	<i>Use:</i> Quantitative shared-face action comparison <i>Change:</i> Retained nonzero omitted-action baseline and both normalization errors	Finite static target; no general monotonicity theorem <i>Next:</i> Additional recoupling and surrounding-link dependence
Static, spectral, and local-limit development / finite-gap		
R15-C1** Cube lower bound is zero at common $\lambda = \alpha/2$; accepted B1 trial moments permit a variational ground-energy upper bound. Section 6.1. <i>Scope:</i> accepted finite-graph side theorem and retained original source gap research/round15/forward/C1/derivation.md	<i>Use:</i> Full untruncated finite-graph gap bound <i>Change:</i> Distinguished physical girth gap from full-link $3\alpha/4$ gap	Cube bound zero at $\lambda = \alpha/2$; global budget grows with volume <i>Next:</i> Variational E0 upper bound with independent E1 lower bound
R15-C2** The independent E1 lower bound and trial E0 upper bound repair the C1 zero-margin fixture; no trial gap was substituted for the full gap. Eqs. (38). <i>Scope:</i> accepted finite physical cube variational improvement research/round15/forward/C2/derivation.md	<i>Use:</i> Repair finite cube zero margin <i>Change:</i> Combined one-sided full E1 and trial E0 bounds correctly	Radical interval encloses lower-bound formula, not actual gap <i>Next:</i> Dense two-cube and complete complement improvements
R17-b1** Improved full physical gap lower bound on the dense eleven-face two-cube graph, positive for common $ \lambda /\alpha < 12/43$; finite volume only. Eqs. (38). <i>Scope:</i> accepted research/round17/forward/b1/report.md	<i>Use:</i> Dense finite two-cube spectral guarantee <i>Change:</i> Rebuilt every repeated-index trial entry on actual graph	Fixed graph; zero margin at $12/43$ is certificate boundary <i>Next:</i> Add actual shared-face adjoint trial channel

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R17-b2** Exact positive full physical gap lower $\alpha \cdot 1388/284767457$ at the dense finite coupling $\lambda/\alpha=12/43$, with a proved open variational-amplitude interval. Section 6.1. <i>Scope:</i> accepted research/round17/forward/b2/report.md	<i>Use:</i> Positive full gap at previous endpoint <i>Change:</i> Actual adjoint trial lowers E0 below independent E1 bound	<i>Amplitude</i> midpoint maximizes numerator, not quotient; no volume theorem <i>Next:</i> Complete low-energy complement
R18-b1** Complete physical complement threshold $9\alpha/2$ and exact QVP Gram on the dense two-cube graph; no finite trial spectrum substituted for the entire omitted space. Section 6.1. <i>Scope:</i> accepted research/round18/forward/b1/report.md	<i>Use:</i> Complete physical complement and exact leakage Gram <i>Change:</i> Classified all states below threshold and retained projection subtraction	Independent Haar faces agree at fourth order; Gaussian fourth moment is false control <i>Next:</i> Codimension-one full excited-energy estimate
R18-b2** Full untruncated physical two-cube signed coefficient-box gap lower bound, separate endpoint ground trial, and bare-coupling domain obstruction. No volume-uniform or continuum claim. Section 6.1. <i>Scope:</i> accepted research/round18/forward/b2/report.md	<i>Use:</i> Full untruncated dense two-cube gap <i>Change:</i> Converted exact complement/Gram into continuous signed-box theorem	Fixed graph; standard bare-coupling membership requires $\hat{g}^4 \geq 32/3$ <i>Next:</i> Larger complete projector and better cross envelope
R19-b1** Goal B Loop 1 only: complete strict-cutoff physical projector and exact full cross Gram on the actual finite two-cube $SU(2)$ graph. No continuous coefficient box, dense limit, continuum Yang-Mills or mass-gap claim is accepted here. Section 6.1. <i>Scope:</i> accepted research/round19/forward/b1/report.md	<i>Use:</i> Exact physical channel and cross-Gram inventory <i>Change:</i> Includes all 36 six-cycles and threshold intertwiner multiplicities	Strict cutoff is full spectral classification, not chosen variational list <i>Next:</i> Continuous signed-box full E1 estimate

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R19-b2** Goal B Loop 2 only: finite two-cube continuous signed coefficient-box gap certificate using the accepted B1 48-channel projector, full compressed PVP ledger and exact ordered W^*W cross Gram. No dense limit, continuum Yang-Mills, Clay mass gap or physical-scale matching claim is accepted. Eqs. (39). <i>Scope:</i> accepted research/round19/forward/b2/report.md	<i>Use:</i> Wider finite signed coefficient box <i>Change:</i> Improved completeness6alpha and exact 47-state row/cross envelopes	$r = 1/2$ fails only this sufficient envelope <i>Next:</i> Sharper fixed-graph envelopes or separate volume method
Static, spectral, and local-limit development / strips		
R17-a2** Volume-uniform physical gap for the pairwise link-disjoint magnetic-support family with a common positive energy scale; dense overlapping family remains open. Section 6.2. <i>Scope:</i> accepted research/round17/forward/a2/report.md	<i>Use:</i> Volume-uniform sparse magnetic family <i>Change:</i> Full-link factorization plus gauge-invariant unique ground	Pairwise link-disjoint support; dense overlaps excluded <i>Next:</i> Internally overlapping dressed strip
R18-a1** Finite dressed three-square bridge; physical gap at least $\alpha/8$ throughout the declared primary family; no homogeneous volume extension. Section 6.2. <i>Scope:</i> accepted research/round18/forward/a1/report.md	<i>Use:</i> Bridge two dressed sparse blocks <i>Change:</i> Free bridge link proves zero reference mean and one-norm improvement	Finite cluster; exact dressed wavefunction unknown and unnecessary <i>Next:</i> Repeat clusters with complete remainder ledger
R18-a2** All finite volumes of the specified spatially decaying inhomogeneous family have physical gap at least $7\alpha_{\min}/64$; boundary reclassification and homogeneous dense/thermodynamic limits remain explicit. Section 6.2. <i>Scope:</i> accepted research/round18/forward/a2/report.md	<i>Use:</i> All-finite-volume inhomogeneous full-support family <i>Change:</i> Repeated link-disjoint overlapping strips plus summable remaining faces	Boundary rule reclassifies coefficients; not yet literal restrictions <i>Next:</i> Fixed infinite assignment and clipped-component theorem

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R19-a1** For every reachable selected clipped component L , M , R , LM , MR or LMR under the frozen coefficient bounds $ \lambda_L /\alpha \leq 1/2$, $ \mu_M /\alpha \leq 1/8$ and $ \lambda_R /\alpha \leq 1/2$, the full-link local gap lower estimate is at least $\alpha/8$. Positive full-link uniqueness and gauge invariance permit the same lower estimate after Gauss-sector restriction for the finite local component result. Section 6.2. <i>Scope:</i> accepted research/round19/forward/a1/report.md	<i>Use:</i> Literal finite restrictions of one infinite assignment <i>Change:</i> Removes Round18 coefficient reclassification	Finite gap statement alone does not construct thermodynamic representation <i>Next:</i> Direct product representation and summable perturbation
R19-a2** Goal A Loop 2 only: direct incomplete product-representation construction for a summable omitted-face perturbation of the accepted A1 complete-strip/free reference. This is not finite clipped-restriction convergence, homogeneous dense stability, continuum Yang-Mills or a mass-gap theorem. Eqs. (40), (41). <i>Scope:</i> accepted research/round19/forward/a2/report.md	<i>Use:</i> Explicit infinite-volume summable Hamiltonian <i>Change:</i> Closed product-reference form and rank-one spectral isolation	Initially no finite clipped-limit convergence; fixed representation and spacing <i>Next:</i> Quantitative finite-factor and literal-box convergence
R20-d1** Finite-factor completion gives norm-resolvent convergence with explicit dyadic tail on the exact A2 exterior-reference lift. Section 6.2. <i>Scope:</i> accepted research/round20/forward/d1/report.md	<i>Use:</i> Quantitative finite-factor completion <i>Change:</i> Correct exterior-reference lift preserves infinite generator	Finite factors remain infinite-dimensional; no literal arbitrary-phase theorem yet <i>Next:</i> Ground energies, projections and states
R20-d2** Ground energy, rank-one projections and bounded observables converge with explicit dyadic errors for D1 lifts. Section 6.2. <i>Scope:</i> accepted research/round20/forward/d2/report.md	<i>Use:</i> Ground-state convergence with explicit dyadic errors <i>Change:</i> Zero trial mean and rank-one residual give sharp denominator	Requires $\beta < \delta$ and correct embedding <i>Next:</i> Literal geometric boxes and energy origins

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R20-e1** Literal aligned boxes $N=4m+3$ converge after explicit reference-energy subtraction and exact exterior-reference lift. Section 6.2. <i>Scope:</i> accepted <code>research/round20/forward/e1/report.md</code>	<i>Use:</i> Aligned literal-box convergence <i>Change:</i> Retained exact strip energy subtraction and orientation-specific tail	Unshifted absolute energy/resolvent convergence not proved <i>Next:</i> All boundary phases in local-state topology
R20-e2** All upper-boundary phases of literal growing rectangular boxes converge to the A2 ground state on bounded local observables. Section 6.2. <i>Scope:</i> accepted <code>research/round20/forward/e2/report.md</code>	<i>Use:</i> All-phase rectangular local-state limit <i>Change:</i> Observable support included in common complete central factors	No global vector or arbitrary-phase norm-resolvent conclusion <i>Next:</i> Local dynamics and GNS identification
Static, spectral, and local-limit development / conditional		
R17-c1** Complete four-adjoint Haar tensor and realized four-cube conditional boundary; no bulk or spectral claim. Section 5.3. <i>Scope:</i> accepted <code>research/round17/forward/c1/report.md</code>	<i>Use:</i> Complete central four-adjoint Haar tensor <i>Change:</i> Three invariant channels and negative inverse-Gram entries retained	One central link with 32 others fixed <i>Next:</i> Nonzero-action joint observable with actual boundary
R17-c2** Same-action conditional four-face expectations with complete rational numerator and partition errors; action-only observable closure rejected, full bulk and spectrum open. Section 5.3. <i>Scope:</i> accepted <code>research/round17/forward/c2/report.md</code>	<i>Use:</i> Reject action-only observable closure <i>Change:</i> Equal partition and action coexist with different joint observables	Conditional central result; surrounding marginal not integrated <i>Next:</i> Sufficient complete action-observable coordinates
R18-c1** Complete joint-Gram sufficiency for the declared central S3 observable/action class, exact rational admissibility including singular cases, and independently matched degree 0–6 diagnostics. No minimality, finite-tail, surrounding measure or mass-gap conclusion. Eqs. (36). <i>Scope:</i> accepted <code>research/round18/forward/c1/report.md</code>	<i>Use:</i> Coordinate sufficiency for central integral <i>Change:</i> Handles rank-deficient, reflected and zero-action data exactly	Sufficiency is not minimality or surrounding measure closure <i>Next:</i> Integrate actual surrounding Haar link

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R18-c2** Full normalized two-link conditional six-face integral at the declared signed fixtures, exact tails, actual surrounding Gram/measure connection and preserved omission/symmetry controls. Thirty-one links remain fixed; no physical spectrum or full bulk/continuum claim. Section 5.3. <i>Scope:</i> accepted research/round18/forward/c2/report.md	<i>Use:</i> Two-link normalized six-face conditional integral <i>Change:</i> Restored all surrounding-only weights and partition reweighting	31links fixed; consistent V->Vdagger is a valid integrated exception <i>Next:</i> Three-link shared-middle chain
R19-c1** Goal C Loop 1 only: exact finite static U-V-W chain integral on the actual four-cube graph with all other links fixed. This accepts a static mathematical checkpoint and common-V failed-independence discriminator; it does not accept physical energy/time-scale matching, dense convergence, continuum Yang-Mills, a spectral gap theorem or the Clay mass gap. Section 5.3. <i>Scope:</i> accepted research/round19/forward/c1/report.md	<i>Use:</i> Actual three-link conditional chain integral <i>Change:</i> Shared-middle dependence retained and freezeW regression checked	Thirty surrounding links remain fixed; static scale unmatched <i>Next:</i> Continuous signed-kappa result
R19-c2** Static common-V integral only; physical energy/time matching remains open. No Hamiltonian spectral gap or continuum claim. Eqs. (37). <i>Scope:</i> accepted research/round19/forward/c2/report.md	<i>Use:</i> Certified continuous positivity and sign asymmetry <i>Change:</i> Converted exact coefficients into normalized all-interval inequalities	Degree8 boundary failures do not prove nonpositivity; static model only <i>Next:</i> More surrounding variables or independent physical matching
Static, spectral, and local-limit development / clock		
R20-f1** Even among local reversible diffusions the C1 static law does not identify an energy/time coefficient: $H_{\text{kappa},c=c} A_{\text{kappa}}$. Section 8.1. <i>Scope:</i> accepted research/round20/forward/f1/report.md	<i>Use:</i> Demonstrate static clock non-identifiability <i>Change:</i> Same static density supports independently scaled reversible dynamics	Added conditional generator, not physical lattice derivation <i>Next:</i> Full derivative closure and independently observed slope

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R20-f2** Exact shared-middle gradient closure, action range[-5,7] and conditional diffusion gap $c/6$ on $ \kappa \leq 1/8$. Section 8.1. <i>Scope:</i> accepted <code>research/round20/forward/f2/report.md</code>	<i>Use:</i> Exact chosen-generator transform and conditional clock inverse <i>Change:</i> Retained actual common-V derivative and improved action range	No physical slope measured; c unmatched to α <i>Next:</i> Mobility deformations and identifying observables
R21-k1** A positive state-dependent mobility preserves the static density but changes reversible dynamics; two chosen Haar correlation slopes identify c and ζ within this conditional family. Section 8.1. <i>Scope:</i> accepted <code>research/round21/forward/k1/report.md</code>	<i>Use:</i> Two-rate affine-mobility identification <i>Change:</i> Static law independent of ζ ; gradmobility drift retained	Haar inverse only known $\kappa = 0$; synthetic rates only <i>Next:</i> Uniform nonzero- κ rank and conditioning
R21-k2** A tunable observable gives a uniform two-rate inverse across the declared static interval, while a positive blind cubic proves that two slopes do not identify arbitrary mobility. Section 8.1. <i>Scope:</i> accepted <code>research/round21/forward/k2/report.md</code>	<i>Use:</i> Robust rank certificate in declared affine family <i>Change:</i> Monotone double-integral determinant and blind cubic counter-model	Finite-family injectivity does not identify arbitrary mobility <i>Next:</i> Third rate detects hidden cubic
R21-l1** A third observable detects the previously hidden cubic mobility and gives an exact three-parameter inverse in the declared conditional family. Eqs. (48). <i>Scope:</i> accepted <code>research/round21/forward/l1/report.md</code>	<i>Use:</i> Three-parameter conditional inverse <i>Change:</i> Third observable exposes p_3 invisible to old pair	Inverse at known $\kappa = 0$; polynomial itself classical <i>Next:</i> Finite-data nonidentifiability beyond chosen family
R21-l2** Any finite list of initial correlation slopes leaves strictly positive mobility freedom; a second derivative detects the explicit hidden fifth-degree direction. Section 8.1. <i>Scope:</i> accepted <code>research/round21/forward/l2/report.md</code>	<i>Use:</i> Exact finite-slope limitation and held-out curvature <i>Change:</i> Explicit positive p_5 mobility keeps 3 slopes but changes generator	Does not show complete-time-curve ambiguity <i>Next:</i> Specify finite basis and reserve dynamical data

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
Static, spectral, and local-limit development / gns		
R20-g2** The E2 limiting local state GNS on the full factor algebra is the A2 representation with canonical generator H-e and its inherited gap. Section 6.3. <i>Scope:</i> accepted <code>research/round20/forward/g2/report.md</code>	<i>Use:</i> Identify physical energy origin of full-algebra limiting state <i>Change:</i> Irreducibility and cyclicity proved on dense finite-excitation core	Gauss-only algebra not yet identified; $\beta < \delta$ restored <i>Next:</i> Nonvacuous invariant local algebra and physical GNS
R21-j1** An actual omitted Wilson loop has a nonzero gauge-invariant fluctuation in the perturbed dyadic vacuum, with a uniform rational variance floor. Section 6.3. <i>Scope:</i> accepted <code>research/round21/forward/j1/report.md</code>	<i>Use:</i> Nonzero physical Wilson fluctuation <i>Change:</i> Transfers actual reference variance to perturbed dyadic ground	Invariant cyclic space only inclusion at this stage <i>Next:</i> Prove equality with invariant Hilbert subspace
R21-j2** Physical vacuum GNS space equals the gauge-invariant subspace of this representation, with the restricted shifted energy generator and a nonzero imaginary-time Wilson correlator. Eqs. (42). <i>Scope:</i> accepted <code>research/round21/forward/j2/report.md</code>	<i>Use:</i> Physical GNS space and imaginary-time spectrum <i>Change:</i> Finite gauge averaging proves reverse cyclic inclusion	No real-time magnitude decay or measured glueball mass <i>Next:</i> Different homogeneous/continuum representation requires separate proof
Static, spectral, and local-limit development / profile		
R20-h1** The continuous omitted-face profile has an exact budget and a rigorously delimited sufficient gap certificate at fixed physical scale. Eqs. (44). <i>Scope:</i> accepted <code>research/round20/forward/h1/report.md</code>	<i>Use:</i> Continuous spatial decay budget <i>Change:</i> Exact class sums locate sufficient budget boundary	qcrit is not physical transition; q1 infinite sum diverges <i>Next:</i> Constant-budget rays and actual ground-state behavior
R20-h2** Every strict uniform global perturbation-budget ray returns the actual ground projection to the selected-strip reference, with an exact vacuum variance and a quantitative rate. Eqs. (45). <i>Scope:</i> accepted <code>research/round20/forward/h2/report.md</code>	<i>Use:</i> Ground-state return along fixed-budget rays <i>Change:</i> Exact two-free-link variance beats generic overlap bound	Endpoint retains selected strips; fixed omitted coefficients vanish <i>Next:</i> Strong operator and correlation limits

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R21-m1** The fixed-budget summable perturbation converges strongly to zero, giving strong-resolvent and compact-time strong dynamics convergence to the selected-strip reference. Section 7.1. Scope: accepted research/round21/forward/m1/report.md	<i>Use:</i> Strong-resolvent and compact-time strong dynamics limit <i>Change:</i> Complete-factor commutator extends vacuum residual to dense set	Norm resolvent/dynamics status unresolved; endpoint is selected strip <i>Next:</i> Connected correlations on growing time windows
Static, spectral, and local-limit development / windows		
R21-m2** Connected physical correlations converge on explicitly growing time windows in the canonical summable profile, with a nonzero actual physical Wilson fluctuation. Section 7.2. Scope: accepted research/round21/forward/m2/report.md	<i>Use:</i> Connected physical correlations with $\gamma < 3/2$ <i>Change:</i> All centering, phases and factor supports retained	Endpoint failure is failure of this upper certificate <i>Next:</i> Use stationary Heisenberg comparison
R22-n1** Stationary Heisenberg comparison proves the sharper local connected-correlation bound in the canonical summable model. Eqs. (46). Scope: accepted research/round22/reverse/n1/report.md	<i>Use:</i> Improved fixed-support growing-time certificate <i>Change:</i> Stationarity removes unnecessary time-times-vacuum-residual term	Actual γ_3 behavior unresolved; no real-time decay assertion <i>Next:</i> Moving complete-factor support budgets
R22-n2** Complete tail-block cubes have an exact weighted incident-face budget, yielding a joint support/time convergence certificate. Eqs. (47). Scope: accepted research/round22/reverse/n2/report.md	<i>Use:</i> Joint support and time design certificate <i>Change:</i> Exact incident budget including outgoing links and origin boundary	Necessity only for nonnegative certificate; translated cubes differ <i>Next:</i> Actual endpoint or stronger dynamics estimate
Static, spectral, and local-limit development / inverse		
R22-o2** The bare-reference elimination has explicit all-support bounds, but its retained-diagonal loss majorants cannot certify an infinite iteration at nonzero coupling. Eqs. (53). Scope: limited research/round22/reverse/o2/report.md	<i>Use:</i> Precise failed iteration certificate <i>Change:</i> Retained diagonal term forces eventual failure of stated loss majorant	Not actual algorithm failure; regrouped1/64 differs indexed $(9N+16)/(64(3N+1))$ <i>Next:</i> Structure-sensitive interacting inverse and full boundary terms

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R22-r1** The actual initial diagonal has a uniform local inverse, while its local homological generator leaves an exact nonzero SU2 crossing-boundary defect. Eqs. (54). Scope: accepted research/round22/reverse/r1/report.md	<i>Use:</i> Initial interacting local inverse and boundary audit <i>Change:</i> Absorbs interior diagonal while proving actual SU2 crossing-defect	Diagnostic source not identified as generated residual; initial D only <i>Next:</i> Boundary-complete inverse response with domain control
R22-r2** The actual initial boundary-complete vacuum response converges with explicit spatial and graph tails; its source-sector and specified volume-weight limits are exact. Eqs. (55). Scope: accepted research/round22/reverse/r2/report.md	<i>Use:</i> Initial boundary-complete response with spatial/graph tails <i>Change:</i> Proves actual response regularity and exact source-sector mismatch	Global vacuum cancellation not local identity extension; cardinality upper sum fails <i>Next:</i> Full-source cancellation plus useful weighted decomposition and later-stage gaps
Static, spectral, and local-limit development / tree		
R22-p1** The prescribed section is nonclosable and its fitted clock fails the reserved correlation. Section 8.2. Scope: accepted research/round22/reverse/p1/report.md	<i>Use:</i> Falsify proposed physical section and clock <i>Change:</i> Nonclosable map and held-out full-time discrepancy on actual graph	Rejects this section only; other reductions remain possible <i>Next:</i> Genuine Haar maximal-tree isometry
R22-p2** A genuine Haar reduction exactly transports the finite electric dynamics; the declared conditional family fails the reserved data. Eqs. (49). Scope: accepted research/round22/reverse/p2/report.md	<i>Use:</i> Exact pure-electric selected reduction <i>Change:</i> Unitary16chord map, Haar adjoint, full tree/mixed derivatives and domains	Different observable completion from P1; conditional family still fails <i>Next:</i> Restore all magnetic faces and quantify leakage
Static, spectral, and local-limit development / memory		
R22-q1** The full magnetic action has correlated positive conditional variance, proving selected-image leakage and a quantitative nonautonomous dynamic defect. Eqs. (50). Scope: accepted research/round22/reverse/q1/report.md	<i>Use:</i> Exact actual-model magnetic leakage <i>Change:</i> Three correlated faces rule out autonomous selected compression	Haar is not interacting ground; no operator-norm Taylor promotion <i>Next:</i> Exact projected memory and complementary spectrum

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
R22-q2** The actual magnetic projection has exact memory/resolvent identities, seven-energy fixed-channel leading matrices and explicit cubic full-Hilbert operator remainders. Eqs. (51). <i>Scope:</i> accepted research/round22/reverse/q2/report.md	<i>Use:</i> Projected dynamics with explicit kernel and self-energy errors <i>Change:</i> Full return factor, offdiagonal weight1/4 and complete leading spectrum	No autonomous2-channel closure; absolute not late-time relative accuracy <i>Next:</i> Controlled interacting projected approximation with original clock
Full-source, observable, and computed-heat results		
recent01** Nonzero indexed cubic source Eqs. (56), (57). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/s1/report.md	<i>Use:</i> Provides the actual order- τ^3 off-diagonal source for the later filter and loading tests. <i>Change:</i> The cubic term cannot be dismissed as algebraically zero; it is an indexed term rather than the full grouped cubic remainder.	Initial retained diagonal and one specified indexed term only; $\tau=0$ is the trivial zero control. No all-sector inverse follows. <i>Next:</i> Control its complete crossing defect and the residual in the declared interaction norm.
recent02** Interior inverse and complete connected crossing defect Eqs. (58), (59). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/s1/report.md	<i>Use:</i> One-step full-source inverse-with-defect and weighted connected-family certificate. <i>Change:</i> The local vacuum inverse is valid only after its boundary defect is included; reverse improves 192 Mell $e^{\wedge}(7\mu)$ to 168 Mell $e^{\wedge}(7\mu)$.	The nonzero value of the actual crossing defect and an actual nonzero SU(2) resonance are not evaluated. No nonlinear induction or all-order summability. <i>Next:</i> Remove or contract the full defect without changing the action or replacing exterior identity by a vacuum projector.
recent03** Compact-time graph-domain commutator with retained sinc residual Eqs. (60), (61). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/s2/report.md	<i>Use:</i> Bounded regulated inverse on the actual initial graph domain. <i>Change:</i> The commutator equation has a residual, including its entire zero-frequency block.	$0 < \Theta \leq 1/16$ at the historical coupling cap. Θ is a dimensionless proof parameter, not a calibrated physical time. No residual contraction is claimed here. <i>Next:</i> Quantify the residual on this particular cubic source, then preserve locality while reducing it.
recent04** Connected weighted bounds and infinite initial-operator identification Eqs. (62), (63), (64), (65). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/s2/report.md	<i>Use:</i> Complete weight-2 short-time interaction bounds and a qualified infinite-volume initial-stage construction. <i>Change:</i> Finite-volume formulas acquire an actual infinite initial-generator interpretation in the forward proof; the extensive translated sum is still not a bounded Hilbert-space operator.	Forward-only infinite identification; no global norm unitary limit, later-diagonal iteration or all-time positive-weight bound. <i>Next:</i> Establish the topology and locality required for a later renormalized diagonal, with its own domain control.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
recent05** The short-time residual cannot be made small at the admitted cap Eqs. (66), (67). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round24/forward/w1/report.md	<i>Use:</i> Discriminating lower certificate against deleting the short-filter residual. <i>Change:</i> Smaller proof time preserves most of the source; this is an actual-source failure certificate rather than a generic toy resonance.	The statement concerns the original cubic source and admitted short-filter range. It does not establish a nonzero zero-frequency atom or forbid longer or different filters. <i>Next:</i> Use a filter with a controlled long-time or smooth-kernel mechanism.
recent06** Iterated filter converges on the vacuum column but retains resonant blocks Eqs. (68), (69), (70). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round24/forward/w2/report.md	<i>Use:</i> Certified column inverse and a precise surviving-block target for all-sector work. <i>Change:</i> Column convergence and strong operator convergence are different results; neither compact resolvent nor pointwise spectral damping gives an operator-norm inverse.	Old weighted estimates close only for $n \leq 3$; no nonzero actual resonance block is proved and no all-order weighted inverse follows. <i>Next:</i> Bound $D_{\text{G A}}$ directly and find a filter that contracts the complete source in a usable interaction norm.
recent07** The resonant block and spectral windows are boundary controlled Eqs. (71), (72). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/ab1/report.md	<i>Use:</i> Quantifies possible resonance from the complete connected boundary defect. <i>Change:</i> A full source can have a much smaller resonant block than its short-filter residual; this supplies an upper bound, not a nonzero resonance proof.	The historical gate remains limited because actual crossing resonance values and all-sector inversion are unresolved. <i>Next:</i> Compute an actual resonance block or contract the entire source without assuming that the block vanishes.
recent08** Gaussian boundary identity gives strict operator residual contraction Eqs. (73). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/ab2/report.md	<i>Use:</i> Strict full operator residual contraction for the original cubic source at the old coupling cap. <i>Change:</i> Removes the W1 short-time obstruction through a different proof filter while keeping the same Hamiltonian.	Operator contraction does not itself give the original weighted interaction norm or contraction of later transformed sources. <i>Next:</i> Localize the Gaussian map quantitatively and obtain a compatible weighted contraction theorem.
recent09** Gaussian collar locality and graph-domain truncation error Eqs. (74), (75). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/ae1/report.md	<i>Use:</i> Explicit finite-collar approximations in operator norm and, by the reverse extension, the graph domain. <i>Change:</i> Quasi-locality is established in radius/diameter; this does not close an exponential penalty in support cardinality.	The displayed upper budget 2^{-R} cannot absorb $\exp(\mu O(R^3))$; failure of that upper estimate is not a proof that the weighted norm diverges. <i>Next:</i> Change the combinatorial summation or kernel to establish finiteness in the original weighted norm.
recent10** Compact residual contraction and original weighted finiteness Eqs. (76), (77), (78). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/ae2/report.md	<i>Use:</i> First simultaneous original-source operator contraction and original weight-2 finiteness under a stricter coupling cap. <i>Change:</i> The old cap cannot satisfy the present contraction and series-radius inequalities; shrinking the declared parameter range resolves this method-level incompatibility.	Weighted finiteness is not weighted contraction, and the theorem is only for the original cubic source. No nonlinear homogeneous gap proof. <i>Next:</i> Prove contraction in the actual weighted interaction norm and propagate it through the nonlinear update.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
recent11** Full-space leading-memory return model and unshifted heat error Eqs. (79), (80), (81), (82). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/t1/report.md	<i>Use:</i> Unique full-space Volterra approximation with fixed-time cubic and uniform absolute quadratic coupling errors. <i>Change:</i> The two-channel second derivative has a nonzero omitted defect; direct selected propagation is not the full magnetic heat.	Unshifted absolute error only at T1; fixed finite graph, constant coupling, Euclidean heat. No relative, real-time, graph-size or continuum theorem. <i>Next:</i> Center each full-space evolution at its own minimum and certify a finite implementation.
recent12** Separately centered full-selected heat remains uniformly close Eqs. (83). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/t2/report.md	<i>Use:</i> Time-uniform absolute comparison of two individually centered full-space heat models. <i>Change:</i> Correct energy centering is part of the observable approximation, not a harmless plotting convention.	T2 uses correlated same-author derivations and skeptical self-review. No relative theorem for arbitrary inputs and no finite executable truncation at this stage. <i>Next:</i> Control a concrete finite physical basis and specify input preparation before claiming relative accuracy.
recent13** Electric-energy cutoff and isolated-ground truncation theorem Eqs. (84), (85), (86). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round24/forward/x1/report.md	<i>Use:</i> Explicit delayed uniform truncation and numerical perturbation criteria. <i>Change:</i> A mathematically adequate cutoff can be enormous and unassembled; the theorem alone is not an executed solver.	The R=1024 matrix is not assembled. No claim of an arbitrary retained-block theorem from external real-time truncation literature. <i>Next:</i> Assemble a smaller physical basis and exploit its actual residual.
recent14** Representation-degree path truncation and the assembled 21-state seed Eqs. (87), (88). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round24/forward/x1/report.md	<i>Use:</i> A second truncation mechanism plus the explicit 21-dimensional Ritz model. <i>Change:</i> Large theoretical cutoffs and the actually assembled small basis are different deliverables; the latter has nonzero omitted coupling.	X1 remains limited; its assembled 21-state space does not yet have the later residual and all-time relative certificates. <i>Next:</i> Compute all omitted channels and their norm, rather than infer accuracy from matrix size.
recent15** Complete residual of the 21-state ground trial Eqs. (89), (90). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round24/forward/x2/report.md	<i>Use:</i> Exact model-specific residual norm and spectral distribution for ground and heat certification. <i>Change:</i> The residual is a physical channel calculation, not a norm estimate based solely on discarded basis count.	The electric spectral weights concern this residual vector; they are not the full interacting spectrum. <i>Next:</i> Use both its norm and energy moment to bound the ground shift and late heat.
recent16** Two-sided ground enclosure and evaluated late heat Eqs. (91), (92). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round24/forward/x2/report.md	<i>Use:</i> Certified late-time absolute full-space heat and evaluated output in the 21-state seed. <i>Change:</i> A two-sided energy interval proves strict Ritz bias, enabling the later relative-error falsifier.	Absolute late-time theorem; arbitrary nearly ground-null input can still have unbounded relative error. <i>Next:</i> Declare a preparation class and bound the true output denominator.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
recent17** Uniform relative heat accuracy fails on ground-null inputs Eqs. (93). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round24/forward/z1/report.md	<i>Use</i> : Discriminating falsifier against extending an absolute semigroup estimate to arbitrary-input relative accuracy. <i>Change</i> : Ground overlap and denominator convention must be part of the theorem; exact agreement at $\lambda=0$ or $\sigma=0$ does not repair late-time failure.	Constructed bad inputs are not the later normalized vacuum-neighborhood preparation class. <i>Next</i> : Prove relative accuracy on an explicit input class whose true output has a certified floor.
recent18** Exact omitted Gram matrix and all-time relative repair for prepared inputs Eqs. (94), (95), (96), (97). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round24/forward/z2/report.md	<i>Use</i> : Full-time same-input true-output-relative error with stated preparation hypotheses. <i>Change</i> : The obstruction is repaired by an input contract and exact dynamic loading, not by redefining away preparation bias.	Different forward/reverse preparation sets must not be conflated. No arbitrary-ground-null-input relative claim. <i>Next</i> : Include the complete omitted image as new physical channels and check whether the error actually improves.
recent19** The 293-state physical channel enlargement Eqs. (98), (99). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round26/forward/ac1/report.md	<i>Use</i> : An actually assembled sparse gauge-invariant 293-dimensional Galerkin matrix. <i>Change</i> : Exact zero leakage on the old subspace is a first-step property; it is not invariance under the enlarged retained evolution.	The 293-state space is not a complete electric-energy shell. Channel count alone does not prove the desired heat improvement. <i>Next</i> : Bound delayed occupation of the new channels before applying their remaining omitted coupling.
recent20** Improved ground certificate does not yet prove an improved heat bound Eqs. (100). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round26/forward/ac1/report.md	<i>Use</i> : Ground-state improvement and a documented failed coarse all-time error target. <i>Change</i> : The historical AC1 gate correctly remains limited despite successful matrix assembly and a tighter ground certificate.	No implication from static ground improvement to the requested dynamic relative improvement. <i>Next</i> : Use the initially empty new channels and their actual loading equation.
recent21** Delayed channel loading closes the improved all-time relative bound Eqs. (101), (102), (103), (104), (105). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round26/forward/ac2/report.md	<i>Use</i> : Strict improvement over the Z2 physical all-time preparation-class certificate. <i>Change</i> : The necessary improvement comes from the time ordering of physical channels, not the dimension 293 itself.	The input remains in the original P21 class. No all-input relative bound, real-time theorem, graph-size uniformity or executed solver for every coupling. <i>Next</i> : Execute the finite heat action with rigorous ground centering and numerical output error.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
recent22** Actual 293-coordinate heat at one declared time Eqs. (106). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/af1/report.md	<i>Use:</i> Executed 293-coordinate output with Haar metric, polynomial, centering and export errors accounted for. <i>Change:</i> The heat theorem becomes a finite evaluated artifact at the fixed parameter-time-input contract.	AF1 executes one coupling, time and vacuum input only; it is not the all-time evaluator. <i>Next:</i> Prevent approximate centering from causing secular error and cover the full declared preparation class.
recent23** Piecewise all-time computed heat and the fixed numerical scope Eqs. (107), (108), (109). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/af2/report.md	<i>Use:</i> Complete executed all-time finite-graph heat approximation on the declared preparation class. <i>Change:</i> The early/late switch makes the error budget uniform while retaining a finite computation at requested rational times.	Fixed lambda=0.01 only. Reverse CLI accepts real rational 293-vectors, with 11e-30 rounding; complex operator bounds and forward complex demonstrations use the separate 15e-40 allowance, not an arbitrary-complex CLI. A tiny switch discontinuity is allowed within budget. <i>Next:</i> Validate graph-size or observable dependence and an explicit physical scale-identification contract; do not infer continuum or calibration results.
recent24** Canonical rank observable has an actual eight-link first-order resonance Eqs. (110), (111), (112), (113). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/u1/report.md	<i>Use:</i> A concrete nonzero secular coefficient for the canonical summable family. <i>Change:</i> This supplies a physical-model resonance mechanism rather than a two-level analogy, while leaving higher bulk terms unresolved at U1.	Rank-two bounded operator, not Wilson multiplication. U1 is limited and has only same-author self-review. No exact endpoint limit yet. <i>Next:</i> Bound all connected higher orders of the full model uniformly on the stated endpoint window.
recent25** Full-system endpoint witness from connected-word tails Eqs. (114), (115), (116). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round23/forward/u2/report.md	<i>Use:</i> A full-model counterexample to extending the earlier subcritical window to the critical cubic time exponent for every bounded local probe. <i>Change:</i> The gamma=3 endpoint obstruction is established in the frozen canonical model; it is not merely a formal secular term.	Existence of a suitable probe, not a universal statement for all observables. No exact trajectory at U2, Wilson measurement or physical scale calibration. Same-author self-review. <i>Next:</i> Provide an operational readout and prove an actual scalar limit using controlled spatial collars.
recent26** Finite-outcome and coherent readout of the complex correlation Eqs. (117), (118). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round24/forward/v1/report.md	<i>Use:</i> An ideal coherent measurement contract with explicit operator approximation tolerance. <i>Change:</i> Finite outcome count does not establish a finite laboratory implementation; complex order must be measured coherently.	Ideal bounded operators and controlled unitaries are assumed, not physically calibrated or implemented. <i>Next:</i> Include clock, state, gate and statistical errors in the same correlation budget.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
recent27** Full-Haar operator-norm obstruction to multiplying-observable replacement Eqs. (119). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round24/forward/v1/report.md	<i>Use</i> : Rules out an unjustified identification of the original rank observable with Wilson multiplication. <i>Change</i> : Same vacuum-created vector or a formal compressed agreement does not imply that the complete observables are equal.	This full-space norm obstruction does not forbid separately analyzing a Wilson multiplier or a declared restricted norm. <i>Next</i> : Construct and certify a Wilson observable on its own operator and state contract.
recent28** Clock and finite-sample error budget for the coherent protocol Eqs. (120), (121). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round24/forward/v2/report.md	<i>Use</i> : A complete ideal-protocol error and shot-count certificate at explicit finite q . <i>Change</i> : Operational tolerances are mathematically quantified and can be extremely demanding even when the ideal signal is nonzero.	Sufficient theoretical budget, not evidence of experimental attainability. State, gate and timing conventions differ between directions and are not interchangeable. <i>Next</i> : Find feasible apparatus-specific approximations with a justified physical scale calibration.
recent29** Complete-factor spatial collars with factorial connected tails Eqs. (122), (123). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round24/forward/y1/report.md	<i>Use</i> : Finite spatial reduction with explicit error that preserves the endpoint signal. <i>Change</i> : A controlled physical collar replaces an arbitrary geometric crop; its small tail does not by itself justify averaging the regional spectrum.	Collars and counts depend on the stated forward/reverse construction. No regional matrix or infinite limiting generator is computed at this stage. <i>Next</i> : Establish strong averaging on fixed energy blocks and then pass the scalar limit through the collar bound.
recent30** Actual scalar endpoint limit from strong block averaging Eqs. (124), (125), (126), (127), (128). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round24/forward/y2/report.md	<i>Use</i> : An actual limiting endpoint function with quantitative small-window location. <i>Change</i> : U2 nonconvergence is strengthened to existence of the relevant scalar limit; the whole energy shell and the $q=1$ global generator are not thereby constructed.	Fixed small window, scalar limit and strong block topology. Compact resolvent is not a global operator-norm averaging theorem. <i>Next</i> : Compute a reducing reached component or else retain the full regional energy block.
recent31** Exact sixteen-cycle compression and spectral measure Eqs. (129). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round25/solo/aa1/report.md	<i>Use</i> : An exact finite spectral candidate for the endpoint function. <i>Change</i> : An autonomous two-state approximation is replaced by the correct sixteen-state compression, while exterior closure remains an explicit missing hypothesis at AA1.	AA1 gate is limited. The cube compression alone does not transfer to the surrounding or full model. Same-author serial self-review, not independent review. <i>Next</i> : Prove that exterior faces cannot return resonantly into the shell and identify a reducing reached component.
recent32** Exterior selection proves the reached component reduces Eqs. (130), (131). <i>Scope</i> : model-specific derivation, obstruction, or certified computation research/round25/solo/aa2/report.md	<i>Use</i> : Transfers the exact spectral measure to the actual canonical endpoint scalar on the admitted window. <i>Change</i> : AA1 compression becomes a full-model result because exterior resonant escape is excluded, not because the entire energy shell is finite.	Only the reached component of the averaged generator is reducing; instantaneous evolution can leave it. No large- z stationary-limit claim follows from the analytic formula. Same-author review. <i>Next</i> : Attach finite- q and arithmetic errors to the full correlation and analyze simpler multiplier probes.

Continued on next page

Contribution and derivation	Application and change	Limitation and next obligation
recent33** Finite-q approach and numerical endpoint enclosure Eqs. (132). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round25/solo/aa2/report.md	<i>Use:</i> An explicit finite-q scalar enclosure for the original rank observable. <i>Change:</i> The endpoint spectral formula is accompanied by an actual full-model finite-q error budget.	Numerical point is extremely close to $q=1$ and has an enormous uncalibrated physical duration. No experimental feasibility or larger- z theorem. <i>Next:</i> Check the same full finite-q transfer for actual Wilson multipliers with their own operator norms.
recent34** Nonzero-variance elementary Wilson probe is endpoint-null Eqs. (133). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/ad1/report.md	<i>Use:</i> A physical-model null control against inferring a slow coupling from a constant phase-removed scalar. <i>Change:</i> Nonzero variance and an apparently clean signal are insufficient for scale identification; this probe has no endpoint sensitivity to that coupling.	AD1 remains limited as a rate-bridge target. A constant endpoint does not rule out subleading finite-q dependence or sensitivity of another observable. <i>Next:</i> Choose a Wilson multiplier whose created vector reaches a nontrivial averaged block and prove the full scalar transfer.
recent35** Informative single-cycle and sum Wilson endpoints Eqs. (134), (135). <i>Scope:</i> model-specific derivation, obstruction, or certified computation research/round26/forward/ad2/report.md	<i>Use:</i> Actual Wilson-multiplication probes with quadratic or linear small-window endpoint sensitivity. <i>Change:</i> The rank observable is no longer silently identified with multiplication; different physical observables can share an endpoint spectral component yet differ as operators.	Fixed canonical summable model and $0 < z \leq 1e-6$. The coefficient is not an empirical coupling calibration, and no large- z limit or homogeneous/continuum transfer is proved. <i>Next:</i> Declare the observable, clock, physical scales and comparison norm in an actual parameter-identification experiment.

B Complete historical record

This table preserves all 121 recorded study or loop entries. It is an index of the history, not a list of 121 new discoveries. Historical verdicts retain their original scope. The contribution and skeptic ledgers identify the fresh manuscript checks; inclusion here does not claim that every old production simulation was rerun. Complete source paths and SHA-256 bindings are in `audit/history-ledger.json`.

Table 3: All recorded entries, grouped by round.

Record	Historical entry	Verdict and first source
R3 PRODUCTION	Source-off electric reversal	historical study — scope below evidence/qeg-research/round3/README.md
R3 REFINEMENT	Regulator and node refinements	historical study — scope below evidence/qeg-research/round3/README.md
R3 CONTROLS	Matching and independent formulations	historical study — scope below evidence/qeg-research/round3/README.md
R3 BENCHMARK	Retained precision failure	historical study — scope below evidence/qeg-research/round3/README.md

Continued on next page

Record	Historical entry	Verdict and first source
R4 RESPONSE	Source-amplitude tangent	historical study — scope below evidence/qeg-research/round4/README.md
R4 AUDIT	Source and report reconciliation	historical study — scope below evidence/qeg-research/round4/README.md
R5 THEOREM	Finite-model existence and response	historical study — scope below evidence/qeg-research/round5/README.md
R5 PLANNER	Two-front proof planning	historical study — scope below evidence/qeg-research/round5/README.md
R5 GRAVITY	Bianchi-I constraint bridge	historical study — scope below evidence/qeg-research/round5/README.md
R6 COEFFICIENT	Exact finite-window coefficient	historical study — scope below evidence/qeg-research/round6/README.md
R6 COFINAL	Cofinal cutoff obstruction	historical study — scope below evidence/qeg-research/round6/README.md
R6 TANGENT	Stationary second variation	historical study — scope below evidence/qeg-research/round6/README.md
R6 REPAIR	Core and coefficient audit	historical study — scope below evidence/qeg-research/round6/README.md
R7 FINITE- SCALAR	Scalar-coupled finite model	historical study — scope below evidence/qeg-research/round7/README.md
R7 GRAVITY- SCALAR	Reduced gravitational simulation	historical study — scope below evidence/qeg-research/round7/README.md
R7 SUBTRACTION	Reference-subtraction and bounded-addition tests	historical study — scope below evidence/qeg-research/round7/README.md
R8 SPECTRAL	Spectral and finite-volume counterexamples	historical study — scope below research/round8/README.md
R8 DELAYED	Delayed scalar-response study	historical study — scope below research/round8/README.md
R8 PROOF	Conditional proof replay	historical study — scope below research/round8/README.md
R9 HAAR	Nonabelian identity hierarchy	historical study — scope below research/round9/README.md
R9 WILSON	Four-dimensional Wilson lattice runs	historical study — scope below research/round9/README.md
R9 TRANSFER	Central-convolution spectrum	historical study — scope below research/round9/README.md

Continued on next page

Record	Historical entry	Verdict and first source
R10 GAP	Certified full one-plaquette gap	historical study — scope below research/round10/README.md
R10 DRIVE	Driven one-square diagnostics	historical study — scope below research/round10/README.md
R10 QUOTIENT	Three-trace reconstruction	historical study — scope below research/round10/README.md
R11 OPERATOR	Physical two-plaquette operator	historical study — scope below research/round11/README.md
R11 GAP	Continuous two-plaquette gap certificate	historical study — scope below research/round11/README.md
R11 DYNAMICS	Driven checks and falsifiers	historical study — scope below research/round11/README.md
R12 DYNAMIC-TAIL	Rigorous representation-error theorem	historical study — scope below research/round12/README.md
R12 VOLUME	Growing-graph bound and counter-benchmark	historical study — scope below research/round12/README.md
R12 CLOSURE	Compact hierarchy audit	historical study — scope below research/round12/README.md
R13 LOCALITY	Local finite-time volume limit	historical study — scope below research/round13/README.md
R13 STABILITY	Qualitative homogeneous stability	historical study — scope below research/round13/README.md
R13 MOMENTS	Compact moment certificates and uniqueness	historical study — scope below research/round13/README.md
R13 RESPONSE	Exact scalar response	historical study — scope below research/round13/README.md
R14 LOOP1	Exact conditional Haar reduction with removable zero field.	accepted within declared finite-model scope research/round14/advisor/loop1_gate.json
R14 LOOP2	23 exact finite-measure covariance enclosures independently replayed using characters.	accepted within declared finite-model scope research/round14/advisor/loop2_gate.json
R15 A1	All8 positive exact midpoints, all8 coarse transported lower bounds insufficient.	accepted correctly diagnosed insufficiency research/round15/advisor/loop-gates/A1.json
R15 A2	Refined failing cell radii rather than increasing already sufficient point precision.	accepted exact complete positive cover research/round15/advisor/loop-gates/A2.json
R15 B1	Closed cube has six-way dependence while any five face class observables factorize; use actual character gluing and distinct face derivatives in B2.	accepted exact finite cube graph and Haar identities research/round15/advisor/loop-gates/B1.json
R15 B2	Character dimension and separate face derivatives retained; coarse degree failures remain explicit.	accepted exact finite cube integral and rejected factorization research/round15/advisor/loop-gates/B2.json

Continued on next page

Record	Historical entry	Verdict and first source
R15 C1	Cube lower bound is zero at common $\lambda=\alpha/2$; accepted B1 trial moments permit a variational ground-energy upper bound.	accepted finite-graph side theorem and retained original source gap research/round15/advisor/loop-gates/C1.json
R15 C2	The independent E1 lower bound and trial E0 upper bound repair the C1 zero-margin fixture; no trial gap was substituted for the full gap.	accepted finite physical cube variational improvement research/round15/advisor/loop-gates/C2.json
R16 LOOP1	Exact finite two-cube Haar/fusion identities and separate conditional physical energy-scale audit.	accepted research/round16/advisor/loop1-gate.e.json
R16 LOOP2	Positive normalized all-eleven observable difference for outer1/8 and shared1/8 versus0, exact width $\leq 1e-12$.	accepted research/round16/advisor/loop2-gate.e.json
R17 A1	Exact local projector and incidence bounds; physical counterexample to the full pure relative-form shortcut. Dense uniform threshold remains open.	accepted research/round17/advisor/a1-gate.json
R17 A2	Volume-uniform physical gap for the pairwise link-disjoint magnetic-support family with a common positive energy scale; dense overlapping family remains open.	accepted research/round17/advisor/a2-gate.json
R17 B1	Improved full physical gap lower bound on the dense eleven-face two-cube graph, positive for common $ \lambda /\alpha < 12/43$; finite volume only.	accepted research/round17/advisor/b1-gate.json
R17 B2	Exact positive full physical gap lower $\alpha \cdot 1388/284767457$ at the dense finite coupling $\lambda/\alpha=12/43$, with a proved open variational-amplitude interval.	accepted research/round17/advisor/b2-gate.json
R17 C1	Complete four-adjoint Haar tensor and realized four-cube conditional boundary; no bulk or spectral claim.	accepted research/round17/advisor/c1-gate.json
R17 C2	Same-action conditional four-face expectations with complete rational numerator and partition errors; action-only observable closure rejected, full bulk and spectrum open.	accepted research/round17/advisor/c2-gate.json
R18 A1	Finite dressed three-square bridge; physical gap at least $\alpha/8$ throughout the declared primary family; no homogeneous volume extension.	accepted research/round18/advisor/a1-gate.json
R18 A2	All finite volumes of the specified spatially decaying inhomogeneous family have physical gap at least $7\alpha_{\min}/64$; boundary reclassification and homogeneous dense/thermodynamic limits remain explicit.	accepted research/round18/advisor/a2-gate.json

Continued on next page

Record	Historical entry	Verdict and first source
R18 B1	Complete physical complement threshold $9\alpha/2$ and exact QVP Gram on the dense two-cube graph; no finite trial spectrum substituted for the entire omitted space.	accepted research/round18/advisor/b1-gate.json
R18 B2	Full untruncated physical two-cube signed coefficient-box gap lower bound, separate endpoint ground trial, and bare-coupling domain obstruction. No volume-uniform or continuum claim.	accepted research/round18/advisor/b2-gate.json
R18 C1	Complete joint-Gram sufficiency for the declared central S3 observable/action class, exact rational admissibility including singular cases, and independently matched degree0–6 diagnostics. No minimality, finite-tail, surrounding measure or mass-gap conclusion.	accepted research/round18/advisor/c1-gate.json
R18 C2	Full normalized two-link conditional six-face integral at the declared signed fixtures, exact tails, actual surrounding Gram/measure connection and preserved omission/symmetry controls. Thirty-one links remain fixed; no physical spectrum or full bulk/continuum claim.	accepted research/round18/advisor/c2-gate.json
R19 A1	For every reachable selected clipped component L, M, R, LM, MR or LMR under the frozen coefficient bounds $ \lambda_L /\alpha \leq 1/2$, $ \mu_M /\alpha \leq 1/8$ and $ \lambda_R /\alpha \leq 1/2$, the full-link local gap lower estimate is at least $\alpha/8$. Positive full-link uniqueness and gauge invariance permit the same lower estimate after Gauss-sector restriction for the finite local component result.	accepted research/round19/advisor/a1-gate.json
R19 A2	Goal A Loop 2 only: direct incomplete product-representation construction for a summable omitted-face perturbation of the accepted A1 complete-strip/free reference. This is not finite clipped-restriction convergence, homogeneous dense stability, continuum Yang-Mills or a mass-gap theorem.	accepted research/round19/advisor/a2-gate.json
R19 B1	Goal B Loop 1 only: complete strict-cutoff physical projector and exact full cross Gram on the actual finite two-cube SU(2) graph. No continuous coefficient box, dense limit, continuum Yang-Mills or mass-gap claim is accepted here.	accepted research/round19/advisor/b1-gate.json

Continued on next page

Record	Historical entry	Verdict and first source
R19 B2	Goal B Loop 2 only: finite two-cube continuous signed coefficient-box gap certificate using the accepted B1 48-channel projector, full compressed PVP ledger and exact ordered W^*W cross Gram. No dense limit, continuum Yang-Mills, Clay mass gap or physical-scale matching claim is accepted.	accepted research/round19/advisor/b2-gate.json
R19 C1	Goal C Loop 1 only: exact finite static U-V-W chain integral on the actual four-cube graph with all other links fixed. This accepts a static mathematical checkpoint and common-V failed-independence discriminator; it does not accept physical energy/time-scale matching, dense convergence, continuum Yang-Mills, a spectral gap theorem or the Clay mass gap.	accepted research/round19/advisor/c1-gate.json
R19 C2	Static common-V integral only; physical energy/time matching remains open. No Hamiltonian spectral gap or continuum claim.	accepted research/round19/advisor/c2-gate.json
R20 D1	Finite-factor completion gives norm-resolvent convergence with explicit dyadic tail on the exact A2 exterior-reference lift.	accepted research/round20/advisor/d1-gate.json
R20 D2	Ground energy, rank-one projections and bounded observables converge with explicit dyadic errors for D1 lifts.	accepted research/round20/advisor/d2-gate.json
R20 E1	Literal aligned boxes $N=4m+3$ converge after explicit reference-energy subtraction and exact exterior-reference lift.	accepted research/round20/advisor/e1-gate.json
R20 E2	All upper-boundary phases of literal growing rectangular boxes converge to the A2 ground state on bounded local observables.	accepted research/round20/advisor/e2-gate.json
R20 F1	Even among local reversible diffusions the C1 static law does not identify an energy/time coefficient: $H_{\kappa,c}=c A_{\kappa}$.	accepted research/round20/advisor/f1-gate.json
R20 F2	Exact shared-middle gradient closure, action range $[-5,7]$ and conditional diffusion gap $c/6$ on $\kappa \leq 1/8$.	accepted research/round20/advisor/f2-gate.json
R20 G1	Literal-box real-time dynamics converges in operator norm on each local observable at fixed physical time, with a strongly continuous Hilbert implementation.	accepted research/round20/advisor/g1-gate.json
R20 G2	The E2 limiting local state GNS on the full factor algebra is the A2 representation with canonical generator H_e and its inherited gap.	accepted research/round20/advisor/g2-gate.json

Continued on next page

Record	Historical entry	Verdict and first source
R20 H1	The continuous omitted-face profile has an exact budget and a rigorously delimited sufficient gap certificate at fixed physical scale.	accepted research/round20/advisor/h1-gate.json
R20 H2	Every strict uniform global perturbation-budget ray returns the actual ground projection to the selected-strip reference, with an exact vacuum variance and a quantitative rate.	accepted research/round20/advisor/h2-gate.json
R21 I1	Complete 24-link blocking gives 21 omitted anchor faces and exact normalized local interaction norm $7 \tau $; qualitative fixed-spacing stability applies with unevaluated source constants.	limited research/round21/advisor/i1-gate.json
R21 I2	Exact one-star vacuum rotation cancels first-order vacuum mixing and retains a bounded quadratic remainder; a bare pure relative bound for the original interaction fails.	limited research/round21/advisor/i2-gate.json
R21 J1	An actual omitted Wilson loop has a nonzero gauge-invariant fluctuation in the perturbed dyadic vacuum, with a uniform rational variance floor.	accepted research/round21/advisor/j1-gate.json
R21 J2	Physical vacuum GNS space equals the gauge-invariant subspace of this representation, with the restricted shifted energy generator and a nonzero imaginary-time Wilson correlator.	accepted research/round21/advisor/j2-gate.json
R21 K1	A positive state-dependent mobility preserves the static density but changes reversible dynamics; two chosen Haar correlation slopes identify c and ζ within this conditional family.	accepted research/round21/advisor/k1-gate.json
R21 K2	A tunable observable gives a uniform two-rate inverse across the declared static interval, while a positive blind cubic proves that two slopes do not identify arbitrary mobility.	accepted research/round21/advisor/k2-gate.json
R21 L1	A third observable detects the previously hidden cubic mobility and gives an exact three-parameter inverse in the declared conditional family.	accepted research/round21/advisor/l1-gate.json
R21 L2	Any finite list of initial correlation slopes leaves strictly positive mobility freedom; a second derivative detects the explicit hidden fifth-degree direction.	accepted research/round21/advisor/l2-gate.json
R21 M1	The fixed-budget summable perturbation converges strongly to zero, giving strong-resolvent and compact-time strong dynamics convergence to the selected-strip reference.	accepted research/round21/advisor/m1-gate.json

Continued on next page

Record	Historical entry	Verdict and first source
R21 M2	Connected physical correlations converge on explicitly growing time windows in the canonical summable profile, with a nonzero actual physical Wilson fluctuation.	accepted research/round21/advisor/m2-gate.json
R22 N1	Stationary Heisenberg comparison proves the sharper local connected-correlation bound in the canonical summable model.	accepted research/round22/advisor/n1-gate.json
R22 N2	Complete tail-block cubes have an exact weighted incident-face budget, yielding a joint support/time convergence certificate.	accepted research/round22/advisor/n2-gate.json
R22 O1	Complete simultaneous homogeneous rotation has explicit weighted remainder with preserved unbounded operator domain.	limited research/round22/advisor/o1-gate.json
R22 O2	The bare-reference elimination has explicit all-support bounds, but its retained-diagonal loss majorants cannot certify an infinite iteration at nonzero coupling.	limited research/round22/advisor/o2-gate.json
R22 P1	The prescribed section is nonclosable and its fitted clock fails the reserved correlation.	accepted research/round22/advisor/p1-gate.json
R22 P2	A genuine Haar reduction exactly transports the finite electric dynamics; the declared conditional family fails the reserved data.	accepted research/round22/advisor/p2-gate.json
R22 Q1	The full magnetic action has correlated positive conditional variance, proving selected-image leakage and a quantitative nonautonomous dynamic defect.	accepted research/round22/advisor/q1-gate.json
R22 Q2	The actual magnetic projection has exact memory/resolvent identities, seven-energy fixed-channel leading matrices and explicit cubic full-Hilbert operator remainders.	accepted research/round22/advisor/q2-gate.json
R22 R1	The actual initial diagonal has a uniform local inverse, while its local homological generator leaves an exact nonzero SU2 crossing-boundary defect.	accepted research/round22/advisor/r1-gate.json
R22 R2	The actual initial boundary-complete vacuum response converges with explicit spatial and graph tails; its source-sector and specified volume-weight limits are exact.	accepted research/round22/advisor/r2-gate.json
R23 S1	The genuine cubic indexed O1 source is nonzero; its initial interior inverse preserves the graph domain and leaves an explicitly bounded connected crossing interaction.	limited research/round23/advisor/s1-gate.json

Continued on next page

Record	Historical entry	Verdict and first source
R23 S2	A compact-time regulated full-source commutator map preserves the actual graph domain and has complete connected weight2 bounds; its sinc residual is retained exactly.	accepted research/round23/advisor/s2-gate.json
R23 T1	The full selected magnetic evolution is approximated by the unique leading-memory return equation with cubic fixed-time and uniform absolute quadratic coupling errors.	accepted research/round23/advisor/t1-gate.json
R23 T2	Separately ground-centered full-selected heat evolutions differ by at most $60 \lambda^2$ uniformly for all $t \geq 0$ on the frozen finite physical graph.	accepted research/round23/advisor/t2-gate.json
R23 U1	Actual eight-link gauge-invariant resonance and first-order endpoint coefficient $C \eta/84$; higher-order endpoint behavior unresolved in U1.	limited research/round23/advisor/u1-gate.json
R23 U2	Actual full-system endpoint nonconvergence for the frozen bounded local gauge-invariant autocorrelation: $\liminf \text{error} > C \eta/168$ for fixed $0 < C \eta \leq 10^{-6}$.	accepted research/round23/advisor/u2-gate.json
R24 V1	Three outcomes with a uniform error budget	accepted_within_scope research/round24/advisor/v1-gate.json
R24 V2	Finite confidence, with explicit resource costs	accepted_within_scope research/round24/advisor/v2-gate.json
R24 W1	The actual short-filter residual stays cubic	accepted_within_scope research/round24/advisor/w1-gate.json
R24 W2	What repeated filtering actually converges to	limited research/round24/advisor/w2-gate.json
R24 X1	An actual 21-state gauge sector, with its leakage	limited research/round24/advisor/x1-gate.json
R24 X2	A small computation certifies the full graph	accepted_within_scope research/round24/advisor/x2-gate.json
R24 Y1	A fixed spatial collar retains the endpoint signal	accepted_within_scope research/round24/advisor/y1-gate.json
R24 Y2	The endpoint correlation has an actual scalar limit	accepted_within_scope research/round24/advisor/y2-gate.json
R24 Z1	An actual obstruction to uniform relative accuracy	accepted_within_scope research/round24/advisor/z1-gate.json

Continued on next page

Record	Historical entry	Verdict and first source
R24 Z2	Useful relative accuracy for specified inputs, at every heat time	accepted_within_scope research/round24/advisor/z2-gate.json
R25 AA1	Complete physical 16-state cube shell, compressed matrix and U spectral measure	limited research/round25/advisor/aa1-gate.json
R25 AA2	Proved 16-state observable-reachable reducing component; exact canonical scalar endpoint and evaluated finite-q error	accepted_within_scope research/round25/advisor/aa2-gate.json
R26 AB1	AB1 / Locate the resonant source in the boundary defect	limited research/round26/advisor/ab1-gate.json
R26 AB2	AB2 / Contract the full source at finite spectral resolution	proved-in-model research/round26/advisor/ab2-gate.json
R26 AC1	AC1 / Retain every old residual channel in 293 physical states	limited research/round26/advisor/ac1-gate.json
R26 AC2	AC2 / Control delayed loading and sharpen all-time heat accuracy	proved-in-model research/round26/advisor/ac2-gate.json
R26 AD1	AD1 / Identify an elementary Wilson observable with a null slow endpoint	limited research/round26/advisor/ad1-gate.json
R26 AD2	AD2 / Certify nontrivial endpoints for actual six-cycle Wilson observables	proved-in-model research/round26/advisor/ad2-gate.json
R26 AE1	AE1 / Localize the Gaussian correction with all-time control	proved-in-model research/round26/advisor/ae1-gate.json
R26 AE2	AE2 / Combine source contraction with the original support-size norm	proved-in-model research/round26/advisor/ae2-gate.json
R26 AF1	AF1 / Compute and certify the actual 293-coordinate heat vector	proved-in-model research/round26/advisor/af1-gate.json
R26 AF2	AF2 / Certify heat evaluation at every time on the original input class	proved-in-model research/round26/advisor/af2-gate.json

C Reproduction and manuscript build

The accompanying source bundle contains this manuscript’s L^AT_EX files, bibliography, vector figures, calculator code, machine-readable historical and contribution ledgers, and independent audit scripts. The standalone `calculators.html` runs locally without a server. Use a fresh Python environment for the companion requirements; plot rendering dependencies are listed separately by the calculator guide. The companion was exercised with Python 3.12.14, mpmath 1.3.0, NumPy 2.3.5, and Matplotlib 3.10.8. Its default working precision is 70 decimal digits.

Exact rational outputs use Python’s standard-library fraction arithmetic; a 70-digit diagnostic is not thereby an interval enclosure.

Representative commands, executed from the unpacked manuscript directory, are:

```
python3 -m pip install -r calculators/requirements.txt
python3 calculators/ym_calculators.py jacobi --ratio 2 --dimension 24
python3 calculators/ym_calculators.py static --kappa 1
python3 calculators/ym_calculators.py filter --kind gaussian \
    --M 35/1664 --duration 4
python3 calculators/ym_calculators.py wilson --z 1/1000000
python3 calculators/ym_calculators.py heat-certificates
python3 calculators/test_calculators.py --output /tmp/ym-calculator-tests.json
python3 audit/skeptic_checks.py --output /tmp/ym-skeptic-checks
python3 audit/skeptic_early_checks.py --output /tmp/ym-early-checks
python3 audit/skeptic_middle_checks.py --output /tmp/ym-middle-checks
python3 audit/skeptic_recent_checks.py --output /tmp/ym-recent-checks
node audit/test_web_calculators.cjs
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The skeptic commands require fresh output directories; use new names on a second run. The heat companion computes certificate budgets; it is not a replacement for the original 293-coordinate solver. To obtain a certified vector, clone the workbench, check out the exact commit in Section 1, and use a fresh absolute output directory:

```
python3 -B research/round26/reverse/af2/check.py \
    --sigma 7/2 --output /absolute/new/heat-at-seven-halves
```

The optional `--input-vector PATH` takes a JSON list of 293 real rational strings and checks exact normalization, original P_{21} support, and the radius-0.01 condition. The reverse command interface does not accept arbitrary complex coordinates. Forward exports include complex demonstrations under a separate export-error allowance. The operator estimate is broader than the implemented input interface; those scopes must remain distinct.

The workbench’s original historical guides remain authoritative for their production commands and dependencies. Some early programs write outputs in place, so run them only in an isolated checkout. Never remove a bound source hash or reinterpret a missing test as a proved premise to make a replay pass. The new source bundle adds no research loop and changes no frozen historical record.